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Unit Six Interview Ernesto Gianoli.

Chapter 35 **35.1 tree** Raimund Linke/Photodisc/Getty Images; **seedling** Beata Becia/Shutterstock; **leaf cross section** P&R Fotos/AGE Fotostock/Alamy Stock Photo; **chloroplasts** John Durham/Science Source; **tube-shaped cells** Science Photo Library/Alamy Stock Photo; **root hairs** Scenics & Science/Alamy Stock Photo; **35.3** Jeremy Burgess/Science Source; **35.4 buttress roots** Karl Weidmann/Science Source; **prop roots** Natalie Bronstein; **beet** Rob Walls/Alamy Stock Photo; **pneumatophores** Bjorn Svensson/AGE Fotostock/Alamy Stock Photo; **strangling roots** Dana Tezarr/Photodisc/Getty Images; **35.5 rhizomes** Maureen Spuhler/Seelelver.com; **stolons** Dorling Kindersley Ltd/Alamy Stock Photo; **tubers** Toshihiko Watanabe/Aflo/Alamy Stock Photo; **Scientific Skills Exercise** Dorling Kindersley Ltd/Alamy Stock Photo; **35.7 tendrils** Neil Cooper/Alamy Stock Photo; **spines** Martin Ruegner/Photodisc/Getty Images; **storage leaves** Dmytro Skorobogatov/123RF; **reproductive leaves** Godunova Tatiana/Shutterstock; **35.9** Steve Gschmeissner/SPL/AGE Fotostock; **35.10 parenchyma** M I (Spike) Walker/Alamy Stock Photo; **collenchyma** Keith Wheeler/Science Source; **scleireid** Graham Kent/Pearson Education; **fiber** Graham Kent/Pearson Education; **tracheids and vessels** N.C. Brown Center for Ultrastructure Studies; **sieve-tube element (TEM)** From: *Plant Cell Biology* on DVD: Information for students and a resource for teachers Springer-Verlag 2009, by B Gunning; **sieve-tube element (LM)** Ray F. Evert; **sieve plate** Graham Kent/Pearson Education; **35.13** From: ABA-mediated ROS in mitochondria regulate root meristem activity by controlling PLETHORA expression in *Arabidopsis*. Yang L. *PLoS Genet.* 2014 Dec 18;10(12):e1004791. doi: 10.1371/journal.pgen.1004791. eCollection 2014 Dec. Figure 6G; **35.14a, b** Ed Reschke; **35.14a inset** Chuck Brown/Science Source; **35.15** Michael Clayton; **35.16** Ed Reschke/Stone/Getty Images; **35.17, 35.18** Ed Reschke/Stone/Getty Images; **35.20 left** Michael Clayton; **right** Alison W. Roberts; **35.23** University of Southern California; **35.25** Edu Boer, NVWA, NL; **35.26** From: Natural variation in *Arabidopsis*: from molecular genetics to ecological genomics. D. Weigel. *Plant Physiol.* 2012 Jan;158(1):2-22. doi: 10.1104/pp.111.189845. Epub 2011 Dec 6. Fig. 1; **Chory** Micheline Pelletier/Courtesy of Joanne Chory; **35.28** From: Microtubule plus-ends reveal essential links between intracellular polarization and localized modulation of endocytosis during division-plane establishment in plant cells. P. Dhonukshe. *BMC Biol.* 2005 Apr 14;3:11. Fig4B; **35.28 inset** B. Wells and K. Roberts; **35.29** Bart-Jan Janssen, Lance Lund, Neelima Sinha, Overexpression of a Homeobox Gene, LeT6, Reveals Indeterminate Features in the Tomato Compound Leaf, *Plant Physiology*, Volume 117, Issue 3, July 1998, Pages 771–786, <https://doi.org/10.1104/pp.117.3.771>; **35.31** Lawrence Jensen; **35.32** From: Genetic interactions among floral homeotic genes of *Arabidopsis*. J.L. Bowman, DR Smyth, EM Meyerowitz. *Development*. 1991 May;112(1):1-20; Fig. 1A; **Walbot** Courtesy of Virginia Walbot; **Test Your Understanding woody eudicot** From: Anatomy of the vessel network within and between tree rings of *Fraxinus lanuginosa* (Oleaceae). P. B. Kitin et al. *American Journal of Botany*. 2004;91:779-788. Fig. 1; **tea leaves** Volodymyr Burdiak/Shutterstock; **tea leaf cross section** Keith Wheeler/Science Source; **iris leaves** Rob Stark/Shutterstock; **iris leaf cross section** www.willemsmicroscope.com; **bicycle** Janet Horton/Alamy Stock Photo; **Synthesize Your Knowledge** Biophoto Associates/Science Source.

Chapter 36 **36.1** velislava/Alamy Stock Photo; **36.3** Rolf Rutishauser and Evelin Pfeifer; **plants** Nigel Cattlin/Alamy Stock Photo; **36.8** Benjamin Blonder and David Elliott; **36.10** Scott Camazine/Science Source; **36.13** age fotostock/Alamy Stock Photo; **36.14** Power and Syred/Science Source; **36.15 leafless ocotillo** mike lane/Alamy Stock Photo; **green ocotillo** Rick & Nora Bowers/Alamy Stock Photo; **inset** Steven J. Baskauf <http://bioimages.vanderbilt.edu>; **oleander cross section** Natalie Bronstein; **inset** SutidaS/Shutterstock; **cactus** Danita Delimont/Alamy Stock Photo; **36.18** M. H. Zimmerman/Harvard Forest; **36.19** From: A coiled-coil interaction mediates cauliflower mosaic virus cell-to-cell movement. L. Stavalone et al. *Proc Natl Acad Sci U S A*. 2005 Apr 26;102(17):6219-24. Epub 2005 Apr 18. Fig. 5c; **Zambryski** Patricia C. Zambryski; **Synthesize Your Knowledge** Catalin Petolea/Alamy Stock Photo.

Chapter 37 **37.1** Visuals Stock/Alamy Stock Photo; **37.2** ARS/USDA; **37.4** National Oceanic and Atmospheric Administration (NOAA); **37.5** Menlo Park/USGS; **37.6** Chronicle/Alamy Stock Photo; **Coruzzi** Courtesy of Gloria M. Coruzzi; **37.8 healthy** View Stock RF/AGE Fotostock; **nitrogen-deficient** Guillermo Roberto Pugliese/International Plant Nutrition Institute; **phosphorus-deficient** C. Witt/International Plant Nutrition Institute; **potassium-deficient** Rupinder singh 0071/Shutterstock; **Scientific Skills Exercise** Sarah2/Shutterstock; **37.9 lichen** David T. Webb, University of Montana; **lichen section** Courtesy of Ralf Wagner; **puffer** Andrey Nekrasov/Pixtal/AGE Fotostock; **azolla** Daniel L. Nickrent; **ant** Juan Carlos Vindas/Moment Open/Getty Images; **fungal garden** Martin Dohrn/Nature Picture Library; **root** Yoshihiro Kobae; **nectar** Oxford Scientific/The Image Bank/Getty Images; **37.10** Sarah Lydia Lebeis; **37.12** Scimat/Science Source; **37.14 sheath** Hugues B. Massicotte/University of Northern British Columbia Ecosystem and Management Program, Prince George, BC, Canada; **cells, arbuscules** Mark Brundrett; **37.15 fern** David Wall/Alamy Stock Photo; **mistletoe** Peter Lane/Alamy Stock Photo; **dodder** Emilio Erez/Alamy Stock Photo; **ghost pipe** Martin Shields/Alamy Stock Photo; **sundew** Fritz Polking/Frank Lane Picture Agency Limited W. Rolfes/Arco Images GmbH/Alamy Stock Photo; **pitcher plants** Dorling Kindersley Ltd/Alamy Stock Photo; **ant** Paul Zahl/Science Source; **Venus flytrap** Chris Mattison/Nature Picture Library/Alamy Stock Photo; **Synthesize Your Knowledge** Wayne Forward/Mode Images/Alamy Stock Photo.

Chapter 38 **38.1** blinkwinkel/Alamy Stock Photo; **inset** Nicolas J. Vereecken; **38.4 hazel carpellate** Friedhelm Adam/imageBROKER/Getty Images; **hazel staminate** Wildlife GmbH/Alamy Stock Photo; **dandelions** © Björn Rørslett/NN/Samfoto/Sipa USA; **moth**

Doug Backlund/WildPhotosPhotography.com; **blowfly** Kjell B. Sandved/Science Source; **bat** Danita Delimont/Shutterstock; **hummingbird** Rolf Nussbaumer Photography/Alamy Stock Photo; **38.5** W. Barthlott and W.Rauh/Nees Institute for Biodiversity of Plants; **38.6 top to bottom** Michael Clayton/Botany Dept., University of Wisconsin; Ed Reschke/Stone/Getty Images; **38.10** Blickwinkel/Alamy Stock Photo; **38.12 coconut** Kevin Schafer/Alamy Stock Photo; **Alsomitra macrocarpa** adrian daves/Alamy Stock Photo; **maple seed** AlessandroZocc/Shutterstock; **dandelion** Steve Bloom Images/Alamy Stock Photo; **tumbleweed** Nurlan Kalchinov/Alamy Stock Photo; **Tribulus terrestris** California Department of Food and Agriculture's Plant Health and Pest Prevention Services; **feces** Sergei Prokhorov/Shutterstock; **squirrel** Alan Williams/Alamy Stock Photo; **ant** Benoit Guénard; **38.13** Dennis Frates/Alamy Stock Photo; **Scientific Skills Exercise monkeyflower** Ken Barber/Alamy Stock Photo; **hummingbird** Dec Hogan/Shutterstock; **38.14a** Marcel Dorken; **38.14b** Nobumitsu Kawakubo; **38.15** Meriel G. Jones, University of Liverpool School of Biological Sciences; **38.16** Dorling Kindersley Ltd/Alamy Stock Photo; **Herrera-Estrella** Michael Starghill Photography; **38.17** Gary P. Munkvold; **38.18** ton koene/Alamy Stock Photo; **Gonsalves** Courtesy of Dennis Gonsalves; **Synthesize Your Knowledge** Pictorial Press Ltd/Alamy Stock Photo.

Chapter 39 **39.1** Christopher Ison/Alamy Stock Photo; **39.2** Natalie Bronstein; **39.6** From: Regulation of polar auxin transport by AtPIN1 in *Arabidopsis* vascular tissue. L. Gälweiler et al. *Science*. 1998 Dec 18;282(5397):2226-30; Fig. 4; **39.9a** Richard Amasino; **39.9b** Fred Jensen, Kearney Agricultural Center; **39.11 mangrove** Nublee bin Shamsu Bahar/Shutterstock; **maize** Karen E. Koch; **39.13a** Kurt Stepnitz; **39.13b** Joseph J. Kieber; **39.14** Ed Reschke/Stone/Getty Images; **39.16** Nigel Cattlin/Alamy Stock Photo; **39.18** Martin Shields/Alamy Stock Photo; **Satter** Robin Heyden/Pearson Education; **39.22** Michael L. Evans/Ohio State University; **39.23** From the cover of *Cell*, Volume 60, Issue 3, 9 February 1990. Janet Braam, Ronald W. Davis. Used by permission, Copyright ©1990 Cell Press. Image courtesy of Elsevier Sciences, Ltd; **39.24** Martin Shields/Alamy Stock Photo; **39.25** J. L. Basq/M. C. Drew; **39.26** New York State Agricultural Experiment Station/Cornell University College of Agriculture and Life Sciences; **39.27** poppy seed De Meester Johan/Arterra Picture Library/Alamy Stock Photo; **taro plant** David T. Webb; **olive leaf** Science Photo Library/Alamy Stock Photo; **cactus spines** Susumu Nishinaga/Science Source; **snowflake plant** Giuseppe Mazza; **passion flower** Lawrence E. Gilbert/University of Texas-Austin; **hummingbird** Danny Kessler; **bamboo plants** Kim Jackson/Mode Images/Alamy Stock Photo; **wasp, cocoons** Custom Life Science Images/Alamy Stock Photo; **Dangl** Courtesy of Jeff Dangl; **Synthesize Your Knowledge** Gary Crabbe/Alamy Stock Photo.

Unit Seven Interview Bratcher Adrienne Bratcher; **physiocatch** University of Louisville.

Chapter 40 **40.1 penguins** Doug Allan/Bluegreen Pictures/Alamy Stock Photo; **form** CNRS/IPEV/IPHC, France; **behavior** Nature Picture Library/Alamy Stock Photo; **40.2 seal** Dave Fleetham/Alamy Stock Photo **penguin** WILDLIFE GmbH/Alamy Stock Photo; **tuna** ArteSub/Alamy Stock Photo; **40.4 intestine** Eye of Science/Science Source; **lung, kidney** Susumu Nishinaga/Science Source; **40.5 epithelia** Steve Downing/Pearson Education; **loose connective tissue, adipose tissue, bone, skeletal muscle** Nina Zanetti/Pearson Education; **blood** Jarun Ontakrai/Shutterstock; **cartilage** Chuck Brown/Science Source; **fibrous connective tissue, smooth muscle, cardiac muscle** Ed Reschke/Photolibrary/Getty Images; **neuron** James Cavallini/Science Source; **glia** Thomas Deerinck; **40.7 otter** Kaufung Agency/blickwinkel/Alamy Stock Photo; **bass** Roderick Paul Walker/Alamy Stock Photo; **40.11** Meiqianbao/Shutterstock; **40.12a** Paul Souders/Danita Delimont Creative/Alamy Stock Photo; **40.12b** Bill Gozansky/Alamy Stock Photo; **40.15** Mirko Graul/Shutterstock; **40.16** From: Assessment of oxidative metabolism in brown fat using PET imaging. Otto Muzik, Thomas J. Mangner and James G. Granneman. *Front. Endocrinol.*, 08 February 2012 | <http://dx.doi.org/10.3389/fendo.2012.00015> Fig. 2; **40.20** Jeff Rodman/Alamy Stock Photo; **40.22** FLPA/Alamy Stock Photo; **40.24 sunflowers** PhiloPhotos/iStock/Getty Images Plus; **fly** WildPictures/Alamy Stock Photo; **plants** Irin-K/Shutterstock; **fox** Jo Rabhans/Panther Media GmbH/Alamy Stock Photo; **sprouts** Bogdan Wankowicz/Shutterstock; **molting** Nature's Images/Science Source; **plant vessels** Last Refuge/Robert Harding Picture Library Ltd/Alamy Stock Photo; **blood vessels** Susumu Nishinaga/Science Source; **root hairs** FLPA/Alamy Stock Photo; **intestinal lining** David M. Martin/Science Source; **peas** Scott Rothstein/Shutterstock; **pigs** steven goodier/Alamy Stock Photo; **mesophyll** Steve Gschmeissner/SPL/Getty Images; **alveoli** David M. Phillips/Science Source; **Bartholomew** Robin Heyden **Synthesize Your Knowledge** mapman/Shutterstock.

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Chapter 42 **42.1** John Cancalosi/Alamy Stock Photo; **42.2a** Reinhard Dirscherl/mauritus images GmbH/Alamy Stock Photo; **42.2b** Eric Grave/Science Source; **sea spider** cbimages/Alamy Stock Photo; **42.9 artery** Jose Luis Calvo/Shutterstock; **capillary** Ed Reschke/Stone/Getty Images; **Yanagisawa** Skeeter Hagler; **42.18** Eye of Science/Science Source; **42.19** Image Source Plus/Alamy Stock Photo; **Scientific Skills Exercise** Jarun Ontakrai/Shutterstock; **42.21a** © DeepSeaPhotography.com; **42.21b** Olysha/Shutterstock; **42.21c** Greg Apmptman/Shutterstock; **42.23c** Prepared by Dr. Hong Y. Yan, University of Kentucky and Dr. Peng Chai, University of Texas; **42.24** Motta and Macchiarelli, Anatomy Dept., Univ. La Sapienza, Rome/Science Source; **42.27** Hans-Rainer Duncker, Institute of Anatomy and Cell Biology, Justus-Liebig-University Giessen; **42.32** Doug Allan/Nature Picture Library/Alamy Stock Photo; **Synthesize Your Knowledge** CB2/ZOB/WENN.com/Newscom.

Chapter 43 **43.1 macrophage** SPL/Science Source; **virus** James Cavallini/BSIP SA/Alamy Stock Photo; **bacterium** Chris Bjornberg/Science Source; **fungus** Callista Images/Cultura Creative (RF)/Alamy Stock Photo; **influenza** Kateryna Kon/Shutterstock; **43.15** Steve Gschmeissner/Science Source; **43.28** CNRI/Science Source; **Scientific Skills Exercise** Eye of Science/Science Source; **Wong-Staal** Bill Branson/NIH; **43.30** Stephen C. Harrison/The Laboratory of Structural Cell Biology/Harvard Medical School; **zur Hausen** Tobias Schwerdt; **vaccine** Tatan Yuflana/AP Images.

Chapter 44 **44.1** David Wall/Alamy Stock Photo; **44.2** Mark Conlin/VWPICS/Visual&Written SL/Alamy Stock Photo; **44.4** Eye of Science/Science Source; **Scientific Skills Exercise** Gerhard Koertner/Avalon.red/Alamy Stock Photo; **44.6 fish** GeorgePeters/E+/Getty Images; **squirrel** Eric Isselée/Shutterstock; **bird** Maksym Gorpenyuk/Shutterstock; **44.7** Stephane Bidouze/Shutterstock; **44.12** Steve Gschmeissner/Science

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Chapter 45 **45.1 top** G&M Therin-Weise/robertharding/Alamy Stock Photo; **bottom** Craig K. Lorenz/Science Source; **45.3** Volker Witte/Ludwig-Maximilians-Universität München; **45.11** Cathy Keifer/123RF; **45.17** Vintage_Space/Alamy Stock Photo; **Problem-Solving Exercise** angellodeco/Shutterstock; **45.22 tadpole** Blickwinkel/Alamy Stock Photo; **frog** Rick & Nora Bowers/Alamy Stock Photo; **Synthesize Your Knowledge** Eric Roubos.

Chapter 46 **46.1 coral** NOAA/USGS; **hydra** Roland Birke/Okapia/Science Source; **nudibranchs** Colin Marshall/FLPA/Alamy Stock Photo; **frogs** Andy Sands/Nature Picture Library; **sperm** Don W. Fawcett/Science Source; **cardinals** William Leaman/Alamy Stock Photo; **zebras** Mike Taylor/Alamy Stock Photo; **46.2** Colin Marshall/FLPA/Alamy Stock Photo; **46.3a** P. de Vries/David Crews; **46.5** Andy Sands/Nature Picture Library; **46.6** John Cancalosi/Alamy Stock Photo; **Scientific Skills Exercise** Andrew Syred/Science Source; **46.12** Design Pics Inc/Alamy Stock Photo; **46.17** LENNART NILSSON, TT/SCIENCE PHOTO LIBRARY **46.21** M.I. Walker/Science Source; **Synthesize Your Knowledge** Dave Thompson/AP Images.

Chapter 47 **47.1** Brad Smith/Stamps School of Art & Design, University of Michigan; **inset** Oxford Scientific/Getty Images; **47.3 top to bottom** Victor D. Vacquier; From: Wave of free calcium at fertilization in the sea urchin egg visualized with fura-2. M. Hafner et al. *Cell Motil Cytoskeleton*. 1988;9(3):271-7. Fig. 1; **47.6** George von Dassow; **47.7 top to bottom** Jürgen Berger/Max Planck Institute for Developmental Biology, Tübingen Germany; Andrew J. Ewald, Johns Hopkins Medical School; **47.13b** Alejandro Diaz Diez/AGE Fotostock/Alamy Stock Photo; **47.14a** P. Huw Williams and Jim Smith, The Wellcome Trust/Cancer Research UK Gurdon Institute; **47.14c** Thomas Poole, SUNY Health Science Center; **47.15b** Keith Wheeler/Science Source; **47.18b** From: Cell lineage analysis in ascidian embryos by intracellular injection of a tracer enzyme. III. Up to the tissue restricted stage. H. Nishida. *Dev Biol*. 1987 Jun;121(2):526-41. Fig. 1. Reprinted by permission of Academic Press; **47.19** SDym Photography/Alamy Stock Photo; **47.20** Susan Strome; **47.21** Susan Strome; **47.24** From: Dorsal-ventral patterning and neural induction in *Xenopus* embryos. E. M. De Robertis and H. Kuroda. *Annu Rev Cell Dev Biol*. 2004;20:285-308. Fig. 1; **47.25a** Kathryn Tosney, University of Michigan; **47.26** Based on Honig and Summerbell, courtesy of Lawrence S. Honig; **Synthesize Your Knowledge** FABRICE COFFRINI/AFP/Getty Images.

Chapter 48 **48.1** Franco Banfi/Science Source; **48.2** Edwin R. Lewis; **Olivera** From: QnAs with Baldomero M. Olivera. Beth Azar. *PNAS* August 21, 2012 109 (34) 13470; <https://doi.org/10.1073/pnas.1211581109>. Fig. 1; **48.5** Thomas Deerinck/National Center for Microscopy and Imaging Research, University of California, San Diego; **48.14** Alan Peters; **Synthesize Your Knowledge** Eric B.A.E./Alamy Stock Photo.

Chapter 49 **49.10** Sergi Reboredo/Alamy Stock Photo; **49.11** MriMan/Shutterstock; **49.15** From: A functional MRI study of happy and sad affective states induced by classical music. M. T. Mitterschiffthaler et al. *Hum Brain Mapp*. 2007 Nov. 28(11):1150-62. Fig. 1; **49.18** Marcus E. Raichle, Washington University Medical Center. From research based on "Positron emission tomographic studies of the cortical anatomy of single-word processing". S.E. Petersen et al. *Nature* 331:585-589 (1988); **49.19** National Library of Medicine (NLM); **Jarvis** Walter Oleksy/Alamy Stock Photo; **p. 1103 Heberlein** Matthew Staley; **49.25** Martin M. Rotker/Science Source; **Synthesize Your Knowledge** Eric Delmar/Getty Images.

Chapter 50 **50.1** Gary Meszaros/Science Source; **50.6** CSIRO Publishing; **inset** R. A. Steinbrecht, Max Planck Institute; **50.7a** Laura Morse/NOAA; **50.7b** Grischa Georgiew/Panther Media/AGE Fotostock; **50.9** ALLIMDI.NET/Matthias Lenke/Alamy Stock Photo; **50.10 SPL**/Science Source; **50.16a** APHIS Animal and Plant Health Inspection Service/USDA; **50.17** Steve Gschmeissner/Science Source; **50.21** Neitz Laboratories; **50.26** Clara Franzini-Armstrong; **50.27** H. E. Huxley; **50.34** Joe Quinn/Alamy Stock Photo; **50.39** Dave Watts/Alamy Stock Photo; **Dawson** Terence Dawson; **Scientific Skills Exercise** Vance A. Tucker; **Synthesize Your Knowledge** msgrafixx/Shutterstock.

Chapter 51 **51.1** Robert Koss/Shutterstock; **51.3** Manamana/Shutterstock; **51.5b** Scott Camazine/Alamy Stock Photo; **spider** Ian Fletcher/Shutterstock; **twins** Dustin Finkelstein/Getty Images for SXSW; **51.7** Thomas D. McAvoy/The LIFE Picture Collection/Shutterstock; **51.9** Lincoln Brower/Sweet Briar College; **51.11** Fiona Rogers/Nature Picture Library/Alamy Stock Photo; **51.12** Richard Wragham; **inset** Mike Korostelev www.mkorostelev.com/Moment Open/Getty Images; **Scientific Skills Exercise** Matt Goff; **51.14a** Matt T. Lee; **51.14b** David Osborn/Alamy Stock Photo; **51.14c** imageBROKER/David Tipling/Alamy Stock Photo; **51.15** Fotograferen.net/Alamy Stock Photo; **51.16** Gerald S. Wilkinson; **51.17** Juniors Bildarchiv/F300/Alamy Stock Photo; **51.20** Martin Harvey/Photolibrary/Getty Images; **51.21** Erik Svensson/Lund University, Sweden; **51.22** Lowell Getz; **51.23** Rory Doolin; **51.25** Jennifer Jarvis, University of Cape Town; **51.27** Fred van Wijk/Alamy Stock Photo; **51.28** Jupiterimages/Creatas/Thinkstock/Getty Images; **Wilson** Michael Dwyer/Alamy Stock Photo; **Synthesize Your Knowledge** William Leaman/Alamy Stock Photo.

Unit Eight Interview **Bonaccorso** Universidad San Francisco de Quito; **fieldwork** Paúl Tito; **inset** Paul Molina photography/Shutterstock.

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Chapter 54 **54.1** eel Andi de Boer/Shutterstock; **coral reef** Jan Wlodarczyk/Alamy Stock Photo; **triggerfish** imageBROKER/Alamy Stock Photo; **shark** Andrey Armyagov/Alamy Stock Photo; **coral bleaching** Reinhard Dirscherl/Alamy Stock Photo; **54.2 left to right** Joseph T. Collins/Science Source; Eladio Fernandez/Science Source; **54.4** imageBROKER/Frank W Lane/Alamy Stock Photo; **Scientific Skills Exercise** Johan Larson/Shutterstock; **54.6a** Tony Heald/Nature Picture Library; **54.6b** Tom Brakefield/Getty Images; **54.6c** Dirk Ercken/Shutterstock; **54.6d** Barry Mansell/Nature Picture Library; **54.6e left to right** Photoshot/Avalon.red/Alamy Stock Photo; Andrew DuBois/Alamy Stock Photo; **54.6f left to right** David J Martin/Shutterstock; Lightwriter1949/Alamy Stock Photo; **54.7a** David Fleetham/Alamy Stock Photo; **54.7b** Ethan Daniels/Alamy Stock Photo; **54.7c** Georgette Apol/Alamy Stock Photo; **Langkilde** Patrick Mansell/PennState; **54.8** Doug Perrine/Nature Picture Library; **54.9a** Bazzano Photography/Alamy Stock Photo; **54.9b** Nicholas Smythe/Science Source; **54.10** Arco/G. Lacz/imageBROKER.com GmbH & Co. KG/Alamy Stock Photo; **54.11a** Sally D. Hacker; **54.13** Gary W. Saunders; **54.14** Dung Vu Trung/Science Source; **54.15** Cedar Creek Ecosystem Science Reserve, University of Minnesota; **54.19** Genny Anderson; **54.20** Adam Welz; **54.24** National Park Service (NPS.gov); **Turner** From: Tracking butterfly movements with harmonic radar reveals an effect of population age on movement distance. O. Ovaskainen et al. *Proc Natl Acad Sci U S A*. 2008 Dec 9;105(49):19090-5. doi: 10.1073/pnas.0802066105. Epub 2008 Dec 5. Fig. 1; **54.26 (1)** Charles D. Winters/Science Source; **(2)** Keith Boggs; **(3)** Don Paulson Photography/Purestock/SuperStock/Alamy Stock Photo; **(4)** Glacier Bay National Park and Preserve; **54.27 left to right** Charles D. Winters/Science Source; Keith Boggs; Don Paulson Photography/Purestock/SuperStock/Alamy Stock Photo; Glacier Bay National Park and Preserve; **54.28 left to right** R. Grant Gilmore/NOAA; Lance Horn/National Undersea Research Center/University of North Carolina-Wilmington/NOAA; **54.32** Tim Laman/National Geographic/Getty Images; **54.34** Nelish Pradhan/Bates College/Lewiston, ME; **54.35** Josh Spice; **Synthesize Your Knowledge** Jim Holden/Alamy Stock Photo.

Chapter 55 **55.1** Steven Kazlowski/RGB Ventures/SuperStock/Alamy Stock Photo; **55.2** Stone Nature Photography/Alamy Stock Photo; **55.3 left to right** Scimat/Science Source; Justus de Cuveland/imageBROKER/AGE Fotostock; **55.5** MODIS Science Team/Earth Observatory/NASA; **55.8** A. T. Willett/Alamy Stock Photo; **Problem-Solving Exercise** tree Steven Katovich/USDA Forest Service; **mountain pine beetles** Keith Douglas/Alamy Stock Photo; **55.9** Thala_JSR Photography/Shutterstock; **Scientific Skills Exercise** David R. Frazier Photolibrary, Inc./Alamy Stock Photo; **55.15** Hubbard Brook Research Foundation/USDA Forest Service; **Likens** Courtesy of Gene Likens; **55.16** Mark Gallagher/Princeton Hydro, LLC/Ringoes, NJ; **55.17 top to bottom** Kissimmee Division/South Florida Water Management District; Adam Wilson/Alamy Stock Photo; Supapixx/Alamy Stock Photo; From: Species richness accelerates marine ecosystem restoration in the Coral Triangle. S. L. Williams et al. *Proc Natl Acad Sci U S A*. 2017 Nov 7;114(45):11986-11991. doi: 10.1073/pnas.1707962114. Epub 2017 Oct 24. Fig. 1a. Photos courtesy of D. Trockel, University of California, Davis, CA; **55.18** U.S. Department of Energy; **Synthesize Your Knowledge** Bill Gozansky/Alamy Stock Photo.

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Steitz, Yale University, New Haven; **Figure 30.9** Paul Atkinson/Shutterstock; **Ch. 35 Test Your Understanding 11** From: Anatomy of the vessel network within and between tree rings of *Fraxinus lanuginosa* (Oleaceae). Peter B. Kitin, Tomoyuki Fujii, Hisashi Abe and Ryo Funada. *American Journal of Botany*. 2004;91:779-788.

Appendix B Bacteria, Archaea Eye of Science/Science Source; **diatoms** M I Walker/NHPA/Photoshot/Newscom; **lily** Howard Rice/Dorling Kindersley, Ltd./Alamy Stock Photo; **fungus** 635926/Shutterstock; **chimpanzees** Anup Shah/Stone/Getty Images;

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Chapter 3 3.7 map based on NOAA Fisheries, Bowhead Whale (*Balaena mysticetus*); sea ice extent from National Snow and Ice Data Center (<https://nsidc.org/arcticseaicenews/>); **3.9** Based on Simulating Water and the Molecules of Life by Mark Gerstein and Michael Levitt, from *Scientific American*, November 1998; **Scientific Skills Exercise** Data from C. Langdon et al., Effect of Calcium Carbonate Saturation State on the Calcification Rate of an Experimental Coral Reef, *Global Biogeochemical Cycles* 14:639–654 (2000).

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Appendix A Figure 5.11 Wallace/Sanders/Ferli, *Biology: The Science of Life*, 3rd Ed., © 1991. Reprinted and electronically reproduced by permission of Pearson Education, Inc., Upper Saddle River, New Jersey.

Useful Reference: The Metric System

Metric Prefixes: 10^{15} = peta (P) 10^6 = mega (M) 10^{-2} = centi (c) 10^{-9} = nano (n)
 10^{12} = tera (T) 10^3 = kilo (k) 10^{-3} = milli (m) 10^{-12} = pico (p)
 10^9 = giga (G) 10^{-1} = deci (d) 10^{-6} = micro (μ) 10^{-15} = femto (f)

Measurement	Unit and Abbreviation	Metric Equivalent	Metric-to-English Conversion Factor	English-to-Metric Conversion Factor
Length	1 kilometer (km)	= 1,000 (10^3) meters	1 km = 0.62 mile	1 mile = 1.61 km
	1 meter (m)	= 100 (10^2) centimeters = 1,000 millimeters	1 m = 1.09 yards 1 m = 3.28 feet 1 m = 39.37 inches	1 yard = 0.914 m 1 foot = 0.305 m
	1 centimeter (cm)	= 0.01 (10^{-2}) meter	1 cm = 0.394 inch	1 foot = 30.5 cm 1 inch = 2.54 cm
	1 millimeter (mm)	= 0.001 (10^{-3}) meter	1 mm = 0.039 inch	
	1 micrometer (μ m) (formerly micron, μ)	= 10^{-6} meter (10^{-3} mm)		
	1 nanometer (nm) (formerly millimicron, m μ)	= 10^{-9} meter (10^{-3} μ m)		
	1 angstrom ($\text{\AA}$)	= 10^{-10} meter (10^{-4} μ m)		
Area	1 hectare (ha)	= 10,000 square meters	1 ha = 2.47 acres	1 acre = 0.405 ha
	1 square meter (m^2)	= 10,000 square centimeters	1 m^2 = 1.196 square yards 1 m^2 = 10.764 square feet	1 square yard = 0.8361 m^2 1 square foot = 0.0929 m^2
	1 square centimeter (cm^2)	= 100 square millimeters	1 cm^2 = 0.155 square inch	1 square inch = 6.4516 cm^2
Mass	1 metric ton (t)	= 1,000 kilograms	1 t = 1.103 tons	1 ton = 0.907 t
	1 kilogram (kg)	= 1,000 grams	1 kg = 2.205 pounds	1 pound = 0.4536 kg
	1 gram (g)	= 1,000 milligrams	1 g = 0.0353 ounce 1 g = 15.432 grains	1 ounce = 28.35 g
	1 milligram (mg)	= 10^{-3} gram	1 mg = approx. 0.015 grain	
	1 microgram (μ g or mcg)	= 10^{-6} gram		
Volume (solids)	1 cubic meter (m^3)	= 1,000,000 cubic centimeters	1 m^3 = 1.308 cubic yards 1 m^3 = 35.315 cubic feet	1 cubic yard = 0.7646 m^3 1 cubic foot = 0.0283 m^3
	1 cubic centimeter (cm^3 or cc)	= 10^{-6} cubic meter	1 cm^3 = 0.061 cubic inch	1 cubic inch = 16.387 cm^3
	1 cubic millimeter (mm^3)	= 10^{-9} cubic meter = 10^{-3} cubic centimeter		
Volume (liquids and gases)	1 kiloliter (kL or kl)	= 1,000 liters	1 kL = 264.17 gallons	
	1 liter (L or l)	= 1,000 milliliters	1 L = 0.264 gallon 1 L = 1.057 quarts	1 gallon = 3.785 L 1 quart = 0.946 L
	1 milliliter (mL or ml)	= 10^{-3} liter = 1 cubic centimeter (cc)	1 mL = 0.034 fluid ounce 1 mL = approx. $\frac{1}{4}$ teaspoon 1 mL = approx. 15–16 drops (gtt.)	1 quart = 946 mL 1 pint = 473 mL 1 fluid ounce = 29.57 mL 1 teaspoon = approx. 5 mL
	1 microliter (μ L or μ l)	= 10^{-6} liter (10^{-3} milliliter)		
Pressure	1 megapascal (MPa)	= 1,000 kilopascals	1 MPa = 10 bars	1 bar = 0.1 MPa
	1 kilopascal (kPa)	= 1,000 pascals	1 kPa = 0.01 bar	1 bar = 100 kPa
	1 pascal (Pa)	= 1 newton/ m^2 (N/ m^2)	1 Pa = 1.0×10^{-5} bar	1 bar = 1.0×10^5 Pa
Time	1 second (s or sec)	= $\frac{1}{60}$ minute		
	1 millisecond (ms or msec)	= 10^{-3} second		
Temperature	Degrees Celsius ($^{\circ}\text{C}$) (0 K [Kelvin] = -273.15°C)		$^{\circ}\text{F} = \frac{9}{5}^{\circ}\text{C} + 32$	$^{\circ}\text{C} = \frac{5}{9} (^{\circ}\text{F} - 32)$

Useful Reference: The Periodic Table of the Elements

Atomic number
(number of protons)

6

Element symbol

C

Atomic mass
(number of protons plus
number of neutrons averaged
over all isotopes)

12.01

Metals

Metalloids

Nonmetals

Representative elements

Groups: Elements in a vertical column have the same number of electrons in their valence (outer) shell and thus have similar chemical properties.

Periods: Each horizontal row contains elements with the same total number of electron shells. Across each period, elements are ordered by increasing atomic number.

Alkali metals

Alkaline earth metals

1 Group 1A

2 Group 2A

13 Group 3A

14 Group 4A

15 Group 5A

16 Group 6A

17 Group 7A

18 Group 8A

Transition elements

Halogens

Noble gases

1	1	2											13	14	15	16	17	18
1	1	2											13	14	15	16	17	18
1	1	2											13	14	15	16	17	18
2	3	4											5	6	7	8	9	10
2	3	4											5	6	7	8	9	10
3	11	12	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
3	11	12	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
4	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
4	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
5	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54
5	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54
6	55	56	57*	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86
6	55	56	57*	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86
7	87	88	89†	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118
7	87	88	89†	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118

*Lanthanides

†Actinides

Name (Symbol)	Atomic Number	Name (Symbol)	Atomic Number	Name (Symbol)	Atomic Number	Name (Symbol)	Atomic Number	Name (Symbol)	Atomic Number
Actinium (Ac)	89	Copper (Cu)	29	Iron (Fe)	26	Osmium (Os)	76	Silicon (Si)	14
Aluminum (Al)	13	Curium (Cm)	96	Krypton (Kr)	36	Oxygen (O)	8	Silver (Ag)	47
Americium (Am)	95	Darmstadtium (Ds)	110	Lanthanum (La)	57	Palladium (Pd)	46	Sodium (Na)	11
Antimony (Sb)	51	Dubnium (Db)	105	Lawrencium (Lr)	103	Phosphorus (P)	15	Strontium (Sr)	38
Argon (Ar)	18	Dysprosium (Dy)	66	Lead (Pb)	82	Platinum (Pt)	78	Sulfur (S)	16
Arsenic (As)	33	Einsteinium (Es)	99	Lithium (Li)	3	Plutonium (Pu)	94	Tantalum (Ta)	73
Astatine (At)	85	Erbium (Er)	68	Livermorium (Lv)	116	Polonium (Po)	84	Technetium (Tc)	43
Barium (Ba)	56	Europium (Eu)	63	Lutetium (Lu)	71	Potassium (K)	19	Tellurium (Te)	52
Berkelium (Bk)	97	Fermium (Fm)	100	Magnesium (Mg)	12	Praseodymium (Pr)	59	Tennessine (Ts)	117
Beryllium (Be)	4	Flerovium (Fl)	114	Manganese (Mn)	25	Promethium (Pm)	61	Terbium (Tb)	65
Bismuth (Bi)	83	Fluorine (F)	9	Meitnerium (Mt)	109	Protactinium (Pa)	91	Thallium (Tl)	81
Bohrium (Bh)	107	Francium (Fr)	87	Mendelevium (Md)	101	Radium (Ra)	88	Thorium (Th)	90
Boron (B)	5	Gadolinium (Gd)	64	Mercury (Hg)	80	Radon (Rn)	86	Thulium (Tm)	69
Bromine (Br)	35	Gallium (Ga)	31	Molybdenum (Mo)	42	Rhenium (Re)	75	Tin (Sn)	50
Cadmium (Cd)	48	Germanium (Ge)	32	Moscovium (Mc)	115	Rhodium (Rh)	45	Titanium (Ti)	22
Calcium (Ca)	20	Gold (Au)	79	Neodymium (Nd)	60	Roentgenium (Rg)	111	Tungsten (W)	74
Californium (Cf)	98	Hafnium (Hf)	72	Neon (Ne)	10	Rubidium (Rb)	37	Uranium (U)	92
Carbon (C)	6	Hassium (Hs)	108	Neptunium (Np)	93	Ruthenium (Ru)	44	Vanadium (V)	23
Cerium (Ce)	58	Helium (He)	2	Nickel (Ni)	28	Rutherfordium (Rf)	104	Xenon (Xe)	54
Cesium (Cs)	55	Holmium (Ho)	67	Nihonium (Nh)	113	Samarium (Sm)	62	Ytterbium (Yb)	70
Chlorine (Cl)	17	Hydrogen (H)	1	Niobium (Nb)	41	Scandium (Sc)	21	Yttrium (Y)	39
Chromium (Cr)	24	Indium (In)	49	Nitrogen (N)	7	Seaborgium (Sg)	106	Zinc (Zn)	30
Cobalt (Co)	27	Iodine (I)	53	Nobelium (No)	102	Selenium (Se)	34	Zirconium (Zr)	40
Copernicium (Cn)	112	Iridium (Ir)	77	Oganesson (Og)	118				

Glossary

Pronunciation Key

ā	ace
a/ah	ash
ch	chose
ē	meet
e/eh	bet
g	game
ī	ice
i	hit
ks	box
kw	quick
ng	song
ō	robe
o	ox
oy	boy
s	say
sh	shell
th	thin
ū	boot
u/uh	up
z	zoo

' = primary accent

' = secondary accent

5' cap A modified form of guanine nucleotide added onto the 5' end of a pre-mRNA molecule.

Anaerobic respiration (an-er-ō'-bik) A catabolic pathway in which inorganic molecules other than oxygen accept electrons at the "downhill" end of electron transport chains.

ABC hypothesis A model of flower formation identifying three classes of organ identity genes that direct formation of the four types of floral organs.

Abiotic (ā'-bī-ot'-ik) Nonliving; referring to the physical and chemical properties of an environment.

Abortion The deliberate termination of a pregnancy in progress.

Abscisic acid (ABA) (ab-sis'-ik) A plant hormone that slows growth, often antagonizing the actions of growth hormones. Two of its many effects are to promote seed dormancy and facilitate drought tolerance.

Absorption The third stage of food processing in animals: the uptake of small nutrient molecules by an organism's body.

Absorption spectrum The range of a pigment's ability to absorb various wavelengths of light; also a graph of such a range.

Abyssal zone (uh-bis'-ul) The part of the ocean's benthic zone between 2,000 and 6,000 m deep.

Acanthodians (ak'-an-thō'-dē-un) A group of ancient jawed aquatic vertebrates from the Silurian and Devonian periods.

Accessory fruit A fruit, or assemblage of fruits, in which the fleshy parts are derived largely or entirely from tissues other than the ovary.

Acclimatization (uh-klī'-muh-tī-zā'-shun) Physiological adjustment to a change in an environmental factor.

Acetyl CoA Acetyl coenzyme A; the entry compound for the citric acid cycle in cellular respiration, formed from a two-carbon fragment of pyruvate attached to a coenzyme.

Acetylcholine (as'-uh-til-kō'-lēn) One of the most common neurotransmitters; functions by binding to receptors and altering the permeability of the postsynaptic membrane to specific ions, either depolarizing or hyperpolarizing the membrane.

Acid A substance that increases the hydrogen ion concentration of a solution.

Acquired immunodeficiency syndrome (AIDS) The symptoms and signs present during the late stages of HIV infection, defined by a specified reduction in the number of T cells and the appearance of characteristic secondary infections.

Acrosomal reaction (ak'-ruh-sōm'-ul) The discharge of hydrolytic enzymes from the acrosome, a vesicle in the tip of a sperm, when the sperm approaches or contacts an egg.

Acrosome (ak'-ruh-sōm) A vesicle in the tip of a sperm containing hydrolytic enzymes and other proteins that help the sperm reach the egg.

Actin (ak'-tin) A globular protein that links into chains, two of which twist helically about each other, forming microfilaments (actin filaments) in muscle and other kinds of cells.

Action potentials Electrical signals that propagate (travel) along the membrane of a neuron or other excitable cell as a nongraded (all-or-none) depolarization.

Action spectrum A graph that profiles the relative effectiveness of different wavelengths of radiation in driving a particular process.

Activation energy The amount of energy that reactants must absorb before a chemical reaction will start; also called free energy of activation.

Activator A protein that binds to DNA and stimulates gene transcription. In prokaryotes, activators bind in or near the promoter; in eukaryotes, activators generally bind to control elements in enhancers.

Active immunity Long-lasting immunity conferred by the action of B cells and T cells and the resulting B and T memory cells specific for a pathogen. Active immunity can develop as a result of natural infection or immunization.

Active site The specific region of an enzyme that binds the substrate and that forms the pocket in which catalysis occurs.

Active transport The movement of a substance across a cell membrane against its concentration or electrochemical gradient, mediated by specific transport proteins and requiring an expenditure of energy.

Adaptations Inherited characteristics of an organism that enhance its survival and reproduction in a specific environment.

Adaptive evolution A process in which traits that enhance survival or reproduction tend to increase in frequency over time, resulting in a better match between organisms and their environment.

Adaptive immunity A vertebrate-specific defense that is mediated by B lymphocytes (B cells) and T lymphocytes (T cells) and that exhibits specificity, memory, and self-nonself recognition; also called acquired immunity.

Adaptive radiations Periods of evolutionary change in which groups of organisms form many new species whose adaptations allow them to fill different ecological roles in their communities.

Addition rule A rule of probability stating that the probability of any one of two or more mutually exclusive events occurring can be determined by adding their individual probabilities.

Adenosine triphosphate (a-den'-ō-sēn trī-fos'-fāt) An adenine-containing nucleoside triphosphate that releases free energy when its phosphate bonds are hydrolyzed. This energy is used to drive endergonic reactions in cells.

Adenylyl cyclase (uh-den'-uh-lil) An enzyme that converts ATP to cyclic AMP in response to an extracellular signal.

Adhesion The clinging of one substance to another, such as water to plant cell walls, in this case by means of hydrogen bonds.

Adipose tissue A connective tissue that insulates the body and serves as a fuel reserve; contains fat-storing cells called adipose cells.

Adrenal gland (uh-drē'-nul) One of two endocrine glands located adjacent to the kidneys in mammals. Endocrine cells in the outer portion (cortex) respond to adrenocorticotrophic hormone (ACTH) by secreting steroid hormones that help maintain homeostasis during long-term stress. Neurosecretory cells in the central portion (medulla) secrete epinephrine and

norepinephrine in response to nerve signals triggered by short-term stress.

Aerobic respiration A catabolic pathway for organic molecules, using oxygen as the final electron acceptor in an electron transport chain and ultimately producing ATP. This is the most efficient catabolic pathway and is carried out in most eukaryotic cells and many prokaryotic organisms.

Age structure The relative number of individuals of each age in a population.

Aggregate fruit A fruit derived from a single flower that has more than one carpel.

AIDS (acquired immunodeficiency syndrome) The symptoms and signs present during the late stages of HIV infection, defined by a specified reduction in the number of T cells and the appearance of characteristic secondary infections.

Alcohol fermentation Glycolysis followed by the reduction of pyruvate to ethyl alcohol, regenerating NAD⁺ and releasing carbon dioxide.

Algae (singular, *alga*) A general term for any species of photosynthetic protist, including both unicellular and multicellular forms. Algal species are included in several eukaryote supergroups.

Alimentary canal (al'-uh-men'-tuh-rē) A complete digestive tract, consisting of a tube running between a mouth and an anus.

Alkaline vents Deep-sea hydrothermal vents that release water that is warm (40 to 90 degrees C) rather than hot and that has a high pH (is basic). These vents consist of tiny pores lined with iron and other catalytic minerals that some scientists hypothesize might have been the location of the earliest abiotic synthesis of organic compounds.

Allele (uh-lē'-ul) Any of the alternative versions of a gene that may produce distinguishable phenotypic effects.

Allopatric speciation (al'-uh-pat'-rik) The formation of new species in populations that are geographically isolated from one another.

Allopolyploid (al'-ō-pol'-ē-ploid) A fertile individual that has more than two chromosome sets as a result of two different species interbreeding and combining their chromosomes.

Allosteric regulation The binding of a regulatory molecule to a protein at one site that affects the function of the protein at a different site.

α helix (al'-fuh hē'-liks) A coiled region constituting one form of the secondary structure of proteins, arising from a specific pattern of hydrogen bonding between atoms of the polypeptide backbone (not the side chains).

Alternation of generations A life cycle in which there is both a multicellular diploid form, the sporophyte, and a multicellular haploid form, the gametophyte; characteristic of plants and some algae.

Alternative RNA splicing A type of eukaryotic gene regulation at the RNA-processing level in which different mRNA molecules are produced from the same primary transcript, depending on which RNA segments are treated as exons and which as introns.

Altruism (al'-trū-iz-um) Selflessness; behavior that reduces an individual's fitness while increasing the fitness of another individual.

Alveolates (al-vē'-uh-lets) One of the three major supergroups for which the SAR eukaryotic supergroup is named. This clade arose by secondary endosymbiosis, and its members have membrane-enclosed sacs (alveoli) located just under the plasma membrane.

Alveolus (al-vē'-uh-lus) (plural, *alveoli*) One of the dead-end air sacs where gas exchange occurs in a mammalian lung.

Alzheimer's disease (alts'-hī-merz) An age-related dementia (mental deterioration) characterized by confusion and memory loss.

Amino acid (uh-mēn'-ō) An organic molecule possessing both a carboxyl and an amino group. Amino acids serve as the monomers of polypeptides.

Amino group (uh-mēn'-ō) A chemical group consisting of a nitrogen atom bonded to two hydrogen atoms; can act as a base in solution, accepting a hydrogen ion and acquiring a positive charge.

Aminoacyl-tRNA synthetases Enzymes that join each amino acid to the appropriate tRNA.

Ammonia A small, toxic molecule (NH subscript 3) produced by nitrogen fixation or as a metabolic waste product of protein and nucleic acid metabolism.

Ammonites A group of shelled cephalopods that were important marine predators for hundreds of millions of years until their extinction at the end of the Cretaceous period (65.5 million years ago).

Amniocentesis (am'-nē-ō-sen-tē'-sis) A technique associated with prenatal diagnosis in which amniotic fluid is obtained by aspiration from a needle inserted into the uterus. The fluid and the fetal cells it contains are analyzed to detect certain genetic and congenital defects in the fetus.

Amniotes (am'-nē-ōt) A clade of tetrapods named for a key derived character, the amniotic egg, which contains specialized membranes, including the fluid-filled amnion, that protect the embryo. Amniotes include mammals as well as birds and other reptiles.

Amniotic egg An egg that contains specialized membranes that function in protection, nourishment, and gas exchange. The amniotic egg was a major evolutionary innovation, allowing embryos to develop on land in a fluid-filled sac, thus reducing the dependence of tetrapods on water for reproduction.

Amoebas (uh-mē'-buhs) Protists characterized by the presence of pseudopodia.

Amoebocytes (uh-mē'-buh-sits') Amoeba-like cells that move by pseudopodia and are found in most animals. Depending on the species, amoebocytes may digest and distribute food, dispose of wastes, form skeletal fibers, fight infections, or change into other cell types.

Amoebozoan (uh-mē'-buh-zō'-an) A protist in a clade that includes many species with lobe- or tube-shaped pseudopodia.

Amorphea An extremely diverse supergroup of eukaryotes that includes animals, fungi, and some protists. There are two major clades of amorphea, the amoebozoans (tubulinids and close protist relatives) and the opisthokonts (animals, fungi, and closely related protist groups). This supergroup was formerly called Unikonta.

Amphibians The clade of tetrapods that includes salamanders, frogs, and caecilians.

Amphipathic (am'-fē-path'-ik) Having both a hydrophilic region and a hydrophobic region.

Amplification The strengthening of stimulus energy during transduction.

Amygdala (uh-mig'-duh-luh) A structure in the temporal lobe of the vertebrate brain that has a major role in the processing of emotions.

Amylase (am'-uh-lās') An enzyme that hydrolyzes starch (a glucose polymer from plants) and glycogen (a glucose polymer from animals) into smaller polysaccharides and the disaccharide maltose.

Anabolic pathways (an'-uh-bol'-ik) Metabolic pathways that consume energy to synthesize a complex molecule from simpler molecules.

Analogous Having characteristics that are similar because of convergent evolution, not homology.

Analogy (an-al'-uh-jē) Similarity between two species that is due to convergent evolution rather than to descent from a common ancestor with the same trait.

Anaphase The fourth stage of mitosis, in which the chromatids of each chromosome have separated and the daughter chromosomes are moving to the poles of the cell.

Anatomy The structure of an organism.

Anchorage dependence The requirement that a cell must be attached to a substratum in order to initiate cell division.

Androgens (an'-drō-jens) Steroid hormones, such as testosterone, that stimulate the development and maintenance of the male reproductive system and secondary sex characteristics.

Aneuploidy (an'-yū-ploy'-dē) A chromosomal aberration in which one or more chromosomes are present in extra copies or are deficient in number.

Angiosperms (an'-jē-ō-sperm) Flowering plants, which form seeds inside a protective chamber called an ovary.

Anhydrobiosis (an-hī'-drō-bī-ō'-sis) A dormant state involving loss of almost all body water.

Animal pole The point at the end of an egg in the hemisphere where the least yolk is concentrated; opposite of vegetal pole.

Anion (an'-i-on) A negatively charged ion.

Anterior pituitary A portion of the pituitary gland that develops from nonneural tissue; consists of endocrine cells that synthesize and secrete several tropic and nontropic hormones.

Anther In an angiosperm, the terminal pollen sac of a stamen, where pollen grains containing sperm-producing male gametophytes form.

Antheridia (an-thuh-rid'-ē-uh) (singular, **antheridium**) In plants, the male gametangium, a moist chamber in which gametes develop.

Anthropoids (an'-thruh-poyd) Members of a primate group made up of the monkeys and the apes (gibbons, orangutans, gorillas, chimpanzees, bonobos, and humans).

Antibody A protein secreted by plasma cells (differentiated B cells) that binds to a particular antigen; also called immunoglobulin. All antibodies have the same Y-shaped structure and in their monomer form consist of two identical heavy chains and two identical light chains.

Anticodon (an'-ti-kō'-don) A nucleotide triplet at one end of a tRNA molecule that base-pairs with a particular complementary codon on an mRNA molecule.

Antidiuretic hormone (ADH) (an'-ti-dī-yū-ret'-ik) A peptide hormone, also called vasopressin, that promotes water retention by the kidneys. Produced in the hypothalamus and released from the posterior pituitary, ADH also functions in the brain.

Antigen (an'-ti-jen) A substance that elicits an immune response by binding to receptors of B or T cells.

Antigen presentation (an'-ti-jen) The process by which an MHC molecule binds to a fragment of an intracellular protein antigen and carries it to the cell surface, where it is displayed and can be recognized by a T cell.

Antigen-presenting cell (an'-ti-jen) A cell that upon ingesting pathogens or internalizing pathogen proteins generates peptide fragments that are bound by class II MHC molecules and subsequently displayed on the cell surface to T cells. Macrophages, dendritic cells, and B cells are the primary antigen-presenting cells.

Antigen receptor (an'-ti-jen) The general term for a surface protein, located on B cells and T cells, that binds to antigens, initiating adaptive immune responses. The antigen receptors on B cells are called B cell receptors, and the antigen receptors on T cells are called T cell receptors.

Antiparallel Referring to the arrangement of the sugar-phosphate backbones in a DNA double helix (they run in opposite 5' to 3' directions).

Aphotic zone (ā'-fō'-tik) The part of an ocean or lake beneath the photic zone, where light does not penetrate sufficiently for photosynthesis to occur.

Apical bud (ā'-pik-ul) A bud at the tip of a plant stem; also called a terminal bud.

Apical dominance (ā'-pik-ul) Tendency for growth to be concentrated at the tip of a plant shoot because the apical bud partially inhibits axillary bud growth.

Apical ectodermal ridge (AER) (ā'-pik-ul) A thickened area of ectoderm at the tip of a limb bud that promotes outgrowth of the limb bud.

Apical meristems (ā'-pik-ul mār'-uh-stems) Localized regions at a growing tip of a plant body where one or more cells divide repeatedly. The dividing cells of an apical meristem enable the plant to grow in length.

Apicomplexans (ap'-ē-kom-pleks'-un) A group of alveolate protists, this clade includes many species that parasitize animals. Some apicomplexans cause human disease.

Apomixis (ap'-uh-mik'-sis) The ability of some plant species to reproduce asexually through seeds without fertilization by a male gamete.

Apoplast (ap'-ō-plast) Everything external to the plasma membrane of a plant cell, including cell walls, intercellular spaces, and the space within dead structures such as xylem vessels and tracheids.

Apoptosis (ā-puh-tō'-sus) A type of programmed cell death, which is brought about by activation of enzymes that break down many chemical components in the cell.

Aposematic coloration (ap'-ō-si-mat'-ik) The bright warning coloration of many animals with effective physical or chemical defenses.

Appendix A small, finger-like extension of the vertebrate cecum; contains a mass of white blood cells that contribute to immunity.

Aquaporins Channel proteins in a cellular membrane that specifically facilitate osmosis, the diffusion of free water across the membrane.

Aqueous solution (ā'-kwē-us) A solution in which water is the solvent.

Arachnids A subgroup of the major arthropod clade Chelicerata. Arachnids have six pairs of appendages, including four pairs of walking legs, and include spiders, scorpions, ticks, and mites.

Arbuscular mycorrhizae (ar-bus'-kyū-lur mī'-kō-rī'-zuh) Association of a fungus with a plant root system in which the fungus causes the invagination of the host (plant) cells' plasma membranes; also called endomycorrhiza.

Arbuscular mycorrhizal fungi (ar-bus'-kyū-lur) Symbiotic fungi whose hyphae grow through the cell wall of plant roots and extend into the root cell (enclosed in tubes formed by invagination of the root cell plasma membrane).

Arbuscules Specialized branching hyphae that are found in some mutualistic fungi and exchange nutrients with living plant cells.

Archaea (ar'-kē'-uh) A domain (major branch of the tree of life); consists of single-celled prokaryotic organisms. (Though recent studies suggest that Eukarya likely evolved within the

Archaea, it is still useful to distinguish the prokaryotic species in Archaea as a group, and so in this text the term Archaea refers to prokaryotic species unless otherwise noted.)

Archaeplastida (ar'-kē-plas'-tid-uh) A supergroup of eukaryotes proposed in a current hypothesis of the evolutionary history of eukaryotes that includes red algae, green algae, and plants; descended from an ancient protistan ancestor that engulfed a cyanobacterium.

Archegonia (ar-ki-gō'-nē-uh) (singular, **archegonium**) In plants, the female gametangium, a moist chamber in which gametes develop.

Archenteron (ar-ken'-tuh-ron) The endoderm-lined cavity, formed during gastrulation, that develops into the digestive tract of an animal.

Archosaurs (ar'-kō-sōr) The reptilian group that includes crocodiles, alligators and dinosaurs, including birds.

Arteries Vessels that carry blood away from the heart to organs throughout the body.

Arterioles (ar-ter'-ē-ōls) Vessels that convey blood between an artery and a capillary bed.

Arthropods Segmented ecdysozoans with a hard exoskeleton and jointed appendages. Familiar examples include insects, spiders, millipedes, and crabs.

Artificial intelligence (AI) The development and use of computer systems that can function like neural networks in the human brain to perform various tasks.

Artificial selection The selective breeding of domesticated plants and animals to encourage the occurrence of desirable traits.

Asci (singular, **ascus**) A saclike spore capsule located at the tip of a dikaryotic hypha of a sac fungus.

Ascocarps The fruiting bodies of a sac fungus (ascomycete).

Ascomycetes (as'-kuh-mī'-sēt) Members of the fungal phylum Ascomycota, commonly called sac fungi. The name comes from the saclike structure in which the spores develop.

Asexual reproduction The generation of offspring from a single parent that occurs without the fusion of gametes. In most cases, the offspring are genetically identical to the parent.

A site One of a ribosome's three binding sites for tRNA during translation. The A site holds the tRNA carrying the next amino acid to be added to the polypeptide chain. (A stands for aminoacyl tRNA.)

Associative learning The acquired ability to associate one environmental feature (such as a color) with another (such as danger).

Atherosclerosis A cardiovascular disease in which fatty deposits called plaques develop in the inner walls of the arteries, obstructing the arteries and causing them to harden.

Atom The smallest unit of matter that retains the properties of an element.

Atomic mass The total mass of an atom, numerically equivalent to the mass in grams of 1 mole of the atom. (For an element with more than one isotope, the atomic mass is the average mass of the naturally occurring isotopes, weighted by their abundance.)

Atomic nucleus An atom's dense central core, containing protons and neutrons.

Atomic number The number of protons in the nucleus of an atom, unique for each element and designated by a subscript.

ATP (adenosine triphosphate) (a-den'-ō-sēn trī-fos'-fāt) An adenine-containing nucleoside triphosphate that releases free energy when its phosphate bonds are hydrolyzed. This energy is used to drive endergonic reactions in cells.

ATP synthase A complex of several membrane proteins that functions in chemiosmosis with adjacent electron transport chains, using the energy of a hydrogen ion (proton) concentration gradient to make ATP. ATP synthases are found in the inner mitochondrial membranes of eukaryotic cells and in the plasma membranes of prokaryotes.

Atria (ā-trē-uh) (singular, atrium) A chamber of the vertebrate heart that receives blood from the veins and transfers blood to a ventricle.

Atrial natriuretic peptide (ANP) (ā'-trē-ul na'-trē-yū-ret'-ik) A peptide hormone secreted by cells of the atria of the heart in response to high blood pressure. ANP's effects on the kidney alter ion and water movement and reduce blood pressure.

Atrioventricular node (AV node) A region of specialized heart muscle tissue between the left and right atria where electrical impulses are delayed for about 0.1 second before spreading to both ventricles and causing them to contract.

Atrioventricular valve (AV valve) A heart valve located between each atrium and ventricle that prevents a backflow of blood when the ventricle contracts.

Atrium (ā-trē-um) (plural, atria) A chamber of the vertebrate heart that receives blood from the veins and transfers blood to a ventricle.

Autocrine Referring to a secreted molecule that acts on the cell that secreted it.

Autoimmune disease An immunological disorder in which the immune system turns against self.

Autonomic nervous system (ot'-ō-nom'-ik) An efferent branch of the vertebrate peripheral nervous system that regulates the internal environment; consists of the sympathetic and parasympathetic divisions and the enteric nervous system.

Autopolyploid (ot'-ō-pol'-ē-ploid) An individual that has more than two chromosome sets that are all derived from a single species.

Autosomes Chromosomes that are not directly involved in determining sex; not sex chromosomes.

Autotrophs (ot'-ō-trōfs) Organisms that obtain organic food molecules without eating other organisms or substances derived from other organisms. Autotrophs use energy from the sun or from inorganic substances to make organic molecules from inorganic ones.

Auxin (ok'-sin) A term that primarily refers to indoleacetic acid (IAA), a natural plant hormone that has a variety of effects, including cell elongation, root formation, secondary growth, and fruit growth.

Axillary bud (ak'-sil-ār-ē) A structure that has the potential to form a lateral shoot, or branch. The bud appears in the angle formed between a leaf and a stem.

Axon (ak'-son) A typically long extension, or process, of a neuron that carries nerve impulses away from the cell body toward target cells.

Bacteria A domain (major branch of the tree of life); consists of single-celled prokaryotic organisms.

Bacteriophages (singular, **bacteriophage**) (bak-tēr'-ē-ō-fāj) A virus that infects bacteria; also called a phage.

Bacteroids A form of the bacterium *Rhizobium* contained within the vesicles formed by the root cells of a root nodule.

Balancing selection Natural selection that maintains two or more phenotypic forms in a population.

Bar graph A graph in which the independent variable represents groups or nonnumerical categories and the values of the dependent variable(s) are shown by bars.

Bark All tissues external to the vascular cambium, consisting mainly of the secondary phloem and layers of periderm.

Barr body A dense object lying along the inside of the nuclear envelope in cells of female mammals, representing a highly condensed, inactivated X chromosome.

Basal body (bā'-sul) A eukaryotic cell structure consisting of a "9 plus 0" arrangement of microtubule triplets. The basal body may organize the microtubule assembly of a cilium or flagellum and is structurally very similar to a centriole.

Basal metabolic rate (BMR) The metabolic rate of a resting, fasting, and nonstressed endotherm at a comfortable temperature.

Base A substance that reduces the hydrogen ion concentration of a solution.

Basidiocarps Elaborate fruiting bodies of a dikaryotic mycelium of a club fungus.

Basidiomycetes (buh-sid'-ē-ō-mī'-sēts) Members of the fungal phylum Basidiomycota, commonly called club fungi. The name comes from the club-like shape of the basidium.

Basidium (plural, **basidia**) (buh-sid'-ē-um, buh-sid'-ē-ah) A reproductive appendage that produces sexual spores on the gills of mushrooms (club fungi).

Batesian mimicry (bāt'-zē-un mim'-uh-krē) A type of mimicry in which a harmless species resembles an unpalatable or harmful species to which it is not closely related.

B cells The lymphocytes that complete their development in the bone marrow and become effector cells for the humoral immune response.

Behavior The internally coordinated responses (actions or inactions) of whole organisms (individuals or groups) to internal and/or external stimuli, excluding responses more easily understood as developmental changes.

Behavioral ecology The study of behavioral interactions between individuals within populations and communities, usually in an evolutionary context.

Benign tumor A mass of abnormal cells with specific genetic and cellular changes such that the cells are not capable of surviving at a new site and generally remain at the site of the tumor's origin.

Benthic zone The bottom surface of an aquatic environment.

Benthos (ben'-thōz) The communities of organisms living in the benthic zone of an aquatic biome.

Beta oxidation A metabolic sequence that breaks fatty acids down to two-carbon fragments that enter the citric acid cycle as acetyl CoA.

β pleated sheet One form of the secondary structure of proteins in which the polypeptide chain folds back and forth. Two regions of the chain lie parallel to each other and are held together by hydrogen bonds between atoms of the polypeptide backbone (not the side chains).

Bilateral symmetry Body symmetry in which a central longitudinal plane divides the body into two equal but opposite halves.

Bilaterians (bī'-luh-ter'-ē-uhns) Members of a clade of animals with bilateral symmetry and three germ layers.

Bile A mixture of substances that is produced in the liver and stored in the gallbladder; enables formation of fat droplets in water as an aid in the digestion and absorption of fats.

Binary fission A method of asexual reproduction in single-celled organisms in which the cell grows to roughly double its size and then divides into two cells. In prokaryotes, binary fission does not involve mitosis, but in single-celled eukaryotes that undergo binary fission, mitosis is part of the process.

Binomial A common term for the two-part, latinized format for naming a species, consisting of the genus and specific epithet; also called a binomen.

Biodiversity hot spot A relatively small area with numerous endemic species and a large number of endangered and threatened species.

Bioenergetics (1) The overall flow and transformation of energy in an organism. (2) The study of how energy flows through organisms.

Biofilm A surface-coating colony of one or more species of unicellular organisms that engage in metabolic cooperation; most known biofilms are formed by prokaryotes.

Biofuels Fuels produced from biomass.

Biogeochemical cycles The various chemical cycles, including the water cycle, carbon cycle, nitrogen cycle, and phosphorus cycle, which involve both biotic and abiotic components of ecosystems.

Biogeography The scientific study of the past and present geographic distributions of species.

Bioinformatics The use of computational tools to store, organize, and analyze biological information from large data sets.

Biological augmentation An approach to restoration ecology that uses organisms to add essential materials to a degraded ecosystem.

Biological clock An internal timekeeper that controls an organism's biological rhythms. The biological clock marks time with or without environmental cues but often requires signals from the environment to remain tuned to an appropriate period; A physiological cycle of about 24 hours that persists even in the absence of external cues.

Biological magnification A process in which retained substances become more concentrated at each higher trophic level in a food chain.

Biological species concept Definition of a species as a group of populations whose members have the potential to interbreed in nature and produce viable, fertile offspring but do not produce viable, fertile offspring with members of other such groups.

Biology The scientific study of life.

Biomass The total mass of organic matter comprising a group of organisms in a particular habitat.

Biomes (bī'-ōms) Earth's major ecosystem types, often classified according to the predominant vegetation for terrestrial biomes and the physical environment for aquatic biomes and characterized by adaptations of organisms to that particular environment.

Bioremediation The use of organisms to detoxify and restore polluted and degraded ecosystems.

Biotechnology The manipulation of organisms or their components to produce useful products.

Biotic (bī-ot'-ik) Pertaining to the living factors—the organisms—in an environment.

Bipolar disorder A depressive mental illness characterized by swings of mood from high to low; also called manic-depressive disorder.

Birth control pills Hormonal contraceptives that inhibit ovulation, retard follicular development, or alter the cervical mucus to prevent sperm from entering the uterus.

Blades (1) Leaflike structures of a seaweed that provide most of the surface area for photosynthesis. (2) The flattened portions of typical leaves.

Blastocoel (blas'-tuh-sēl) The fluid-filled cavity that forms in the center of a blastula.

Blastocoelom A body cavity lined by tissue derived from mesoderm and by tissue derived from endoderm; sometimes called a hemacoel.

Blastocyst (blas'-tuh-sist) The blastula stage of mammalian embryonic development, consisting of an inner cell mass, a cavity, and an outer layer, the trophoblast. In humans, the blastocyst forms 1 week after fertilization.

Blastomeres Early embryonic cells arising during the cleavage stage of an early embryo.

Blastopore (blas'-tō-pōr) In a gastrula, the opening of the archenteron that typically develops into the anus in deuterostomes and the mouth in protostomes.

Blastula (blas'-tyū-luh) A hollow ball of cells that marks the end of the cleavage stage during early embryonic development in animals.

Blood A connective tissue with a fluid matrix called plasma in which red blood cells, white blood cells, and cell fragments called platelets are suspended.

Blue-light photoreceptors Any of several classes of light-absorbing molecules that have physiological effects when activated by blue light.

Body cavity A fluid- or air-filled space between the digestive tract and the body wall.

Body plan In multicellular eukaryotes, a set of morphological and developmental traits that are integrated into a functional whole—the living organism.

Bohr shift A lowering of the affinity of hemoglobin for oxygen, caused by a drop in pH. It facilitates the release of oxygen from hemoglobin in the vicinity of active tissues.

Book lungs Organs of gas exchange in spiders, each consisting of stacked plates contained in an internal chamber.

Bottleneck effect Genetic drift that occurs when the size of a population is reduced, as by a natural disaster or human actions. Typically, the surviving population is no longer genetically representative of the original population.

Bottom-up control A situation in which the abundance of organisms at each trophic level is limited by nutrient supply or the availability of food at lower trophic levels; thus, the supply of nutrients controls plant numbers, which control herbivore numbers, which in turn control predator numbers.

Bowman's capsule (bō'-munz) A cup-shaped receptacle in the vertebrate kidney that is the initial, expanded segment of the nephron, where filtrate enters from the blood.

Brachiopods (bra'-kē-uh-pods') Marine lophotrochozoans with a shell divided into

dorsal and ventral halves; also called lamp shells.

Brain Organ of the central nervous system where information is processed and integrated.

Brainstem A collection of structures in the vertebrate brain, including the midbrain, the pons, and the medulla oblongata; functions in homeostasis, coordination of movement, and conduction of information to higher brain centers.

Branch point The representation on a phylogenetic tree of the divergence of two or more taxa from a common ancestor. A branch point is usually shown as a dichotomy in which a branch representing the ancestral lineage splits (at the branch point) into two branches, one for each of the two descendant lineages.

Brassinosteroids A class of steroid hormones in plants that have a variety of effects, including inducing cell elongation, retarding leaf abscission, and promoting xylem differentiation.

Breathing Ventilation of the lungs through alternating inhalation and exhalation.

Bronchioles (brong'-kē-ōls') Fine branches of the bronchi that transport air to alveoli.

Bronchus (brong'-kus) (plural, **bronchi**) One of a pair of breathing tubes that branch from the trachea into the lungs.

Brown algae Multicellular, photosynthetic protists with a characteristic brown or olive color that results from carotenoids in their plastids. Most brown algae are marine, and some have a plantlike body.

Bryophytes (brī'-uh-fits) An informal name for mosses, liverworts, and hornworts; nonvascular plants that live on land but lack some of the terrestrial adaptations of vascular plants.

Buffer A solution that contains a weak acid and its corresponding base. A buffer minimizes changes in pH when acids or bases are added to the solution.

Bulk feeders Animals that eat relatively large pieces of food.

Bulk flow The movement of a fluid due to a difference in pressure between two locations.

Bundle-sheath cells In C_4 plants, a type of photosynthetic cell arranged into tightly packed sheaths around the veins of a leaf.

C_3 plants Plants that use the Calvin cycle for the initial steps that incorporate carbon dioxide into organic material, forming a three-carbon compound as the first stable intermediate.

C_4 plants Plants in which the Calvin cycle is preceded by reactions that incorporate carbon dioxide into a four-carbon compound, the end product of which supplies carbon dioxide for the Calvin cycle.

Callus A mass of dividing, undifferentiated cells growing at the site of a wound or in culture.

Calorie (cal) The amount of heat energy required to raise the temperature of 1 g of water by

1 degree C; also the amount of heat energy that 1 g of water releases when it cools by 1 degree C. The Calorie (with a capital C), usually used to indicate the energy content of food, is a kilocalorie.

Calvin cycle The second of two major stages in photosynthesis (following the light reactions), involving fixation of atmospheric carbon dioxide and reduction of the fixed carbon into carbohydrate.

CAM plants Plants that use crassulacean acid metabolism, an adaptation for photosynthesis in arid conditions. In this process, carbon dioxide entering open stomata during the night is converted to organic acids, which release carbon dioxide for the Calvin cycle during the day, when stomata are closed.

Cambrian explosion A relatively brief time in geologic history when many present-day phyla of animals first appeared in the fossil record. This burst of evolutionary change occurred about 535–525 million years ago and saw the emergence of the first large, hard-bodied animals.

Canopy The uppermost layer of vegetation in a terrestrial biome.

Capillaries (kap'-il-ār'-ēs) Microscopic blood vessels that penetrate the tissues and consist of a single layer of endothelial cells that allows exchange between the blood and interstitial fluid.

Capillary beds (kap'-il-ār'-ēs) Networks of capillaries in a tissue or organ.

Capsid The protein shell that encloses a viral genome. It may be rod-shaped, polyhedral, or more complex in shape.

Capsule (1) In many prokaryotes, a dense and well-defined layer of polysaccharide or protein that surrounds the cell wall and is sticky, protecting the cell and enabling it to adhere to substrates or other cells. (2) The sporangium of a bryophyte (moss, liverwort, or hornwort).

Carbohydrates (kar'-bō-hī'-drāts) Sugars (monosaccharides) or one of their dimers (disaccharides) or polymers (polysaccharides).

Carbon fixation The initial incorporation of carbon from carbon dioxide into an organic compound by an autotrophic organism (a plant, another photosynthetic organism, or a chemoautotrophic prokaryote).

Carbonyl group (kar'-buh-nīl) A chemical group present in aldehydes and ketones and consisting of a carbon atom double-bonded to an oxygen atom.

Carboxyl group (kar-bok'-sil) A chemical group present in organic acids and consisting of a single carbon atom double-bonded to an oxygen atom and also bonded to a hydroxyl group.

Cardiac cycle (kar'-dē-ak) The alternating contractions and relaxations of the heart.

Cardiac muscle (kar'-dē-ak) A type of striated muscle that forms the contractile wall of the heart. Its cells are joined by intercalated disks

that relay the electrical signals underlying each heartbeat.

Cardiac output (kar'-dē-ak) The volume of blood pumped per minute by each ventricle of the heart.

Cardiovascular system A closed circulatory system with a heart and branching network of arteries, capillaries, and veins. The system is characteristic of vertebrates.

Carnivores Organisms that consume animals for nutrition.

Carotenoids (kuh-rot'-uh-noyds') Accessory pigments, either yellow or orange, in the chloroplasts of plants and in some prokaryotes. By absorbing wavelengths of light that chlorophyll cannot, carotenoids broaden the spectrum of colors that can drive photosynthesis.

Carpels (kar'-puls) The ovule-producing reproductive organs of a flower, consisting of the stigma, style, and ovary.

Carriers In genetics, individuals who are heterozygous at a given genetic locus for a recessively inherited disorder. The heterozygote is generally phenotypically normal for the disorder but can pass on the recessive allele to offspring.

Carrying capacity The maximum population size that can be supported by the available resources, symbolized as *K*.

Casparian strip (ka-spār'-ē-un) A water-impermeable ring of wax in the endodermal cells of plants that blocks the passive flow of water and solutes into the stele by way of cell walls.

Catabolic pathways (kat'-uh-bol'-ik) Metabolic pathways that release energy by breaking down complex molecules to simpler molecules.

Catalysis (kuh-ta'-luh-sis) A process by which a chemical agent called a catalyst selectively increases the rate of a reaction without being consumed by the reaction.

Catalyst (kat'-uh-list) A chemical agent that selectively increases the rate of a reaction without being consumed by the reaction.

Cation (cat'-ī'-on) A positively charged ion.

Cation exchange (cat'-ī'-on) A process in which positively charged minerals are made available to a plant when hydrogen ions in the soil displace mineral ions from the clay particles.

Cecum (sē'-kum) (plural, **ceca**) The blind pouch forming one branch of the large intestine.

Cell Life's fundamental unit of structure and function; the smallest unit of organization that can perform all activities required for life.

Cell body The part of a neuron that houses the nucleus and most other organelles.

Cell cycle An ordered sequence of events in the life of a cell, from its origin in the division of a parent cell until its own division into two. The eukaryotic cell cycle is composed of interphase (including G₁, S, and G₂ phases) and M phase (including mitosis and cytokinesis).

Cell cycle control system A cyclically operating set of molecules in the eukaryotic cell that both triggers and coordinates key events in the cell cycle.

Cell division The reproduction of cells.

Cell fractionation The disruption of a cell and separation of its parts by centrifugation at successively higher speeds.

Cell-mediated immune response The branch of adaptive immunity that involves the activation of cytotoxic T cells, which defend against infected cells.

Cell plate A membrane-bounded, flattened sac located at the midline of a dividing plant cell, inside which the new cell wall forms during cytokinesis.

Cell wall A protective layer external to the plasma membrane in the cells of plants, prokaryotes, fungi, and some protists. Polysaccharides such as cellulose (in plants and some protists), chitin (in fungi), and peptidoglycan (in bacteria) are important structural components of cell walls.

Cellular respiration The catabolic pathways of aerobic and anaerobic respiration, which break down organic molecules and use an electron transport chain for the production of ATP.

Cellulose (sel'-yū-lōs) A structural polysaccharide of plant cell walls, consisting of glucose monomers joined by beta glycosidic linkages.

Central nervous system (CNS) The portion of the nervous system where signal integration occurs; in vertebrate animals, the brain and spinal cord.

Central vacuole In a mature plant cell, a large membranous sac with diverse roles in growth, storage, and sequestration of toxic substances.

Centrioles (sen'-trē-ōls) Structures in the centrosome of an animal cell composed of a cylinder of microtubule triplets arranged in a "9 plus 0" pattern. A centrosome has a pair of centrioles.

Centromere (sen'-trō-mēr) In a duplicated chromosome, the region on each sister chromatid where it is most closely attached to its sister chromatid by proteins that bind to the centromeric DNA. Other proteins condense the chromatin in that region, so it appears as a narrow "waist" on the duplicated chromosome. (An unduplicated chromosome has a single centromere, identified by the proteins bound there.)

Centrosome (sen'-trō-sōm) A structure present in the cytoplasm of animal cells that functions as a microtubule-organizing center and is important during cell division. A centrosome has two centrioles.

Cercozoans A group of amoeboid or flagellated protists that feed with threadlike pseudopodia.

Cerebellum (sār'-ruh-bel'-um) Part of the vertebrate hindbrain located dorsally; functions in unconscious coordination of movement and balance.

Cerebral cortex (suh-rē'-brul) The surface of the cerebrum; the largest and most complex part of the mammalian brain, containing nerve cell bodies of the cerebrum; the part of the vertebrate brain most changed through evolution.

Cerebrum (suh-rē'-brum) The dorsal portion of the vertebrate forebrain, composed of right and left hemispheres; the integrating center for memory, learning, emotions, and other highly complex functions of the central nervous system.

Cervix (ser'-viks) The neck of the uterus, which opens into the vagina.

Chaparral A scrubland biome of dense, spiny evergreen shrubs found at midlatitudes along coasts where cold ocean currents circulate offshore; characterized by mild, rainy winters and long, hot, dry summers.

Character An observable heritable feature that may vary among individuals.

Character displacement The tendency for characteristics to be more divergent in sympatric populations of two species than in allopatric populations of the same two species.

Checkpoint A control point in the cell cycle where stop and go-ahead signals can regulate the cycle.

Cheliceræ (kē-lih'-suh-ruhs) (singular, **chelicera**) A pair of clawlike feeding appendages characteristic of chelicerates.

Chelicerates (kē-lih'-suh'-rātes) Arthropods that have cheliceræ and a body divided into a cephalothorax and an abdomen. Living chelicerates include sea spiders, horseshoe crabs, scorpions, ticks, and spiders.

Chemical bonds Attractions between two atoms, resulting from a sharing of outer-shell electrons or the presence of opposite charges on the atoms. The bonded atoms gain complete outer electron shells.

Chemical energy Energy available in molecules for release in a chemical reaction; a form of potential energy.

Chemical equilibrium In a chemical reaction, the state in which the rate of the forward reaction equals the rate of the reverse reaction, so that the relative concentrations of the reactants and products do not change with time.

Chemical reactions The making and breaking of chemical bonds, leading to changes in the composition of matter.

Chemiosmosis (kem'-ē-oz-mō'-sis) An energy-coupling mechanism that uses energy stored in the form of a hydrogen ion gradient across a membrane to drive cellular work, such as the synthesis of ATP. Under aerobic conditions, most ATP synthesis in cells occurs by chemiosmosis.

Chemoautotroph (kē'-mō-ot'-ō-trōf) An organism that obtains energy by oxidizing inorganic substances and needs only carbon dioxide as a carbon source.

Chemoheterotroph (kē'-mō-het'-er-ō-trōf) An organism that requires organic molecules for both energy and carbon.

Chemoreceptors Sensory receptors that respond to a chemical stimulus, such as a solute or an odorant.

Chiasma (plural, **chiasmata**) (kī-az'-muh, kī-az'-muh-tuh) The X-shaped, microscopically visible region where crossing over has occurred earlier in prophase I between homologous non-sister chromatids. Chiasmata become visible after synapsis ends, with the two homologues remaining associated due to sister chromatid cohesion.

Chitin (kī'-tin) A structural polysaccharide, consisting of amino sugar monomers, found in many fungal cell walls and in the exoskeletons of all arthropods.

Chlorophyll (klōr'-ō-fil) A green pigment located in membranes within the chloroplasts of plants and algae and in the membranes of certain prokaryotes. Chlorophyll *a* participates directly in the light reactions, which convert solar energy to chemical energy.

Chloroplasts (klōr'-ō-plasts) Organelles found in plants and photosynthetic protists that absorb sunlight and use it to drive the synthesis of organic compounds from carbon dioxide and water.

Choanocytes (kō-an'-uh-sīts) Flagellated feeding cells found in sponges. Also called collar cells, they have a collar-like ring that traps food particles around the base of its flagellum.

Cholesterol (kō-les'-tuh-rol) A steroid that forms an essential component of animal cell membranes and acts as a precursor molecule for the synthesis of other biologically important steroids, such as many hormones.

Chondrichthyans (kon-drik'-thē-an) Members of the clade Chondrichthyes, vertebrates with skeletons made mostly of cartilage, such as sharks and rays.

Chordates Members of the phylum Chordata, animals that at some point during their development have a notochord; a dorsal, hollow nerve cord; pharyngeal slits or clefts; and a muscular, post-anal tail.

Chorionic villus sampling (CVS) (kōr'-ē-on'-ik vil'-us) A technique associated with prenatal diagnosis in which a small sample of the fetal portion of the placenta is removed for analysis to detect certain genetic and congenital defects in the fetus.

Chromatin (krō'-muh-tin) The complex of DNA and proteins that makes up eukaryotic chromosomes. When the cell is not dividing, chromatin exists in its dispersed form, as a mass of very long, thin fibers that are not visible with a light microscope.

Chromosome theory of inheritance (krō'-muh-sōm) A basic principle in biology stating that genes are located at specific positions (loci) on

chromosomes and that the behavior of chromosomes during meiosis accounts for inheritance patterns.

Chromosomes (singular, **chromosome**) (krō'-muh-sōm) A cellular structure consisting of one DNA molecule and associated protein molecules. A duplicated chromosome has two DNA molecules. (In some contexts, such as genome sequencing, the term may refer to the DNA alone.) A eukaryotic cell typically has multiple, linear chromosomes, which are located in the nucleus. A prokaryotic cell often has a single, circular chromosome, which is found in the nucleoid, a region that is not enclosed by a membrane; The complex of DNA and proteins that makes up eukaryotic chromosomes. When the cell is not dividing, chromatin exists in its dispersed form, as a mass of very long, thin fibers that are not visible with a light microscope.

Chylomicrons (kī'-lō-mī'-kron) Lipid transport globules composed of fats mixed with cholesterol and coated with proteins.

Chyme (kīm) The mixture of partially digested food and digestive juices formed in the stomach.

Chytrids (kī'-trids) Members of the fungal phylum Chytridiomycota, mostly aquatic fungi with flagellated zoospores.

Ciliates (sil'-ē-its) A group of protists in the SAR supergroup that move by means of cilia.

Cilium (sil'-ē-um) (plural, **cilia**) A short appendage containing microtubules in eukaryotic cells. A motile cilium is specialized for locomotion or moving fluid past the cell; it is formed from a core of nine outer doublet microtubules and two inner single microtubules (the "9 plus 2" arrangement) ensheathed in an extension of the plasma membrane. A primary cilium is usually nonmotile and plays a sensory and signaling role; it lacks the two inner microtubules (the "9 plus 0" arrangement).

Circadian rhythms (ser-kā'-dē-un) Physiological cycles of about 24 hours that persist even in the absence of external cues.

Cis-trans isomers Compounds that have the same molecular formula and covalent bonds between atoms but differ in the spatial arrangements of their atoms owing to the inflexibility of double bonds; also called geometric isomers.

Citric acid cycle A chemical cycle involving eight steps that completes the metabolic breakdown of glucose molecules begun in glycolysis by oxidizing acetyl CoA (derived from pyruvate) to carbon dioxide; occurs within the mitochondrion in eukaryotic cells and in the cytosol of prokaryotes; together with pyruvate oxidation, the second major stage in cellular respiration.

Clades (klād) Groups of species that include an ancestral species and all of its descendants. A clade is equivalent to a monophyletic group.

Cladistics (kluh-dis'-tiks) An approach to systematics in which organisms are placed into

groups called clades based primarily on common descent.

Classes In Linnaean classification, the taxonomic category above the level of orders.

Cleavage (1) The process of cytokinesis in animal cells, characterized by pinching of the plasma membrane. (2) The succession of rapid cell divisions without significant growth during early embryonic development that converts the zygote to a ball of cells.

Cleavage furrow The first sign of cleavage in an animal cell; a shallow groove around the cell in the cell surface near the old metaphase plate.

Climate The long-term prevailing weather conditions at a given place.

Climate change A directional change in temperature, precipitation, or other aspect of the global climate that lasts for three decades or more.

Climograph A plot of the temperature and precipitation in a particular region.

Clitoris (klit'-uh-ris) An organ at the upper intersection of the labia minora that engorges with blood and becomes erect during sexual arousal.

Cloaca (klō-ā'-kuh) A common opening for the digestive, urinary, and reproductive tracts found in many nonmammalian vertebrates but in few mammals.

Clonal selection The process by which an antigen selectively binds to and activates only those lymphocytes bearing receptors specific for the antigen. The selected lymphocytes proliferate and differentiate into a clone of effector cells and a clone of memory cells specific for the stimulating antigen.

Clone (1) A group of genetically identical individuals or cells. (2) In popular usage, an individual that is genetically identical to another individual. (3) As a verb, to make one or more genetic replicas of an individual or cell; The production of multiple copies of a gene.

Cloning vector In genetic engineering, a DNA molecule that can carry foreign DNA into a host cell and replicate there. Cloning vectors include plasmids and bacterial artificial chromosomes (BACs), which move recombinant DNA from a test tube back into a cell, and viruses that transfer recombinant DNA by infection.

Closed circulatory system A circulatory system in which blood is confined to vessels and is kept separate from the interstitial fluid.

Cnidocytes (nī'-duh-sīts) Specialized cells unique to the phylum Cnidaria; contain a capsule-like organelle housing a coiled thread that, when discharged, explodes outward and functions in prey capture or defense.

Cochlea (kok'-lē-uh) The complex, coiled organ of hearing that contains the organ of Corti.

Coding strand Nontemplate strand of DNA, which has the same sequence as the mRNA except it has thymine (T) instead of uracil (U).

Codominance The situation in which the phenotypes of both alleles are exhibited in the

heterozygote because both alleles affect the phenotype in separate, distinguishable ways.

Codons (kō'-don) Three-nucleotide sequences of DNA or mRNA that specify a particular amino acid or termination signal; the basic units of the genetic code.

Coefficient of relatedness The fraction of genes that, on average, are shared by two individuals.

Coelom (sē'-lōm) A body cavity lined by tissue derived only from mesoderm.

Coenocytic fungi (sē'-no-sī'-tic) Fungi that lack septa and hence whose bodies are made up of a continuous cytoplasmic mass that may contain hundreds or thousands of nuclei.

Coenzyme (kō-en'-zīm) An organic molecule serving as a cofactor. Most vitamins function as coenzymes in metabolic reactions.

Coevolution The joint evolution of two interacting species, each in response to selection imposed by the other.

Cofactors Nonprotein molecules or ions that are required for the proper functioning of an enzyme. Cofactors can be permanently bound to the active site or may bind loosely and reversibly, along with the substrate, during catalysis.

Cognition The process of knowing that may include awareness, reasoning, recollection, and judgment.

Cognitive map A neural representation of the abstract spatial relationships between objects in an animal's surroundings.

Cohesion The linking together of like molecules, often by hydrogen bonds.

Cohesion-tension hypothesis The leading explanation of the ascent of xylem sap. It states that transpiration exerts pull on xylem sap, putting the sap under negative pressure, or tension, and that the cohesion of water molecules transmits this pull along the entire length of the xylem from shoots to roots.

Cohort A group of individuals of the same age in a population.

Coleoptile (kō'-lē-op'-tul) The covering of the young shoot of the embryo of a grass seed.

Coleorhiza (kō'-lē-uh-rī'-zuh) The covering of the young root of the embryo of a grass seed.

Collagen A glycoprotein in the extracellular matrix of animal cells that forms strong fibers, found extensively in connective tissue and bone; the most abundant protein in the animal kingdom.

Collecting duct The location in the kidney where processed filtrate, called urine, is collected from the renal tubules.

Collenchyma cells (kō-len'-kim-uh) A flexible plant cell type that occurs in strands or cylinders that support young parts of the plant without restraining growth.

Colon (kō'-len) The largest section of the vertebrate large intestine; functions in water absorption and formation of feces.

Commensalism (kuh-men'-suh-lizm) A positive-neutral ecological interaction that benefits the individuals of one species but neither harms nor helps the individuals of the other species.

Communication (1) In behavior, a process involving the transmission and reception of signals between organisms. (2) Transfer of information from one cell or molecule to another by means of chemical or physical signals.

Community All the organisms that inhabit a particular area; an assemblage of populations of different species living close enough together for potential interaction.

Community structure The number of species found in an ecological community, the particular species that are present, and the relative abundance of these species.

Companion cells Plant cells that are connected to a sieve tube element by many plasmodesmata and whose nucleus and ribosomes may serve one or more adjacent sieve tube elements.

Competitive exclusion The concept that when populations of two similar species compete for the same limited resources, one population will use the resources more efficiently and have a reproductive advantage that will eventually lead to the elimination of the other population.

Competitive inhibitors Substances that reduce the activity of an enzyme by entering the active site in place of the substrate, whose structure it mimics.

Complement system A group of about 30 blood proteins that may amplify the inflammatory response, enhance phagocytosis, or directly lyse extracellular pathogens.

Complementary DNA (cDNA) A double-stranded DNA molecule made *in vitro* using mRNA as a template and the enzymes reverse transcriptase and DNA polymerase. A cDNA molecule corresponds to the exons of a gene.

Complete dominance The situation in which the phenotypes of the heterozygote and dominant homozygote are indistinguishable.

Complete flowers Flowers that have all four basic floral organs: sepals, petals, stamens, and carpels.

Complete metamorphosis The transformation of a larva into an adult that looks very different, and often functions very differently in its environment, than the larva.

Compound A substance consisting of two or more different elements combined in a fixed ratio.

Compound eyes A type of multifaceted eye in insects and crustaceans consisting of up to several thousand light-detecting, focusing ommatidia.

Concentration gradient A region along which the density of a chemical substance increases or decreases.

Conception The fertilization of an egg by a sperm in humans.

Cones Cone-shaped cells in the retina of the vertebrate eye, sensitive to color.

Conformer An animal for which an internal condition conforms to (changes in accordance with) changes in an environmental variable.

Conidia (singular, **conidium**) Haploid spores produced at the tip of a specialized hypha in ascomycetes during asexual reproduction.

Conifers Members of the largest gymnosperm phylum. Most conifers are cone-bearing trees, such as pines and firs.

Conjugation (kon'-jū-gā'-shun) (1) In prokaryotes, the direct transfer of DNA between two cells that are temporarily joined. When the two cells are members of different species, conjugation results in horizontal gene transfer. (2) In ciliates, a sexual process in which two cells exchange haploid micronuclei but do not reproduce.

Connective tissue Animal tissue that functions mainly to bind and support other tissues, having a sparse population of cells scattered through an extracellular matrix.

Conodonts Early, soft-bodied vertebrates with prominent eyes and dental elements.

Conservation biology The integrated study of ecology, evolutionary biology, physiology, molecular biology, and genetics to sustain biological diversity at all levels.

Consumer An organism that feeds on producers, other consumers, or nonliving organic material.

Contraception The deliberate prevention of pregnancy.

Contractile vacuoles Membranous sacs that help move excess water out of certain freshwater protists.

Control center A component of an animal's homeostatic control system that compares the body's sensor inputs with the set point for a given variable, and, if necessary, alters the activity of an effector (such as a muscle, gland, or specific body cell) in a way that brings the level of the variable closer to set point.

Control elements Segments of noncoding DNA that help regulate transcription of a gene by serving as binding sites for a transcription factor. Multiple control elements are present in a eukaryotic gene's enhancer.

Control group In a controlled experiment, a set of subjects that lacks (or does not receive) the specific factor being tested. Ideally, the control group is identical to the experimental group in other respects.

Controlled experiment An experiment designed to compare an experimental group with a control group; ideally, the two groups differ only in the factor being tested.

Convergent evolution The evolution of similar features in independent evolutionary lineages.

Convergent extension A process in which the cells of a tissue layer rearrange themselves in such a way that the sheet of cells becomes narrower (converges) and longer (extends).

Cooperativity A kind of allosteric regulation whereby a shape change in one subunit of a protein caused by substrate binding is transmitted to all the other subunits, facilitating binding of additional substrate molecules to those subunits.

Corepressor A small molecule that binds to a bacterial repressor protein and changes the protein's shape, allowing it to bind to the operator and switch an operon off.

Cork cambium (kam'-bē-um) A cylinder of meristematic tissue in woody plants that replaces the epidermis with thicker, tougher cork cells.

Corpus callosum (kor'-pus kuh-lō'-sum) The thick band of nerve fibers that connects the right and left cerebral hemispheres in mammals, enabling the hemispheres to process information together.

Corpus luteum (kor'-pus lū'-tē-um) A secreting tissue in the ovary that forms from the collapsed follicle after ovulation and produces progesterone.

Cortex (1) The outer region of cytoplasm in a eukaryotic cell, lying just under the plasma membrane, that has a more gel-like consistency than the inner regions due to the presence of multiple microfilaments. (2) In plants, ground tissue that is between the vascular tissue and dermal tissue in a root or eudicot stem.

Cotransport The coupling of the "downhill" diffusion of one substance to the "uphill" transport of another against its own concentration gradient.

Cotyledons (kot'-uh-lē'-duns) Seed leaves of an angiosperm embryo. Some species have one cotyledon, others two.

Countercurrent exchange The exchange of a substance or heat between two fluids flowing in opposite directions. For example, blood in a fish gill flows in the opposite direction of water passing over the gill, maximizing diffusion of oxygen into and carbon dioxide out of the blood.

Countercurrent multiplier systems Countercurrent systems in which energy is expended in active transport to facilitate exchange of materials and generate concentration gradients.

Covalent bond (kō-vā'-lent) A type of strong chemical bond in which two atoms share one or more pairs of valence electrons.

COVID-19 (COrona VIrus Disease 2019) The disease caused by the coronavirus called SARS-CoV-2; first identified in 2019, the disease became pandemic in 2020 and has caused millions of deaths globally.

Crassulacean acid metabolism (CAM) (crass-yū-lā'-shen) An adaptation for photosynthesis in arid conditions, first discovered in the family Crassulaceae. In this process, a plant takes up carbon dioxide and incorporates it into a variety of organic acids at night; during the day, carbon dioxide is released from organic acids for use in the Calvin cycle.

CRISPR-Cas9 system A technique for editing genes in living cells, involving a bacterial protein called Cas9 associated with a guide RNA complementary to a gene sequence of interest.

Cristae (singular, **crista**) (kris'-tē, kris'-tuh) Infoldings of the inner membrane of a mitochondrion. The inner membrane houses electron transport chains and molecules of the enzyme catalyzing the synthesis of ATP (ATP synthase).

Critical load The amount of added nutrient, usually nitrogen or phosphorus, that can be absorbed by plants without damaging ecosystem integrity.

Critical period A limited phase in an animal's development when learning of particular behaviors can take place; sometimes called a sensitive period.

Crop rotation The practice of growing different crops in succession on the same land chiefly to preserve the productive capacity of the soil.

Cross-fostering study A behavioral study in which the young of one species are placed in the care of adults from another species.

Cross-pollination In angiosperms, the transfer of pollen from an anther of a flower on one plant to the stigma of a flower on another plant of the same species.

Crossing over The reciprocal exchange of genetic material between nonsister chromatids during prophase I of meiosis.

Cryptic coloration Camouflage that makes a potential prey difficult to spot against its background.

Cryptomycetes Members of the fungal phylum Cryptomycota, unicellular fungi that have flagellated spores.

Cryptophytes A clade of phytoplankton with two flagella that inhabit freshwater and marine environments; thought to be closely related to Archaeplastida.

Culture A system of information transfer through social learning or teaching that influences the behavior of individuals in a population.

Cuticle (kyū'-tuh-kul) Any of a variety of tough but flexible, non-mineral outer coverings of an organism, or parts of an organism, which provide protection.

Cyclic AMP (cAMP) Cyclic adenosine monophosphate, named because of its ring structure, is a common chemical signal that has a diversity of roles, including as a second messenger in many eukaryotic cells, and as a regulator of some bacterial operons.

Cyclic electron flow A route of electron flow during the light reactions of photosynthesis that involves only one photosystem and that produces ATP but not NADPH or oxygen.

Cyclin (sī'-klin) A cellular protein that occurs in a cyclically fluctuating concentration and that plays an important role in regulating the cell cycle.

Cyclin-dependent kinases (Cdks) Protein kinases that are active only when attached to a particular cyclin.

Cyclostomes (sī'-cluh-stōm) Members of one of the two main clades of vertebrates; cyclostomes lack jaws and include lampreys and hagfishes.

Cystic fibrosis (sis'-tik fi-brō'-sis) A human genetic disease caused by a recessive allele for a chloride channel protein; characterized by an excessive secretion of mucus and consequent vulnerability to infection; fatal if untreated.

Cytochromes (sī'-tō-krōms) Iron-containing proteins that are components of electron transport chains in the mitochondria and chloroplasts of eukaryotic cells and the plasma membranes of prokaryotic cells.

Cytokines (sī'-tō-kins') A group of small proteins secreted by a number of cell types, including macrophages and helper T cells, that regulate the function of other cells, particularly immune cells. Examples include interferons and interleukins.

Cytokinesis (sī'-tō-kuh-nē'-sis) The division of the cytoplasm to form two separate daughter cells immediately after mitosis, meiosis I, or meiosis II.

Cytokininins (sī'-tō-ki'-nins) A class of related plant hormones that retard aging and act in concert with auxin to stimulate cell division, influence the pathway of differentiation, and control apical dominance.

Cytoplasm (sī'-tō-plaz-um) The contents of the cell bounded by the plasma membrane; in eukaryotes, the portion exclusive of the nucleus.

Cytoplasmic determinants Maternal substances, such as protein or RNA, that when placed into an egg influence the course of early development by regulating the expression of genes that affect the developmental fate of cells.

Cytoplasmic streaming A circular flow of cytoplasm, involving interactions of myosin and actin filaments, that speeds the distribution of materials within cells.

Cytoskeleton A network of microtubules, microfilaments, and intermediate filaments that extend throughout the cytoplasm and serve a variety of mechanical, transport, and signaling functions.

Cytosol (sī'-tō-sol) The semifluid portion of the cytoplasm.

Cytotoxic T cells A type of lymphocyte that, when activated, kills infected cells as well as certain cancer cells and transplanted cells.

Dalton A measure of mass for atoms and subatomic particles; the same as the atomic mass unit, or amu.

Damage-associated molecular patterns (DAMP) A set of molecules characteristic of damaged or stressed body cells or tissues that can be recognized by an animal's immune system.

Data Recorded observations.

Day-neutral plants Plants in which flower formation is not controlled by photoperiod or day length.

De-etiolation The changes a plant shoot undergoes in response to sunlight; also known informally as greening.

Decomposers Organisms that absorb nutrients from nonliving organic material such as corpses, fallen plant material, and the wastes of living organisms and convert them to inorganic forms; also called detritivores.

Deductive reasoning A type of logic in which specific results are predicted from a general premise.

Dehydration reaction A chemical reaction in which two molecules become covalently bonded to each other with the removal of a water molecule.

Deletion (1) A deficiency in a chromosome resulting from the loss of a fragment through breakage. (2) A mutational loss of one or more nucleotide pairs from a gene.

Demographic transition In a stable population, a shift from high birth and death rates to low birth and death rates.

Demography The study of changes over time in the vital statistics of populations, especially birth rates and death rates.

Denaturation (dē-nā'-chur-ā'-shun) In proteins, a process in which a protein loses its native shape due to the disruption of weak chemical bonds and interactions, thereby becoming biologically inactive; in DNA, the separation of the two strands of the double helix. Denaturation occurs under extreme (noncellular) conditions of pH, salt concentration, or temperature.

Dendrites (den'-drīt) Short, highly branched, usually numerous extensions of a neuron that receive signals from other neurons.

Dendritic cells Antigen-presenting cells, located mainly in lymphatic tissues and skin, that are particularly efficient in presenting antigens to helper T cells, thereby initiating a primary immune response.

Density The number of individuals per unit area or volume.

Density dependent Referring to any characteristic that varies with population density.

Density-dependent inhibition The phenomenon observed in normal animal cells that causes them to stop dividing when they come into contact with one another.

Density independent Referring to any characteristic that is not affected by population density.

Deoxyribonucleic acid (DNA) (dē-ok'-sē-rī'-bō-nū-klā-ik) A nucleic acid molecule, usually a double-stranded helix, in which each polynucleotide strand consists of nucleotide monomers with a deoxyribose sugar and the nitrogenous bases adenine (A), cytosine (C), guanine (G), and thymine (T); capable of being replicated and determining the inherited structure of a cell's proteins.

Deoxyribose (dē-ok'-si-rī'-bōs) The sugar component of DNA nucleotides, having one fewer hydroxyl group than ribose, the sugar component of RNA nucleotides.

Dependent variable A factor whose value is measured during an experiment or other test to see whether it is influenced by changes in another factor (the independent variable).

Depolarization A change in a cell's membrane potential such that the inside of the membrane is made less negative relative to the outside. For example, a neuron membrane is depolarized if a stimulus decreases its voltage from the resting potential of negative 70 mV in the direction of zero voltage.

Dermal tissue The outer protective covering of plants.

Desmosomes A type of intercellular junction in animal cells that functions like a rivet, fastening cells together.

Determinate cleavage A type of embryonic development in protostomes that rigidly casts the developmental fate of each embryonic cell very early.

Determinate growth A type of growth characteristic of most animals and some plant organs, in which growth stops after a certain size is reached.

Determination The progressive restriction of developmental potential in which the possible fate of each cell becomes more limited as an embryo develops. At the end of determination, a cell is committed to its fate.

Detritus (di-trī'-tus) Dead organic matter.

Deuteromycetes (dū'-tuh-rō-mī'-sēt) Traditional classification for a fungus with no known sexual stage.

Deuterostome development (dū'-tuh-rō-stōm') In animals, a developmental mode distinguished by the development of the anus from the blastopore; often also characterized by radial cleavage and by the body cavity forming as outpockets of mesodermal tissue.

Deuterostomia (dū'-tuh-rō-stōm'-ē-uh) One of the three main lineages of bilaterian animals.

Development The events involved in an organism's changing gradually from a simple to a more complex or specialized form.

Diabetes mellitus (dī'-uh-bē'-tis mel'-uh-tus)

An endocrine disease marked by an inability to maintain glucose homeostasis. The type 1 form results from autoimmune destruction of insulin-secreting cells; treatment usually requires daily insulin injections. The type 2 form most commonly results from reduced responsiveness of target cells to insulin; obesity and lack of exercise are risk factors.

Diacylglycerol (DAG) (dī-a'-sil-glīs'-er-ol) A second messenger produced by the cleavage of the phospholipid PIP₂ in the plasma membrane.**Diaphragm** (dī'-uh-fram') (1) A sheet of muscle that forms the bottom wall of the thoracic cavity in mammals. Contraction of the diaphragm pulls air into the lungs. (2) A dome-shaped rubber cup fitted into the upper portion of the vagina before sexual intercourse. It serves as a physical barrier to the passage of sperm into the uterus.**Diapsids** (dī-ap'-sid) Members of an amniote clade distinguished by having a pair of holes on each side of the skull. Diapsids include the lepidosaurs and archosaurs.**Diastole** (dī-as'-tō-lē) The stage of the cardiac cycle in which a heart chamber is relaxed and fills with blood.**Diastolic pressure** Blood pressure in the arteries when the ventricles are relaxed.**Diatoms** Photosynthetic protists in the stramenopile clade; diatoms have a unique glass-like wall made of silicon dioxide embedded in an organic matrix.**Dicots** A term traditionally used to refer to flowering plants that have two embryonic seed leaves, or cotyledons. Recent molecular evidence indicates that dicots do not form a clade; species once classified as dicots are now grouped into eudicots, magnoliids, and several lineages of basal angiosperms.**Differential gene expression** The expression of different sets of genes by cells with the same genome.**Differentiation** The process by which a cell or group of cells becomes specialized in structure and function.**Diffusion** The random thermal motion of particles of liquids, gases, or solids. In the presence of a concentration or electrochemical gradient, diffusion results in the net movement of a substance from a region where it is more concentrated to a region where it is less concentrated.**Digestion** The second stage of food processing in animals: the breaking down of food into molecules small enough for the body to absorb.**Dihybrid cross** (dī'-hī'-brid) A cross between two organisms that are each heterozygous for both of the characters being followed (or the self-pollination of a plant that is heterozygous for both characters).**Dihybrids** (dī'-hī'-brids) Organisms that are heterozygous with respect to two genes of interest. All the offspring from a cross between parents doubly homozygous for different

alleles are dihybrids. For example, parents of genotypes *AABB* and *aabb* produce a dihybrid of genotype *AaBb*.

Dikaryotic (dī'-kār-ē-ot'-ik) Referring to a fungal mycelium with two haploid nuclei per cell, one from each parent.**Dinoflagellates** (dī'-nō-flaj'-uh-lets) A group of mostly unicellular photosynthetic algae with two flagella situated in perpendicular grooves in cellulose plates covering the cell.**Dinosaurs** Members of an extremely diverse clade of reptiles varying in body shape, size, and habitat. Birds are the only extant dinosaurs.**Diploblastic** Having two germ layers.**Diploid cell** (dip'-loyd) A cell containing two sets of chromosomes ($2n$), one set inherited from each parent.**Directional selection** Natural selection in which individuals at one end of the phenotypic range survive or reproduce more successfully than do other individuals.**Disaccharide** (dī-sak'-uh-rīd) A double sugar, consisting of two monosaccharides joined by a glycosidic linkage formed by a dehydration reaction.**Discoba** A diverse clade of protists that includes predatory heterotrophs, photosynthetic autotrophs, and mixotrophs found in marine, freshwater, and moist terrestrial ecosystems, as well as parasites; this group includes the euglenids.**Dispersal** The movement of individuals or gametes away from their parent location. This movement sometimes expands the geographic range of a population or species.**Dispersion** The pattern of spacing among individuals within the boundaries of a population.**Disruptive selection** Natural selection in which individuals on both extremes of a phenotypic range survive or reproduce more successfully than do individuals with intermediate phenotypes.**Distal tubule** In the vertebrate kidney, the portion of a nephron that helps refine filtrate and empties it into a collecting duct.**Disturbance** A natural or human-caused event that changes a biological community and usually removes organisms from it. Disturbances, such as fires and storms, play a pivotal role in structuring many communities.**DNA (deoxyribonucleic acid)** (dē-ok'-sē-rī'-bō-nū-klā'-ik) A nucleic acid molecule, usually a double-stranded helix, in which each polynucleotide strand consists of nucleotide monomers with a deoxyribose sugar and the nitrogenous bases adenine (A), cytosine (C), guanine (G), and thymine (T); capable of being replicated and determining the inherited structure of a cell's proteins.**DNA cloning** The production of multiple copies of a specific DNA segment.**DNA ligase** (lī'-gās) A linking enzyme essential for DNA replication; catalyzes the covalent bonding

of the 3' end of one DNA fragment (such as an Okazaki fragment) to the 5' end of another DNA fragment (such as a growing DNA chain).

DNA methylation The presence of methyl groups on the DNA bases (usually cytosine) of plants, animals, and fungi. (The term also refers to the process of adding methyl groups to DNA bases.)**DNA microarray assays** A method to detect and measure the expression of thousands of genes at one time. Tiny amounts of a large number of single-stranded DNA fragments representing different genes are fixed to a glass slide and tested for hybridization with samples of labeled cDNA.**DNA polymerases** (puh-lim'-er-ās-es) Enzymes that catalyze the elongation of new DNA (for example, at a replication fork) by the addition of nucleotides to the 3' end of an existing chain. There are several different DNA polymerases; DNA polymerase III and DNA polymerase I play major roles in DNA replication in *E. coli*.**DNA replication** The process by which a DNA molecule is copied; also called DNA synthesis.**DNA sequencing** Determining the complete nucleotide sequence of a gene or DNA segment.**DNA technology** Techniques for sequencing and manipulating DNA.**Domains** (1) Taxonomic category above the kingdom level. In the three-domain hypothesis, the three domains are Bacteria, Archaea, and Eukarya. In the two-domain hypothesis, the domains are Bacteria and Archaea. (2) Discrete structural and functional regions of a protein.**Dominant allele** An allele that is fully expressed in the phenotype of a heterozygote.**Dormancy** A condition typified by extremely low metabolic rate and a suspension of growth and development.**Dorsal** In an animal with bilateral symmetry, pertaining to the top (in most animals) or back (in animals with upright posture) of the body.**Double bond** A double covalent bond; the sharing of two pairs of valence electrons by two atoms.**Double circulation** A circulatory system consisting of separate pulmonary and systemic circuits, in which blood passes through the heart after completing each circuit.**Double fertilization** A mechanism of fertilization in angiosperms in which two sperm cells unite with two cells in the female gametophyte (embryo sac) to form the zygote and endosperm.**Double helix** The form of native DNA, referring to its two adjacent antiparallel polynucleotide strands wound around an imaginary axis into a spiral shape.**Down syndrome** A human genetic disease usually caused by the presence of an extra chromosome 21; characterized by developmental

delays and heart and other defects that are generally treatable or non-life-threatening.

Duchenne muscular dystrophy (duh-shen') A human genetic disease caused by a sex-linked recessive allele; characterized by progressive weakening and a loss of muscle tissue.

Duodenum (dū'-uh-dēn'-um) The first section of the small intestine, where chyme from the stomach mixes with digestive juices from the pancreas, liver, and gallbladder as well as from gland cells of the intestinal wall.

Duplication An aberration in chromosome structure due to fusion with a fragment from a homologous chromosome, such that a portion of a chromosome is duplicated.

Dyneins (dī'-nē-uns) In cilia and flagella, large motor proteins extending from one microtubule doublet to the adjacent doublet. ATP hydrolysis drives changes in dynein shape that lead to bending of cilia and flagella.

Ecdysozoa (ek'-dē-sō-zō-uh) One of the three main lineages of bilaterian animals; many ecdysozoans are molting animals.

Echinoderms (i-ki'-nō-derms) Slow-moving or sessile marine deuterostomes with a water vascular system and, in larvae, bilateral symmetry. Echinoderms include sea stars, brittle stars, sea urchins, feather stars, and sea cucumbers.

Ecological footprint The aggregate land and water area required by a person, city, or nation to produce all of the resources it consumes and to absorb all of the waste it generates.

Ecological niche (nich) The sum of a species' use of the biotic and abiotic resources in its environment.

Ecological species concept Definition of a species in terms of ecological niche, the sum of how members of the species interact with the nonliving and living parts of their environment.

Ecological succession Transition in the species composition of a community following a disturbance; establishment of a community in an area virtually barren of life.

Ecology The study of how organisms interact with each other and their environment.

Ecosystem All the organisms in a given area as well as the abiotic factors with which they interact; one or more communities and the physical environment around them.

Ecosystem engineers Organisms that influence community structure by causing physical changes in the environment.

Ecosystem services Functions performed by an ecosystem that directly or indirectly benefit humans.

Ecotone The transition from one type of habitat or ecosystem to another, such as the transition from a forest to a grassland.

Ectoderm (ek'-tō-durm) The outermost of the three primary germ layers in animal embryos; gives rise to the outer covering and, in some

phyla, the nervous system, inner ear, and lens of the eye.

Ectomycorrhizae (plural, **ectomycorrhizae**) (ek'-tō-mī'-kō-rī'-zuh, ek'-tō-mī'-kō-rī'-zē) Association of a fungus with a plant root system in which the fungus surrounds the roots but does not cause invagination of the host (plant) cell's plasma membrane.

Ectomycorrhizal fungi Symbiotic fungi that form sheaths of hyphae over the surface of plant roots and also grow into extracellular spaces of the root cortex.

Ectoparasites Parasites that feed on the external surface of a host.

Ectopic Occurring in an abnormal location.

Ectoprocts Sessile, colonial lophotrochozoans; also called bryozoans.

Ectothermic Referring to organisms for which external sources provide most of the heat for temperature regulation.

Ediacaran biota (ē'-dē-uh-keh'-run bī-ō'-tuh) An early group of macroscopic, mostly soft-bodied, multicellular eukaryotes known from fossils that range in age from 635 million to 541 million years old.

Effective population size An estimate of the size of a population based on the numbers of females and males that successfully breed; generally smaller than the total population.

Effector (1) A muscle, gland, or specific body cell that carries out the body's response to stimuli as directed by signals from the brain or other processing center of the nervous system. (2) A lymphocyte that has undergone clonal selection and is capable of mediating an adaptive immune response. (3) In plants, pathogen-encoded protein that cripples the host's innate immune system.

Egg The female gamete.

Ejaculation The propulsion of sperm from the epididymis through the muscular vas deferens, ejaculatory duct, and urethra.

Electrocardiogram (ECG or EKG) A record of the electrical impulses that travel through heart muscle during the cardiac cycle.

Electrochemical gradient The diffusion gradient of an ion, which is affected by both the concentration difference of an ion across a membrane (a chemical force) and the ion's tendency to move relative to the membrane potential (an electrical force).

Electrogenic pump An active transport protein that generates voltage across a membrane while pumping ions.

Electromagnetic receptor A receptor of electromagnetic energy, such as visible light, electricity, or magnetism.

Electromagnetic spectrum The entire spectrum of electromagnetic radiation, ranging in wavelength from less than a nanometer to more than a kilometer.

Electrons Subatomic particles with a single negative electrical charge and a mass about 0.5 percent that of a neutron or proton. One or more electrons move around the nucleus of an atom.

Electron microscope (EM) A microscope that uses magnets to focus an electron beam on or through a specimen, resulting in a practical resolution that is 100-fold greater than that of a light microscope using standard techniques. A transmission electron microscope (TEM) is used to study the internal structure of thin sections of cells. A scanning electron microscope (SEM) is used to study the fine details of cell surfaces.

Electron shells Energy levels of electrons at a characteristic average distance from the nucleus of an atom.

Electron transport chain A sequence of electron carrier molecules (membrane proteins) that shuttle electrons down a series of redox reactions that release energy used to make ATP.

Electronegativity The attraction of a given atom for the electrons of a covalent bond.

Electroporation A technique to introduce recombinant DNA into cells by applying a brief electrical pulse to a solution containing the cells. The pulse creates temporary holes in the cells' plasma membranes, through which DNA can enter.

Element Any substance that cannot be broken down to any other substance by chemical reactions.

Elimination The fourth and final stage of food processing in animals: the passing of undigested material out of the body.

Embryo sac (em'-brē-ō) The female gametophyte of angiosperms, formed from the growth and division of the megaspore into a multicellular structure that typically has eight haploid nuclei.

Embryonic lethals Mutations with a phenotype leading to death of an embryo or larva.

Emergent properties New properties that arise with each step upward in the hierarchy of life, owing to the arrangement and interactions of parts as complexity increases.

Emigration The movement of individuals out of a population.

Enantiomers (en-an'-tē-ō-mers) Compounds that are mirror images of each other and that differ in shape due to the presence of an asymmetric carbon.

Endangered species A species that is in danger of extinction throughout all or a significant portion of its range.

Endergonic reaction (en'-der-gon'-ik) A non-spontaneous chemical reaction in which free energy is absorbed from the surroundings.

Endocrine glands (en'-dō-krin) Ductless glands that secrete hormones directly into the interstitial fluid, from which they diffuse into the bloodstream.

Endocrine system (en'-dō-krin) In animals, the internal system of communication involving hormones, the ductless glands that secrete hormones, and the molecular receptors on or in target cells that respond to hormones; functions in concert with the nervous system to effect internal regulation and maintain homeostasis.

Endocytosis (en'-dō-sī-tō'-sis) Cellular uptake of biological molecules and particulate matter via formation of vesicles from the plasma membrane.

Endoderm (en'-dō-durm) The innermost of the three primary germ layers in animal embryos; lines the archenteron and gives rise to the liver, pancreas, lungs, and the lining of the digestive tract in species that have these structures.

Endodermis In plant roots, the innermost layer of the cortex that surrounds the vascular cylinder.

Endomembrane system The collection of membranes inside and surrounding a eukaryotic cell, related either through direct physical contact or by the transfer of membranous vesicles; includes the plasma membrane, the nuclear envelope, the smooth and rough endoplasmic reticulum, the Golgi apparatus, lysosomes, vesicles, and vacuoles.

Endometriosis (en'-dō-mē-trē-ō'-sis) The condition resulting from the presence of endometrial tissue outside of the uterus.

Endometrium (en'-dō-mē'-trē-um) The inner lining of the uterus, which is richly supplied with blood vessels.

Endoparasites Parasites that live within a host.

Endophytes Harmless fungi, or occasionally other organisms, that live between cells of a plant part or multicellular alga.

Endoplasmic reticulum (ER) (en'-dō-plaz'-mik ruh-tik'-yū-lum) An extensive membranous network in eukaryotic cells, continuous with the outer nuclear membrane and composed of ribosome-studded (rough) and ribosome-free (smooth) regions.

Endorphins (en-dōr'-fins) Several hormones produced in the brain and anterior pituitary that inhibit pain perception.

Endoskeleton A hard skeleton buried within the soft tissues of an animal.

Endosperm In angiosperms, a nutrient-rich tissue formed by the union of a sperm with two polar nuclei during double fertilization. The endosperm provides nourishment to the developing embryo in angiosperm seeds.

Endospores Thick-coated, resistant cells produced by some bacterial cells when they are exposed to harsh conditions.

Endosymbiont theory The theory that mitochondria and plastids originated as prokaryotic cells engulfed by a host cell. The engulfed cell and its host cell then evolved into a single organism; A relationship between two species

in which one organism lives inside the cell or cells of another organism.

Endosymbiosis A relationship between two species in which one organism lives inside the cell or cells of another organism; The theory that mitochondria and plastids originated as prokaryotic cells engulfed by a host cell. The engulfed cell and its host cell then evolved into a single organism.

Endothelium (en'-dō-thē'-lē-um) The simple squamous layer of cells lining the lumen of blood vessels.

Endothermic Referring to organisms that are warmed by heat generated by their own metabolism. This heat usually maintains a relatively stable body temperature higher than that of the external environment.

Endotoxins Toxic components of the outer membrane of certain gram-negative bacteria that are released only when the bacteria die.

Energy The capacity to cause change, especially to do work (to move matter against an opposing force).

Energy coupling In cellular metabolism, the use of energy released from an exergonic reaction to drive an endergonic reaction.

Enhancers Segments of eukaryotic DNA containing multiple control elements, usually located far from the gene whose transcription they regulate.

Enteric nervous system Within the autonomic nervous system, a distinct network of neurons that exerts partially independent control over the digestive tract, pancreas, and gallbladder.

Entropy A measure of molecular disorder, or randomness.

Enzyme (en'-zīm) A macromolecule serving as a catalyst, a chemical agent that increases the rate of a reaction without being consumed by the reaction. Most enzymes are proteins.

Enzyme-substrate complex (en'-zīm) A temporary complex formed when an enzyme binds to its substrate molecule(s).

Eosinophils Immune system cells that secrete destructive enzymes and help defend against multicellular pathogens.

Epicotyl (ep'-uh-kot'-ul) In an angiosperm embryo, the embryonic axis above the point of attachment of the cotyledon(s) and below the first pair of miniature leaves.

Epidemic A widespread outbreak of a disease.

Epidermis (1) The dermal tissue of nonwoody plants, usually consisting of a single layer of tightly packed cells. (2) The outermost layer of cells in an animal.

Epididymis (ep'-uh-did'-uh-mus) A coiled tubule located adjacent to the mammalian testis where sperm are stored.

Epigenetics The study of the inheritance of traits transmitted by mechanisms that do not involve the nucleotide sequence.

Epinephrine (ep'-i-nef'-rin) A catecholamine that, when secreted as a hormone by the adrenal medulla, mediates "fight-or-flight" responses to short-term stresses; also released by some neurons as a neurotransmitter; also called adrenaline.

Epiphytes (ep'-uh-fit) A plant that nourishes itself but grows on the surface of another plant for support, usually on the branches or trunks of trees.

Epistasis (ep'-i-stā'-sis) A type of gene interaction in which the phenotypic expression of one gene alters that of another independently inherited gene.

Epithelial tissue (ep'-uh-thē'-lē-ul) Sheets of tightly packed cells that line organs and body cavities as well as external surfaces.

Epitope A small, accessible region of an antigen to which an antigen receptor or antibody binds.

Equilibrium potential (E_{ion}) The magnitude of a cell's membrane voltage at equilibrium; calculated using the Nernst equation.

Erythrocytes (eh-rith'-ruh-sīts) Blood cells that contain hemoglobin, which transports oxygen; also called red blood cells.

Erythropoietin (EPO) (eh-rith'-rō-poy'-uh-tin) A hormone that stimulates the production of erythrocytes. It is secreted by the kidney when body tissues do not receive enough oxygen.

E site One of a ribosome's three binding sites for tRNA during translation. The E site is the place where discharged tRNAs leave the ribosome. (E stands for exit.)

Esophagus (eh-sof'-uh-gus) A muscular tube that conducts food, by peristalsis, from the pharynx to the stomach.

Essential amino acids Amino acids that an animal cannot synthesize itself and must be obtained from food in prefabricated form.

Essential elements Chemical elements required for an organism to survive, grow, and reproduce.

Essential fatty acids Unsaturated fatty acids that an animal needs but cannot make.

Essential nutrients Substances that an organism cannot synthesize from any other material and therefore must absorb in preassembled form.

Estradiol (es'-truh-dī'-ol) A steroid hormone that stimulates the development and maintenance of the female reproductive system and secondary sex characteristics; the major estrogen in mammals.

Estrogens (es'-trō-jens) Steroid hormones, such as estradiol, that stimulate the development and maintenance of the female reproductive system and secondary sex characteristics.

Estrous cycle (es'-trus) A reproductive cycle characteristic of female mammals except humans and certain other primates, in which the endometrium is reabsorbed in the absence of pregnancy and sexual response

occurs only during a mid-cycle point known as estrus.

Ethylene (eth'-uh-lēn) A gaseous plant hormone involved in responses to mechanical stress, programmed cell death, leaf abscission, and fruit ripening.

Etiolation Plant morphological adaptations for growing in darkness.

Euchromatin (yū-krō'-muh-tin) The less condensed form of eukaryotic chromatin that is available for transcription.

Eudicots (yū-dī'-kots) Members of a clade that contains the vast majority of flowering plants that have two embryonic seed leaves, or cotyledons.

Euglenids (yū'-glen-ids) Single-celled eukaryotes, including *Euglena* and its relatives, characterized by an anterior pocket from which one or two flagella emerge.

Eukarya (yū-kār'-ē-uh) The clade that includes all eukaryotic organisms.

Eukaryotic cell (yū'-kār'-ē-ot'-ik) A type of cell with a membrane-enclosed nucleus and membrane-enclosed organelles. Organisms with eukaryotic cells (protists, plants, fungi, and animals) are called eukaryotes.

Eurypterids (yur-ip'-tuh-rids) Extinct carnivorous chelicerates; also called water scorpions.

Eutherians (yū-thēr'-ē-uns) Placental mammals; mammals whose young complete their embryonic development within the uterus, joined to the female parent by the placenta.

Eutrophication A process by which nutrients, particularly phosphorus and nitrogen, become highly concentrated in a body of water, leading to increased growth of organisms such as algae or cyanobacteria.

Evaporative cooling The process in which the surface of an object becomes cooler during evaporation, a result of the molecules with the greatest kinetic energy changing from the liquid to the gaseous state.

Evapotranspiration The total evaporation of water from an ecosystem, including water transpired by plants and evaporated from a landscape, usually measured in millimeters and estimated for a year.

Evo-devo Evolutionary developmental biology; a field of biology that compares developmental processes of different multicellular organisms to understand how these processes have evolved and how changes can modify existing organismal features or lead to new ones.

Evolution Descent with modification; the process by which species accumulate differences from their ancestors as they adapt to different environments over time; also defined as a change in the genetic composition of a population from generation to generation.

Evolutionary lineage The sequence of ancestral organisms leading to a particular taxon; represented by a branch (line) in a phylogenetic tree.

Evolutionary tree A branching diagram that reflects a hypothesis about evolutionary relationships among groups of organisms.

Excavata (ex'-kuh-vah'-tuh) An informal group of eukaryotes that have unique cytoskeletal features; some species have an "excavated" feeding groove on one side of the cell body.

Excitatory postsynaptic potential (EPSP) An electrical change (depolarization) in the membrane of a postsynaptic cell caused by the binding of an excitatory neurotransmitter from a presynaptic cell to a postsynaptic receptor; makes it more likely for a postsynaptic cell to generate an action potential.

Excretion The disposal of nitrogen-containing metabolites and other waste products.

Exergonic reaction (ek'-ser-gon'-ik) A spontaneous chemical reaction in which there is a net release of free energy.

Exocytosis (ek'-sō-sī-tō'-sis) The cellular secretion of biological molecules by the fusion of vesicles containing them with the plasma membrane.

Exons Sequences within a primary transcript that remain in the RNA after RNA processing; the term exon also refers to the region of DNA from which this sequence was transcribed.

Exoskeleton A hard encasement on the surface of an animal, such as the shell of a mollusc or the cuticle of an arthropod, that provides protection and points of attachment for muscles.

Exotoxins (ek'-sō-tok'-sin) Toxic proteins that are secreted by a prokaryote or other pathogen and that produce specific symptoms, even if the pathogen is no longer present.

Expansins Nonenzyme plant proteins that break the cross-links (hydrogen bonds) between cellulose microfibrils and other cell wall constituents, loosening the wall's fabric.

Experiment A scientific test. Often carried out under controlled conditions that involve manipulating one factor in a system in order to see the effects of changing that factor.

Experimental group A set of subjects that has (or receives) the specific factor being tested in a controlled experiment. Ideally, the experimental group is identical to the control group for all other factors.

Exploitation A positive-negative ecological interaction in which individuals of one species benefit by feeding on (and thereby harming) individuals of the other species. Exploitative interactions include predation, herbivory, and parasitism.

Exponential population growth Growth of a population in an ideal, unlimited environment, represented by a J-shaped curve when population size is plotted over time.

Expression vector A cloning vector that contains a highly active bacterial promoter just upstream of a restriction site where a eukaryotic gene can be inserted, allowing the gene to be expressed

in a bacterial cell. Expression vectors are also available that have been genetically engineered for use in specific types of eukaryotic cells.

Extinction vortex A downward population spiral in which inbreeding and genetic drift combine to cause a small population to shrink and, unless the spiral is reversed, become extinct.

Extracellular matrix (ECM) The meshwork surrounding animal cells, consisting of glycoproteins, polysaccharides, and proteoglycans synthesized and secreted by cells.

Extraembryonic membranes The four membranes (yolk sac, amnion, chorion, and allantois) located outside the embryo that support the developing embryo in reptiles and mammals.

Extreme halophiles Organisms that live in a highly saline environment, such as the Great Salt Lake or the Dead Sea.

Extreme thermophiles Organisms that thrive in hot environments (often 60 to 80 degrees C or hotter).

Extremophiles Organisms that live in environmental conditions so extreme that few other species can survive there. Extremophiles include extreme halophiles ("salt lovers") and extreme thermophiles ("heat lovers").

F₁ generation The first filial, hybrid (heterozygous) offspring arising from a parental (P generation) cross.

F₂ generation The offspring resulting from interbreeding (or self-pollination) of the hybrid F₁ generation.

Facilitated diffusion The passage of molecules or ions down their electrochemical gradient across a biological membrane with the assistance of specific transmembrane transport proteins, requiring no energy expenditure.

Facilitation In ecological succession, a process by which early-arriving species make the environment more favorable for the establishment of later-arriving species; for example, by increasing the nutrient content of the soil.

Facultative anaerobes (fak'-ul-tā'-tiv an'-uh-rōbs) Organisms that make ATP by aerobic respiration if oxygen is present but that switch to anaerobic respiration or fermentation if oxygen is not present.

Family In Linnaean classification, the taxonomic category above genus.

Fast-twitch fibers A muscle fiber used for rapid, powerful contractions.

Fat A lipid consisting of three fatty acids linked to one glycerol molecule; also called a triacylglycerol or triglyceride.

Fate maps Territorial diagrams of embryonic development that display the future derivatives of individual cells and tissues.

Fatty acid A carboxylic acid with a long carbon chain. Fatty acids vary in length and in the number and location of double bonds; three fatty acids linked to a glycerol molecule form a fat molecule, also called triacylglycerol or triglyceride.

Feces (fē'-sēz) The wastes of the digestive tract.

Feedback inhibition A method of metabolic control in which the end product of a metabolic pathway acts as an inhibitor of an enzyme within that pathway.

Feedback regulation The regulation of a process by its output or end product.

Fermentation A catabolic process that makes a limited amount of ATP from glucose (or other organic molecules) without an electron transport chain and that produces a characteristic end product, such as ethyl alcohol or lactic acid.

Fertilization (1) The union of haploid gametes to produce a diploid zygote. (2) The addition of mineral nutrients to the soil.

Fetus (fē'-tus) A developing mammal that has all the major structures of an adult. In humans, the fetal stage lasts from the 9th week of gestation until birth.

F factor In bacteria, the DNA segment that confers the ability to form pili for conjugation and associated functions required for the transfer of DNA from donor to recipient. The F factor may exist as a plasmid or be integrated into the bacterial chromosome.

Fibroblasts A type of cell in loose connective tissue that secretes the protein ingredients of the extracellular fibers.

Fibronectin An extracellular glycoprotein secreted by animal cells that helps them attach to the extracellular matrix.

Filament In an angiosperm, the stalk portion of the stamen, the pollen-producing reproductive organ of a flower.

Filter feeders Animals that feed by using a filtration mechanism to strain small organisms or food particles from its surroundings.

Filtrate Cell-free fluid extracted from the body fluid by the excretory system.

Filtration In excretory systems, the extraction of water and small solutes, including metabolic wastes, from the body fluid.

Fimbriae (singular, **fimbria**) Short, hairlike appendages of a prokaryotic cell that help it adhere to the substrate or to other cells.

First law of thermodynamics The principle of conservation of energy: Energy can be transferred and transformed, but it cannot be created or destroyed.

Fission The separation of an organism into two or more individuals of approximately equal size.

Fixed action pattern In animal behavior, a sequence of unlearned acts that is essentially unchangeable and, once initiated, usually carried to completion.

Flaccid (flas'-id) Limp. Lacking turgor (stiffness or firmness), as in a plant cell in surroundings where there is a tendency for water to leave the cell. (A walled cell becomes flaccid if it has a higher water potential than its surroundings, resulting in the loss of water.)

Flagella (fluh-jel'-uh) (singular, **flagellum**) Long cellular appendages specialized for locomotion. Like motile cilia, eukaryotic flagella have a core with nine outer doublet microtubules and two inner single microtubules (the "9 plus 2" arrangement) ensheathed in an extension of the plasma membrane. Prokaryotic flagella have a different structure.

Florigen A flowering signal, probably a protein, that is made in leaves under certain conditions and that travels to the shoot apical meristems, inducing them to switch from vegetative to reproductive growth.

Flower In an angiosperm, a specialized shoot with up to four sets of modified leaves, bearing structures that function in sexual reproduction.

Fluid feeding Method of ingestion in which an animal sucks nutrient-rich fluids from another living organism.

Fluid mosaic model The currently accepted model of cell membrane structure, which envisions the membrane as a mosaic of protein molecules drifting laterally in a fluid bilayer of phospholipids.

Follicle (fol'-uh-kul) A microscopic structure in the ovary that contains the developing oocyte and secretes estrogens.

Follicle-stimulating hormone (FSH) (fol'-uh-kul) A tropic hormone that is produced and secreted by the anterior pituitary and that stimulates the production of eggs by the ovaries and sperm by the testes.

Food chain The pathway along which food energy is transferred from trophic level to trophic level, beginning with producers.

Food vacuoles Membranous sacs formed by phagocytosis of microorganisms or particles to be used as food by the cell.

Food web The interconnected feeding relationships in an ecosystem.

Foot In molluscs, one of three main body parts: a muscular organ mainly used for movement.

Foraging The seeking and obtaining of food.

Forams (foraminiferans) A group of aquatic protists that secrete a hardened shell containing calcium carbonate and that extend pseudopodia through pores in the shell.

Forebrain One of three ancestral and embryonic regions of the vertebrate brain; develops into the thalamus, hypothalamus, and cerebrum.

Fossils Preserved remnants or impressions of an organism that lived in the past.

Foundation species A species that has strong effects on its community as a result of its large size, high abundance, or pivotal role in community dynamics. Foundation species may provide significant habitat or food for other species; they may also be competitively dominant in exploiting key resources.

Founder effect Genetic drift that occurs when a few individuals become isolated from a larger population and form a new population whose gene pool composition is not reflective of that of the original population.

Fovea (fō'-vē-uh) The place on the retina at the eye's center of focus, where cones are highly concentrated.

F plasmid The plasmid form of the F factor.

Fragmentation A means of asexual reproduction whereby a single parent breaks into parts that regenerate into whole new individuals.

Frameshift mutation A mutation occurring when nucleotides are inserted in or deleted from a gene and the number inserted or deleted is not a multiple of three, resulting in the improper grouping of the subsequent nucleotides into codons.

Free energy The portion of a biological system's energy that can perform work when temperature and pressure are uniform throughout the system. The change in free energy of a system (ΔG) is calculated by taking the change in enthalpy (ΔH) and subtracting the product of the temperature (T) and the change in entropy (ΔS). In biological systems, the change in enthalpy is equivalent to total energy, and the change in temperature is the absolute temperature.

Frequency-dependent selection Selection in which the fitness of a phenotype depends on how common the phenotype is in a population.

Fruit A mature ovary of a flower. The fruit protects dormant seeds and often functions in their dispersal.

Functional groups Specific configurations of atoms commonly attached to the carbon skeletons of organic molecules and involved in chemical reactions.

Fusion In evolutionary biology, a process in which gene flow between two species that can form hybrid offspring weakens barriers to reproduction between the species. This process causes their gene pools to become increasingly alike and can cause the two species to fuse into a single species.

G₀ phase A nondividing state occupied by cells that have left the cell cycle, sometimes reversibly.

G₁ phase The first gap, or growth phase, of the cell cycle, consisting of the portion of interphase before DNA synthesis begins.

G₂ phase The second gap, or growth phase, of the cell cycle, consisting of the portion of interphase after DNA synthesis occurs.

Gallbladder An organ that stores bile and releases it as needed into the small intestine.

Gametangia (gam'-uh-tan'-jē-uh) (singular, **gametangium**) Multicellular plant structures in which gametes are formed. Female gametangia are called archegonia, and male gametangia are called antheridia.

Gametes (singular, **gamete**) (gam'-ēt) A haploid reproductive cell, such as an egg or sperm, that is formed by meiosis or is the descendant of cells

formed by meiosis. Gametes unite during sexual reproduction to produce a diploid zygote.

Game theory An approach to evaluating alternative strategies in situations where the outcome of a particular strategy depends on the strategies used by other individuals.

Gametogenesis (guh-mē'-tō-gen'-uh-sis) The process by which gametes are produced.

Gametophytes (guh-mē'-tō-fits) In organisms (plants and some algae) that have alternation of generations, the multicellular haploid forms that produce haploid gametes by mitosis. The haploid gametes unite and develop into sporophytes.

Ganglia (gan'-glē-uh) (singular, **ganglion**) A cluster (functional group) of nerve cell bodies.

Gap junctions A type of intercellular junction in animal cells, consisting of proteins surrounding a pore that allows the passage of materials between cells.

Gas exchange The uptake of molecular oxygen from the environment and the discharge of carbon dioxide to the environment.

Gastric juice A digestive fluid secreted by the stomach.

Gastrovascular cavity A central cavity with a single opening in the body of certain animals, including cnidarians and flatworms, that functions in both the digestion and distribution of nutrients.

Gastrula (gas'-trū-luh) An embryonic stage in animal development encompassing the formation of three layers: ectoderm, mesoderm, and endoderm.

Gastrulation (gas'-trū-lā'-shun) In animal development, a series of cell and tissue movements in which the blastula-stage embryo folds inward, producing a three-layered embryo, the gastrula.

Gated channels Transmembrane protein channels that open or close in response to a particular stimulus.

Gated ion channels Gated channels for a specific ion. The opening or closing of such channels may alter a cell's membrane potential.

Gel electrophoresis (ē-lek'-trō-fōr-ē'-sis) A technique for separating nucleic acids or proteins on the basis of their size and electrical charge, both of which affect their rate of movement through an electric field in a gel made of agarose or another polymer.

Gene annotation Analysis of genomic sequences to identify protein-coding genes and determine the function of their products.

Gene cloning The production of multiple copies of a gene.

Gene drive A process that biases inheritance such that a particular allele is more likely to be inherited than are other alleles, causing the

favored allele to spread (be "driven") through the population.

Gene editing Altering genes in a specific, predictable way.

Gene expression The process by which information encoded in DNA directs the synthesis of proteins or, in some cases, RNAs that are not translated into proteins and instead function as RNAs.

Gene flow The transfer of alleles from one population to another, resulting from the movement of fertile individuals or their gametes.

Gene pool The aggregate of all copies of every type of allele at all loci in every individual in a population. The term is also used in a more restricted sense as the aggregate of alleles for just one or a few loci in a population.

Genes (singular, **gene**) Discrete units of hereditary information consisting of a specific nucleotide sequence in DNA (or RNA, in some viruses).

Gene therapy The introduction of genes into an afflicted individual for therapeutic purposes.

Genetically modified organisms (GMOs) Organisms that have acquired one or more genes by artificial means.

Genetic drift A process in which chance events cause unpredictable fluctuations in allele frequencies from one generation to the next. Effects of genetic drift are most pronounced in small populations.

Genetic engineering The direct manipulation of genes for practical purposes.

Genetic map An ordered list of genetic loci (genes or other genetic markers) along a chromosome.

Genetic profile An individual's unique set of genetic markers, detected most often today by PCR or, previously, by electrophoresis and nucleic acid probes.

Genetic recombination General term for the production of offspring with combinations of traits that differ from those found in either parent.

Genetics The scientific study of heredity and hereditary variation.

Genetic variation Differences among individuals in the composition of their genes or other DNA sequences.

Genome (jē'-nōm) The genetic material of an organism or virus; the complete complement of an organism's or virus's genes along with its noncoding nucleic acid sequences.

Genome-wide association studies (jē'-nōm) Large-scale analyses of the genomes of many people having a certain phenotype or disease, with the aim of finding genetic markers that correlate with that phenotype or disease.

Genomic imprinting (juh-nō'-mik) A phenomenon in which expression of an allele in offspring depends on whether the allele is inherited from the male or female parent.

Genomics (juh-nō'-miks) The systematic study of whole sets of genes (or other DNA) and their interactions within a species, as well as genome comparisons between species.

Genotype (jē'-nō-tīp) The genetic makeup, or set of alleles, of an organism.

Genus (jē'-nus) (plural, **genera**) A taxonomic category above the species level, designated by the first word of a species' two-part scientific name.

Geologic record A standard time scale dividing Earth's history into time periods, grouped into four eons—Hadean, Archaean, Proterozoic, and Phanerozoic—and further subdivided into eras, periods, and epochs.

Germ layers The three main layers in a gastrula that will form the various tissues and organs of an animal body.

Gestation (jes-tā'-shun) Pregnancy; the condition of carrying one or more embryos in the uterus.

Gibberellin (jib'-uh-rel'-in) Any of a class of related plant hormones that stimulate growth in the stem and leaves, trigger the germination of seeds and breaking of bud dormancy, and (with auxin) stimulate fruit development.

Glans In humans, the rounded head of the penis (in males) or of the clitoris (in females); the glans is highly sensitive to stimulation.

Glia (glial cells) Non-neuronal cells of the nervous system that support, regulate, and nourish neurons; they also insulate neuron axons and remove pathogens and dead cells.

Glomerulus (glō-mār'-yū-lus) A ball of capillaries surrounded by Bowman's capsule in the nephron and serving as the site of filtration in the vertebrate kidney.

Glucagon (glū'-kuh-gon) A hormone secreted by the pancreas that raises blood glucose levels. It promotes glycogen breakdown and release of glucose by the liver.

Glucocorticoids Steroid hormones that are secreted by the adrenal cortex and that influence glucose metabolism and immune function.

Glyceraldehyde 3-phosphate (G3P) (glis'-er-al'-de-hīd) A three-carbon carbohydrate that is the direct product of the Calvin cycle; it is also an intermediate in glycolysis.

Glycogen (glī'-kō-jen) An extensively branched glucose storage polysaccharide found in the liver and muscle of animals; the animal equivalent of starch.

Glycolipids Lipids with one or more covalently attached carbohydrates.

Glycolysis (glī-kol'-uh-sis) A series of reactions that ultimately splits glucose into pyruvate. Glycolysis occurs in almost all living cells, serving as the starting point for fermentation or cellular respiration.

Glycoproteins Proteins with one or more covalently attached carbohydrates.

Glycosidic linkage A covalent bond formed between two monosaccharides by a dehydration reaction.

Gnathostomes (na'-thu-stōm) Members of one of the two main clades of vertebrates; gnathostomes have jaws and include sharks and rays, ray-finned fishes, coelacanth, lungfishes, amphibians, reptiles, and mammals.

Golgi apparatus (gol'-jē) An organelle in eukaryotic cells consisting of stacks of flat membranous sacs that modify, store, and route products of the endoplasmic reticulum and synthesize some products, notably noncellulose carbohydrates.

Gonads (gō'-nads) Male or female gamete-producing organs.

G protein A GTP-binding protein that relays signals from a plasma membrane signal receptor, known as a G protein-coupled receptor, to other signal transduction proteins inside the cell.

G protein-coupled receptors (GPCR) Cell-surface receptors in the plasma membrane involved in relaying signals to other signal transduction proteins inside the cell.

Graded potential An electrical response of a cell to a stimulus, consisting of a change in voltage across the membrane proportional to the stimulus strength.

Gradient A difference in a particular quality or measurement between two regions.

Gram-negative Describing the group of bacteria that have a cell wall that is structurally more complex and contains less peptidoglycan than the cell wall of gram-positive bacteria. Gram-negative bacteria are often more toxic than gram-positive bacteria.

Gram-positive Describing the group of bacteria that have a cell wall that is structurally less complex and contains more peptidoglycan than the cell wall of gram-negative bacteria. Gram-positive bacteria are usually less toxic than gram-negative bacteria.

Gram stain A staining method that distinguishes between two different kinds of bacterial cell walls; may be used to help determine medical response to an infection.

Granum (gran'-um) (plural, **grana**) A stack of membrane-bounded thylakoids in the chloroplast. Grana function in the light reactions of photosynthesis.

Gravitropism (grav'-uh-trō'-pizm) A response of a plant or animal to gravity.

Gray matter Regions of clustered neuron cell bodies within the CNS.

Green algae Photosynthetic protists, named for green chloroplasts that are similar in structure and pigment composition to the chloroplasts of plants. Green algae are a paraphyletic group; some members are more closely related to plants than they are to other green algae.

Greenhouse effect The warming of Earth due to the atmospheric accumulation of carbon

dioxide and certain other gases, which absorb reflected infrared radiation and reradiate some of it back toward Earth.

Gross primary production (GPP) The total primary production of an ecosystem.

Ground tissue Plant tissue that is neither vascular nor dermal, fulfilling a variety of functions, such as storage, photosynthesis, and support.

Growth factor (1) A protein that must be present in the extracellular environment (culture medium or animal body) for the growth and normal development of certain types of cells. (2) A local regulator that acts on nearby cells to stimulate cell proliferation and differentiation.

Growth hormone (GH) A hormone that is produced and secreted by the anterior pituitary and that has both direct (nontropic) and tropic effects on a wide variety of tissues.

Guard cells The two cells that flank the stomatal pore and regulate the opening and closing of the pore.

Gustation The sense of taste.

Guttation The exudation of water droplets from leaves, caused by root pressure in certain plants.

Gymnosperms (jim'-nō-sperm) Vascular plants that bear naked seeds—seeds not enclosed in protective chambers.

Hagfishes Marine jawless vertebrates that have highly reduced vertebrae and a skull made of cartilage; most hagfishes are bottom-dwelling scavengers.

Hair cells Mechanosensory cells that alter output to the nervous system when hairlike projections on the cell surface are displaced.

Half-life The amount of time it takes for 50 percent of a sample of a radioactive isotope to decay.

Hamilton's rule The principle that for natural selection to favor an altruistic act, the benefit to the recipient (*B*), devalued by the coefficient of relatedness (*r*) must exceed the cost to the altruist (*C*); expressed as $rB > C$.

Haploid cell (hap'-loid) A cell containing only one set of chromosomes (*n*).

Haptophytes A clade of single-celled algae currently thought to be closely related to the SAR supergroup; haptophytes include coccolithophores, a group of phytoplankton with unique calcified scales around their cells.

Hardy-Weinberg equilibrium The state of a population in which frequencies of alleles and genotypes remain constant from generation to generation, provided that only Mendelian segregation and recombination of alleles are at work.

Heart A muscular pump that uses metabolic energy to elevate the hydrostatic pressure of the circulatory fluid (blood or hemolymph). The fluid then flows down a pressure gradient through the body and eventually returns to the heart.

Heart attack The damage or death of cardiac muscle tissue resulting from prolonged blockage of one or more coronary arteries.

Heart murmur A hissing sound that most often results from blood squirting backward through a leaky valve in the heart.

Heart rate The frequency of heart contraction (in beats per minute).

Heat Thermal energy in transfer from one body of matter to another.

Heat of vaporization The quantity of heat a liquid must absorb for 1 g of it to be converted from the liquid to the gaseous state.

Heat-shock proteins Proteins that help protect other proteins during heat stress. Heat-shock proteins are found in plants, animals, and microorganisms.

Heavy chains One of the two types of polypeptide chains that make up an antibody molecule and B cell receptor; each heavy chain consists of a variable region, which contributes to the antigen-binding site, and a constant region.

Helicases Enzymes that untwist the double helix of DNA at replication forks, separating the two strands and making them available as template strands.

Helper T cell A type of T cell that, when activated, secretes cytokines that promote the response of B cells (humoral response) and cytotoxic T cells (cell-mediated response) to antigens.

Hemoglobin (hē'-mō-glō'-bin) An iron-containing protein in red blood cells that reversibly binds oxygen.

Hemolymph (hē'-mō-limf') In invertebrates with an open circulatory system, the body fluid that bathes tissues.

Hemophilia (hē'-muh-fil'-ē-uh) A human genetic disease caused by a sex-linked recessive allele resulting in the absence of one or more blood-clotting proteins; characterized by excessive bleeding following injury.

Hepatic portal vein A large vessel that conveys nutrient-laden blood from the small intestine to the liver, which regulates the blood's nutrient content.

Herbivores (hur'-bi-vōrs') Animals that eat primarily plants or algae.

Herbivory A positive-negative ecological interaction in which an organism eats part of a plant or alga.

Heredity The transmission of traits from one generation to the next.

Hermaphrodites (hur-maf'-ruh-dits') Individuals that function as both male and female in sexual reproduction by producing both sperm and eggs.

Hermaphroditism (hur-maf'-rō-dī-tizm) A condition in which an individual has both female and male gonads and functions as both a male

and a female in sexual reproduction by producing both sperm and eggs.

Heterochromatin (het'-er-ō-kro'-muh-tin)

Eukaryotic chromatin that remains highly compacted during interphase and is generally not transcribed.

Heterochrony (het'-uh-rok'-ruh-nē)

Evolutionary change in the timing or rate of an organism's development.

Heterocysts (het'-er-ō-sists)

Specialized cells that engage in nitrogen fixation in some filamentous cyanobacteria; also called a heterocyte.

Heterokaryon (het'-er-ō-kār'-ē-un)

A fungal mycelium that contains two or more haploid nuclei per cell.

Heteromorphic (het'-er-ō-mōr'-fik)

Referring to a condition in the life cycle of plants and certain algae in which the sporophyte and gametophyte generations differ in morphology.

Heterosporous (het-er-os'-pōr-us)

Referring to a plant species that has two kinds of spores: microspores, which develop into male gametophytes, and megaspores, which develop into female gametophytes.

Heterotrophs (het'-er-ō-trōfs)

Organisms that obtain organic food molecules by eating other organisms or substances derived from them.

Heterozygote An organism that has two different alleles for a gene (encoding a character).

Heterozygote advantage Greater reproductive success of heterozygous individuals compared with homozygotes; tends to preserve variation in a gene pool.

Heterozygous (het'-er-ō-zī'-gus)

Having two different alleles for a given gene.

Hibernation A long-term physiological state in which metabolism decreases, the heart and respiratory system slow down, and body temperature is maintained at a lower level than normal.

High-density lipoprotein (HDL)

A particle in the blood made up of thousands of cholesterol molecules and other lipids bound to a protein. HDL scavenges excess cholesterol.

Hindbrain One of three ancestral and embryonic regions of the vertebrate brain; develops into the medulla oblongata, pons, and cerebellum.

Histamine (his'-tuh-mēn)

A substance released by mast cells that causes blood vessels to dilate and become more permeable in inflammatory and allergic responses.

Histogram A variant of a bar graph that is made for numeric data by first grouping, or "binning," the variable plotted on the *x*-axis into intervals of equal width. The "bins" may be integers or ranges of numbers. The height of each bar shows the percent or number of experimental subjects whose characteristics can be described by one of the intervals plotted on the *x*-axis.

Histone acetylation (his'-tōn) The attachment of acetyl groups to certain amino acids of histone proteins.

Histones (his'-tōns)

Small proteins with a high proportion of positively charged amino acids that binds to the negatively charged DNA and plays a key role in chromatin structure.

HIV (human immunodeficiency virus)

The infectious agent that causes AIDS. HIV is a retrovirus.

Holdfast A rootlike structure that anchors a seaweed.

Homeobox (hō'-mē-ō-boks')

A 180-nucleotide sequence within homeotic genes and some other developmental genes that is widely conserved in animals. Related sequences occur in plants and yeasts.

Homeostasis (hō'-mē-ō-stā'-sis)

The steady-state physiological condition of the body.

Homeotic genes (hō-mē-ō'-tik)

Master regulatory genes that control placement and spatial organization of body parts in animals, plants, and fungi by controlling the developmental fate of groups of cells.

Hominins (hō'-mi-nins)

A group consisting of humans and the extinct species that are more closely related to humans than to chimpanzees.

Homologous chromosomes (or homologs)

(hō-mol'-uh-gus) A pair of chromosomes of the same length, centromere position, and staining pattern that possess genes for the same characters at corresponding loci. One homologous chromosome is inherited from the organism's male parent, the other from the female parent. Also called a homologous pair.

Homologous structures (hō-mol'-uh-gus)

Structures in different species that are similar because of common ancestry.

Homologs (or homologous chromosomes)

A pair of chromosomes of the same length, centromere position, and staining pattern that possess genes for the same characters at corresponding loci. One homologous chromosome is inherited from the organism's male parent, the other from the female parent. Also called a homologous pair.

Homology (hō-mol'-ō-jē)

Similarity in characteristics resulting from a shared ancestry.

Homosporous (hō-mos'-puh-rus)

Referring to a plant species that has a single kind of spore, which typically develops into a bisexual gametophyte.

Homozygote An organism that has a pair of identical alleles for a gene (encoding a character).

Homozygous (hō'-mō-zī'-gus)

Having two identical alleles for a given gene.

Horizontal gene transfer

The transfer of genes from one genome to another through mechanisms such as transposable elements, plasmid

exchange, viral activity, and perhaps fusions of different organisms.

Hormones (singular, hormone)

In multicellular organisms, one of many types of secreted chemicals that are formed in specialized cells, travel in body fluids, and act on specific target cells in other parts of the organism, changing the target cells' functioning.

Hornworts Small, herbaceous, nonvascular plants that are members of the phylum Anthocrophyta.

Host The larger participant in a symbiotic relationship, often providing a home and food source for the smaller symbiont.

Host range The limited number of species whose cells can be infected by a particular virus.

Human Genome Project An international collaborative effort to map and sequence the DNA of the entire human genome.

Humoral immune response (hyū'-mer-ul)

The branch of adaptive immunity that involves the activation of B cells and that leads to the production of antibodies, which defend against bacteria and viruses in body fluids.

Humus (hyū'-mus) Decomposing organic material that is a component of topsoil.

Huntington's disease A human genetic disease caused by a dominant allele; characterized by uncontrollable body movements and degeneration of the nervous system; usually fatal 10 to 20 years after the onset of symptoms.

Hybridization In genetics, the mating, or crossing, of two true-breeding varieties.

Hybrids Offspring that result from the mating of individuals from two different species or from two true-breeding varieties of the same species.

Hybrid zone A geographic region in which members of different species meet and mate, producing at least some offspring of mixed ancestry.

Hydration shell The sphere of water molecules around a dissolved ion.

Hydrocarbons Organic molecules consisting only of carbon and hydrogen.

Hydrogen bond A type of weak chemical bond that is formed when the slightly positive hydrogen atom of a polar covalent bond in one molecule is attracted to the slightly negative atom of a polar covalent bond in another molecule or in another region of the same molecule.

Hydrogen ion A single proton with a charge of plus one. The dissociation of a water molecule leads to the generation of a hydroxide ion (with a charge of minus one) and a hydrogen ion (with a charge of plus one); in water, the hydrogen ion is not found alone but associates with a water molecule to form a hydronium ion.

Hydrolysis (hī-drol'-uh-sis) A chemical reaction that breaks bonds between two molecules by the addition of water; functions in disassembly of polymers to monomers.

Hydronium ion A water molecule that has an extra proton bound to it, commonly represented as H superscript plus.

Hydrophilic (hī'-drō-fil'-ik) Having an affinity for water.

Hydrophobic (hī'-drō-fō'-bik) Having no affinity for water; tending to coalesce and form droplets in water.

Hydrophobic interactions (hī'-drō-fō'-bik) A type of weak chemical interaction caused when molecules that do not mix with water coalesce to exclude water.

Hydroponic culture A method in which plants are grown in mineral solutions rather than in soil.

Hydrostatic skeleton A skeletal system composed of fluid held under pressure in a closed body compartment; the main skeleton of most cnidarians, flatworms, nematodes, and annelids.

Hydrothermal vents Areas on the seafloor where heated water and minerals from Earth's interior gush into the seawater, producing a dark, hot, oxygen-deficient environment. The producers in a hydrothermal vent community are chemoautotrophic prokaryotes.

Hydroxide ion A water molecule that has lost a proton, represented as OH superscript minus.

Hydroxyl group (hī-drok'-sil) A chemical group consisting of an oxygen atom joined to a hydrogen atom. Molecules possessing this group are soluble in water and are called alcohols.

Hyperpolarization A change in a cell's membrane potential such that the inside of the membrane becomes more negative relative to the outside. Hyperpolarization reduces the chance that a neuron will transmit a nerve impulse.

Hypersensitive response A plant's localized defense response to a pathogen, involving the death of cells around the site of infection.

Hypertension A disease in which blood pressure remains abnormally high.

Hypertonic Referring to a solution that, when surrounding a cell, will cause the cell to lose water.

Hyphae (singular, **hypha**) (hī'-fē, hī'-fuh) One of many connected filaments that collectively make up the mycelium of a fungus.

Hypocotyl (hī'-puh-cot'-ul) In an angiosperm embryo, the embryonic axis below the point of attachment of the cotyledon(s) and above the radicle.

Hypothalamus (hī'-pō-thal'-uh-mus) The ventral part of the vertebrate forebrain; functions in maintaining homeostasis, especially in coordinating the endocrine and nervous systems; secretes hormones of the posterior pituitary and releasing factors that regulate the anterior pituitary.

Hypothesis (hī'-poth'-uh-sis) A testable explanation for a set of observations based on the

available data and guided by inductive reasoning. A hypothesis is narrower in scope than a theory.

Hypotonic Referring to a solution that, when surrounding a cell, will cause the cell to take up water.

Imbibition The uptake of water by a seed or other structure, resulting in swelling.

Immigration The influx of new individuals into a population from other areas.

Immune system An organism's system of defenses against agents that cause disease.

Immunization The process of generating a state of immunity by artificial means. In vaccination, an inactive or weakened form of a pathogen is administered, inducing B and T cell responses and immunological memory. In passive immunization, antibodies specific for a particular pathogen are administered, conferring immediate but temporary protection.

Immunoglobulin (Ig) (im'-yū-nō-glob'-yū-lin) A protein secreted by plasma cells (differentiated B cells) that binds to a particular antigen; also called immunoglobulin. All antibodies have the same Y-shaped structure and in their monomer form consist of two identical heavy chains and two identical light chains.

Imprinting In animal behavior, the formation at a specific stage in life of a long-lasting behavioral response to a specific individual or object; A phenomenon in which expression of an allele in offspring depends on whether the allele is inherited from the male or female parent.

Inclusive fitness The total effect an individual has on proliferating its genes by producing its own offspring and by providing aid that enables other close relatives to increase production of their offspring.

Incomplete dominance The situation in which the phenotype of heterozygotes is intermediate between the phenotypes of individuals homozygous for either allele.

Incomplete flowers Flowers in which one or more of the four basic floral organs (sepals, petals, stamens, or carpels) are either absent or nonfunctional.

Incomplete metamorphosis A type of development in certain insects, such as grasshoppers, in which the young (called nymphs) resemble adults but are smaller and have different body proportions. The nymph goes through a series of molts, each time looking more like an adult, until it reaches full size.

Independent variable A factor whose value is manipulated or changed during an experiment to reveal possible effects on another factor (the dependent variable).

Indeterminate cleavage A type of embryonic development in deuterostomes in which each cell produced by early cleavage divisions retains the capacity to develop into a complete embryo.

Indeterminate growth A type of growth characteristic of plants, in which the organism continues to grow as long as it lives.

Induced fit Caused by entry of the substrate, the change in shape of the active site of an enzyme so that it binds more snugly to the substrate.

Inducer A specific small molecule that binds to a bacterial repressor protein and changes the repressor's shape so that it cannot bind to an operator, thus switching an operon on.

Induction A process in which a group of cells or tissues influences the development of another group through close-range interactions.

Inductive reasoning A type of logic in which generalizations are based on a large number of specific observations.

Inflammatory response An innate immune defense triggered by physical injury or infection of tissue involving the release of substances that promote swelling, enhance the infiltration of white blood cells, and aid in tissue repair and destruction of invading pathogens.

Inflorescences Groups of flowers tightly clustered together on a plant.

Ingestion The first stage of food processing in animals: the act of eating.

Ingroup A species or group of species whose evolutionary relationships are being examined in a given analysis.

Inhibition In ecological succession, a process by which early-arriving species in an area make the environment less favorable for the establishment of later-arriving species.

Inhibitory postsynaptic potential (IPSP) An electrical change (usually hyperpolarization) in the membrane of a postsynaptic neuron caused by the binding of an inhibitory neurotransmitter from a presynaptic cell to a postsynaptic receptor; makes it more difficult for a postsynaptic neuron to generate an action potential.

Innate behavior Animal behavior that is developmentally fixed and under strong genetic control. Innate behavior is exhibited in virtually the same form by all individuals in a population despite internal and external environmental differences during development and throughout their lifetimes.

Innate immunity A form of defense common to all animals that is active immediately upon exposure to a pathogen and that is the same whether or not the pathogen has been encountered previously.

Inner cell mass An inner cluster of cells at one end of a mammalian blastocyst that subsequently develops into the embryo proper and some of the extraembryonic membranes.

Inner ear One of the three main regions of the vertebrate ear; includes the cochlea (which in turn contains the organ of Corti) and the semicircular canals.

Inositol trisphosphate (IP₃) (in-ō'-suh-tol) A second messenger that functions as an intermediate between certain signaling molecules and a subsequent second messenger, a calcium ion (Ca²⁺), causing a rise in cytoplasmic calcium ion concentration.

Inquiry The search for information and explanation, often focusing on specific questions.

In situ hybridization A technique using nucleic acid hybridization with a labeled probe to detect the location of a specific mRNA in an intact organism.

Insulin (in'-suh-lin) A hormone secreted by pancreatic beta cells that lowers blood glucose levels. It promotes the uptake of glucose by most body cells and the synthesis and storage of glycogen in the liver and also stimulates protein and fat synthesis.

Integral proteins Transmembrane proteins with hydrophobic regions that extend into and often completely span the hydrophobic interior of the membrane and with hydrophilic regions in contact with the aqueous solution on one or both sides of the membrane (or lining the channel in the case of a channel protein).

Integrins (in'-tuh-grins) In animal cells, transmembrane receptor proteins with two subunits that interconnect the extracellular matrix and the cytoskeleton.

Integument (in-teg'-yū-ment) Layer of sporophyte tissue that contributes to the structure of an ovule of a seed plant.

Integumentary system The outer covering of a mammal's body, including skin, hair, and nails, claws, or hooves.

Interferons (in'-ter-fēr'-ons) Proteins that have antiviral or immune regulatory functions. For example, interferons secreted by virus-infected cells help nearby cells resist viral infection.

Intermediate disturbance hypothesis The concept that moderate levels of disturbance can foster greater species diversity than low or high levels of disturbance.

Intermediate filaments Components of the cytoskeleton that include filaments intermediate in size between microtubules and microfilaments.

Interneurons Association neurons; nerve cells within the central nervous system that form synapses with sensory and/or motor neurons and integrate sensory input and motor output.

Internodes Segments of a plant stem between the points where leaves are attached.

Interphase The period in the cell cycle when the cell is not dividing. During interphase, cellular metabolic activity is high, chromosomes and organelles are duplicated, and cell size may increase. Interphase often accounts for about 90 percent of the cell cycle.

Intersexual selection A form of natural selection in which individuals of one sex (usually the females) are choosy in selecting their mates from the other sex; also called mate choice.

Interspecific competition A negative-negative interaction that occurs when individuals of different species both use a resource that limits the survival and reproduction of each species.

Interspecific interactions Relationships between individuals of two or more species in a community.

Interstitial fluid The fluid filling the spaces between cells in most animals.

Intertidal zone The shallow zone of the ocean adjacent to land and between the high- and low-tide lines.

Intrasexual selection A form of natural selection in which there is direct competition among individuals of one sex for mates of the opposite sex.

Intrinsic rate of increase In population models, the per capita rate at which an exponentially growing population increases in size at each instant in time; symbolized in this text by r_{max} .

Introduced species A species moved by humans, either intentionally or accidentally, from its native location to a new geographic region; sometimes called a non-native species, exotic species, or invasive species.

Introns (in'-tron) Noncoding, intervening sequences within a primary transcript that are removed from the transcript during RNA processing; the term intron also refers to the region of DNA from which this sequence was transcribed.

Inversion An aberration in chromosome structure resulting from reattachment of a chromosomal fragment in a reverse orientation to the chromosome from which it originated.

Invertebrates Animals without a backbone. Invertebrates make up 95 percent of animal species.

In vitro fertilization (IVF) (vē'-trō) Fertilization of oocytes in laboratory containers followed by artificial implantation of the early embryo in the uterus.

In vitro mutagenesis A technique used to discover the function of a gene by cloning it, introducing specific changes into the cloned gene's sequence, reinserting the mutated gene into a cell, and studying the phenotype of the mutant.

Ion channels (ī'-ons) Transmembrane protein channels that allow a specific ion to diffuse across the membrane down its concentration or electrochemical gradient.

Ionic bond (ī-on'-ik) A chemical bond resulting from the attraction between oppositely charged ions.

Ionic compounds (ī-on'-ik) Compounds resulting from the formation of an ionic bond; also called salts.

Ions (ī'-ons) Individual atoms or groups of atoms that have gained or lost one or more electrons, thus acquiring a charge.

Iris The colored part of the vertebrate eye, formed by the anterior portion of the choroid.

Isomers (ī'-sō-mers) Two or more compounds that have the same numbers of atoms of the same elements but different structures and hence different properties.

Isomorphic Referring to alternating generations in plants and certain algae in which the sporophytes and gametophytes look alike, although they differ in chromosome number.

Isotonic (ī'-sō-ton'-ik) Referring to a solution that, when surrounding a cell, causes no net movement of water into or out of the cell.

Isotopes (ī'-sō-tōps') Different atomic forms of an element, each with the same number of protons but a different number of neutrons, thus differing in atomic mass.

Iteroparity Reproduction in which adults produce offspring over many years; also called repeated reproduction.

Jasmonates A class of plant hormones that regulate a wide range of developmental processes in plants and play a key role in plant defense against herbivores.

Joule (J) A unit of energy: 1 J equals 0.239 cal; 1 cal equals 4.184 J.

Juxtaglomerular apparatus (JGA) (juks'-tuh-glūh-mār'-yū-ler) A specialized tissue in nephrons that releases the enzyme renin in response to a drop in blood pressure or volume.

Juxtamedullary nephrons In mammals and birds, nephrons with a loop of Henle that extends far into the renal medulla.

Karyogamy (kār'-ē-og'-uh-mē) In fungi, the fusion of haploid nuclei contributed by the two parents; occurs as one stage of sexual reproduction, preceded by plasmogamy.

Karyotype (kār'-ē-ō-tīp) A display of the chromosome pairs of a cell arranged by size and shape.

Keystone species A species that is not necessarily abundant in a community yet exerts strong control on community structure by the nature of its ecological role or niche.

Kidney In vertebrates, one of a pair of excretory organs where blood filtrate is formed and processed into urine.

Kilocalorie (kcal) A thousand calories; the amount of heat energy required to raise the temperature of 1 kg of water by 1 degree C.

Kin selection Natural selection that favors altruistic behavior by enhancing the reproductive success of relatives.

Kinetic energy (kuh-net'-ik) The energy associated with the relative motion of objects. Moving matter can perform work by imparting motion to other matter.

Kinetochore (kuh-net'-uh-kōr) A structure of proteins attached to the centromere that links each sister chromatid to the mitotic spindle.

Kingdoms A taxonomic category, the second broadest after domain.

K-selection Selection for life history traits that are sensitive to population density.

Labia majora A pair of thick, padded ridges that enclose and protect the rest of the vulva.

Labia minora A pair of slender skin folds that surround the openings of the vagina and urethra.

Lacteal (lak'-tē-ul) A tiny lymph vessel extending into the core of an intestinal villus and serving as the destination for absorbed chylomicrons.

Lactic acid fermentation Glycolysis followed by the reduction of pyruvate to lactate, regenerating NAD⁺ with no release of carbon dioxide.

Lagging strand A discontinuously synthesized DNA strand that elongates by means of Okazaki fragments, each synthesized in a 5' to 3' direction away from the replication fork.

Lampreys Jawless vertebrates with highly reduced vertebrae that live in freshwater and marine environments. Almost half of extant lamprey species are parasites that feed by clamping their round, jawless mouth onto the flank of a live fish; nonparasitic lampreys are suspension feeders that feed only as larvae.

Lancelets Members of the clade Cephalochordata, small blade-shaped marine chordates that lack a backbone.

Landscape An area containing several different ecosystems linked by exchanges of energy, materials, and organisms.

Large intestine The portion of the vertebrate alimentary canal between the small intestine and the anus; functions mainly in water absorption and the formation of feces.

Larva (lar'-vuh) (plural, **larvae**) A free-living, sexually immature form in some animal life cycles that may differ from the adult animal in morphology, nutrition, and habitat.

Larynx (lār'-inks) The portion of the respiratory tract containing the vocal cords; also called the voice box.

Lateral line system A mechanoreceptor system consisting of a series of pores and receptor units along the sides of the body in fishes and aquatic amphibians; detects water movements made by the animal itself and by other moving objects.

Lateral meristems (mār'-uh-stems) Meristems that thicken the roots and shoots of woody plants. The vascular cambium and cork cambium are lateral meristems.

Lateral roots Roots that arise from the pericycle of an established root.

Lateralization Segregation of functions in the cortex of the left and right cerebral hemispheres.

Law of conservation of mass A physical law stating that matter can change form but cannot be created or destroyed. In a closed system, the mass of the system is constant.

Law of independent assortment Mendel's second law, stating that each pair of alleles segregates, or assort, independently of each

other pair during gamete formation; applies when genes for two characters are located on different pairs of homologous chromosomes or when they are far enough apart on the same chromosome to behave as though they are on different chromosomes.

Law of segregation Mendel's first law, stating that the two alleles in a pair segregate (separate from each other) into different gametes during gamete formation.

Leading strand The new complementary DNA strand synthesized continuously along the template strand toward the replication fork in the mandatory 5' to 3' direction.

Leaf The main photosynthetic organ of vascular plants.

Leaf primordia (singular, **primordium**) Finger-like projections along the flank of a shoot apical meristem, from which a leaf arises.

Learning The modification of behavior as a result of specific experiences.

Lens The structure in an eye that focuses light rays onto the photoreceptors.

Lenticels (len'-ti-sels) Small raised areas in the bark of stems and roots that enable gas exchange between living cells and the outside air.

Lepidosaurs (leh-pid'-uh-sōr) Members of the reptilian group that includes lizards, snakes, and two species of New Zealand animals called tuataras.

Leukocytes (lū'-kō-sīts') Blood cells that function in fighting infections; also called white blood cells.

Lichen The mutualistic association between a fungus and a photosynthetic alga or cyanobacterium.

Life cycle The generation-to-generation sequence of stages in the reproductive history of an organism.

Life history The traits that affect an organism's schedule of reproduction and survival.

Life table A summary of the age-specific survival and reproductive rates of individuals in a population.

Ligand (lig'-und) A molecule that binds specifically to another molecule, usually a larger one.

Ligand-gated ion channel (lig'-und) A transmembrane protein containing a pore that opens or closes as it changes shape in response to a signaling molecule (ligand), allowing or blocking the flow of specific ions; also called an ionotropic receptor.

Light chains One of the two types of polypeptide chains that make up an antibody molecule and B cell receptor; each light chain consists of a variable region, which contributes to the antigen-binding site, and a constant region.

Light microscope (LM) An optical instrument with lenses that refract (bend) visible light to magnify images of specimens.

Light reactions The first of two major stages in photosynthesis (preceding the Calvin cycle). These reactions, which occur on the thylakoid membranes of the chloroplast or on membranes of certain prokaryotes, convert solar energy to the chemical energy of ATP and NADPH, releasing oxygen in the process.

Light-harvesting complex A complex of proteins associated with pigment molecules (including chlorophyll *a*, chlorophyll *b*, and carotenoids) that captures light energy and transfers it to reaction-center pigments in a photosystem.

Lignin (lig'-nin) A strong polymer embedded in the cellulose matrix of the secondary cell walls of vascular plants that provides structural support in terrestrial species.

Limiting nutrient An element that must be added for production to increase in a particular area.

Limnetic zone In a lake, the well-lit, open surface waters in the area of the lake that is too deep for rooted plants to grow.

Line graph A graph in which each data point is connected to the next point in the data set with a straight line.

Linear electron flow A route of electron flow during the light reactions of photosynthesis that involves both photosystems (I and II) and produces ATP, NADPH, and oxygen. The net electron flow is from water to NADP plus.

Linkage map A genetic map based on the frequencies of recombination between markers during crossing over of homologous chromosomes.

Linked genes Genes located close enough together on a chromosome that they tend to be inherited together.

Lipids (lip'-ids) A group of large biological molecules, including fats, phospholipids, and steroids, that mix poorly, if at all, with water.

Littoral zone In a lake or ocean, the shallow, well-lit waters close to shore, where the bottom-dwelling organisms (benthos) are within the photic zone (depth to which light penetrates).

Liver A large internal organ in vertebrates that performs diverse functions, such as producing bile, maintaining blood glucose level, and detoxifying poisonous chemicals in the blood.

Liverworts Small, herbaceous, nonvascular plants that are members of the phylum Hepatophyta.

Loams The most fertile soil type, made up of roughly equal amounts of sand, silt, and clay.

Lobe-fins Members of a clade of osteichthyans having rod-shaped muscular fins. The group includes coelacanth, lungfishes, and tetrapods.

Local regulators Secreted molecules that influence cells near where they are secreted.

Locomotion Active motion from place to place.

Locus (plural, **loci**), (lō'-kus), (lō'-sī) A specific place along the length of a chromosome where a given gene is located.

Logistic population growth Population growth that levels off as population size approaches carrying capacity.

Long-day plants Plants that flower (usually in late spring or early summer) only when the light period is longer than a critical length.

Long noncoding RNAs (lncRNAs) RNA molecules between 200 and hundreds of thousands of nucleotides in length that do not code for protein but are expressed at significant levels.

Long-term memory The ability to hold, associate, and recall information over one's lifetime.

Long-term potentiation (LTP) An enhanced responsiveness to an action potential (nerve signal) by a receiving neuron.

Loop of Henle (hen'-lē) The hairpin turn, with a descending and ascending limb, between the proximal and distal tubules of the vertebrate kidney; functions in water and salt reabsorption.

Lophophore (lof'-uh-fōr) In some lophotrochozoan animals, including brachiopods, a crown of ciliated tentacles that surround the mouth and function in feeding.

Lophotrochozoa (lo-phah--truh-kō-zō'-uh) One of the three main lineages of bilaterian animals; lophotrochozoans include organisms that have lophophores or trochophore larvae. *See also* Deuterostomia and Ecdysozoa.

Low-density lipoprotein (LDL) A particle in the blood made up of thousands of cholesterol molecules and other lipids bound to a protein. LDL transports cholesterol from the liver for incorporation into cell membranes.

Lungs Infolded respiratory surfaces of a terrestrial vertebrate, land snail, or spider that connect to the atmosphere by narrow tubes.

Luteinizing hormone (LH) (lū'-tē-uh-nī'-zing) A tropic hormone that is produced and secreted by the anterior pituitary and that stimulates ovulation in females and androgen production in males.

Lycophytes (lī'-kuh-fits) An informal name for members of the phylum Lycophyta, which includes club mosses, spike mosses, and quillworts.

Lymph The colorless fluid, derived from interstitial fluid, in the lymphatic system of vertebrates.

Lymph nodes Organs located along a lymph vessel that filter lymph and contain cells that attack viruses and bacteria.

Lymphatic system A system of vessels and nodes, separate from the circulatory system, that returns fluid, proteins, and cells to the blood.

Lymphocytes A type of white blood cell that mediates immune responses. The two main classes are B cells and T cells.

Lysogenic cycle (lī'-sō-jen'-ik) A type of phage replicative cycle in which the viral genome becomes incorporated into the bacterial host chromosome as a prophage, is replicated along with the chromosome, and does not kill the host.

Lysosome (lī'-suh-sōm) A membrane-enclosed sac of hydrolytic enzymes found in the cytoplasm of animal cells and some protists.

Lysozyme (lī'-sō-zīm) An enzyme that destroys bacterial cell walls; in mammals, it is found in sweat, tears, and saliva.

Lytic cycle (lit'-ik) A type of phage replicative cycle resulting in the release of new phages by lysis (and death) of the host cell.

Macroevolution Evolutionary change above the species level. Examples of macroevolutionary change include the origin of a new group of organisms through a series of speciation events and the impact of mass extinctions on the diversity of life and its subsequent recovery.

Macromolecules Giant molecules formed by the joining of smaller molecules. Polysaccharides, proteins, and nucleic acids are macromolecules.

Macronutrients Essential elements that an organism must obtain in relatively large amounts.

Macrophages (mak'-rō-fāj-es) Phagocytic cells present in many tissues that function in innate immunity by destroying microorganisms and in acquired immunity as antigen-presenting cells.

Magnoliids Members of the angiosperm clade that is most closely related to the combined eudicot and monocot clades. Extant examples are magnolias, laurels, and black pepper plants.

Major depressive disorder A mood disorder characterized by feelings of sadness, lack of self-worth, emptiness, or loss of interest in nearly all things.

Major histocompatibility complex molecule (MHC) A host protein that functions in antigen presentation. Foreign MHC molecules on transplanted tissue can trigger T cell responses that may lead to rejection of the transplant.

Malignant tumor A cancerous tumor containing cells that have significant genetic and cellular changes and are capable of invading and surviving in new sites. Malignant tumors can impair the functions of one or more organs.

Malnutrition Ingestion of too little or too much of an essential nutrient, or all essential nutrients, over an extended period of time.

Malpighian tubules (mal-pig'-ē-un) Unique excretory organs of insects that empty into the digestive tract, remove nitrogenous wastes from the hemolymph, and function in osmoregulation.

Mammals Members of the clade Mammalia, amniotes that have hair and mammary glands (glands that produce milk).

Mammary glands Exocrine glands that secrete milk for nourishing the young. Mammary glands are characteristic of mammals.

Mantle One of the three main parts of a mollusc; a fold of tissue that drapes over the mollusc's visceral mass and may secrete a shell. *See also* foot and visceral mass.

Mantle cavity A water-filled chamber that houses the gills, anus, and excretory pores of a mollusc.

Map unit A unit of measurement of the distance between genes. One map unit is equivalent to a 1 percent recombination frequency.

Mark-recapture method A sampling technique used to estimate the size of animal populations.

Marsupials (mar-sū'-pē-ul) Mammals, such as a koalas, kangaroos, or opossums, whose young complete their embryonic development inside a maternal pouch called the marsupium.

Mass extinction The elimination of a large number of species throughout Earth, the result of global environmental changes.

Mass number The total number of protons and neutrons in an atom's nucleus.

Mast cells Immune system cells that secrete histamine; play role in inflammatory response and allergies.

Mate-choice copying Behavior in which individuals in a population copy the mate choice of others, apparently due to social learning.

Maternal effect gene A gene that, when mutant in the female parent, results in a mutant phenotype in the offspring, regardless of the offspring's genotype. Maternal effect genes, also called egg-polarity genes, were first identified in *Drosophila melanogaster*.

Matter Anything that takes up space and has mass.

Maximum likelihood As applied to DNA sequence data, a principle that states that when considering multiple phylogenetic hypotheses, one should take into account the hypothesis that reflects the most likely sequence of evolutionary events, given certain rules about how DNA changes over time.

Maximum parsimony The principle that when considering multiple explanations for an observation, one should first investigate the simplest explanation that is consistent with the facts.

Mean The sum of all data points in a data set divided by the number of data points.

Mechanoreceptors Sensory receptors that detect physical deformation in the body's environment associated with pressure, touch, stretch, motion, or sound.

Medulla oblongata (meh-dul'-uh ob'-longgo'-tuh) The lowest part of the vertebrate brain, commonly called the medulla; a swelling of the hindbrain anterior to the spinal cord that controls autonomic, homeostatic functions, including breathing, heart and blood vessel activity, swallowing, digestion, and vomiting.

Medusa (muh-dū'-suh) (plural, **medusae**) The floating, mouth-down form of the cnidarian body plan. The alternate form is the polyp.

Megapascal (MPa) (meg'-uh-pas-kal') A unit of pressure equivalent to about 10 atmospheres of pressure.

Megaphylls (meh'-guh-fils) Leaves with a highly branched vascular system, found in almost all vascular plants other than lycophytes. See also **microphyll**.

Megaspores Spores from a heterosporous plant species that develop into female gametophytes.

Meiosis (mī-ō'-sis) A modified type of cell division in sexually reproducing organisms consisting of two rounds of cell division but only one round of DNA replication. It results in cells with half the number of chromosome sets as the original cell.

Meiosis I (mī-ō'-sis) The first division of a two-stage process of cell division in sexually reproducing organisms that results in cells with half the number of chromosome sets as the original cell.

Meiosis II (mī-ō'-sis) The second division of a two-stage process of cell division in sexually reproducing organisms that results in cells with half the number of chromosome sets as the original cell.

Melanocyte-stimulating hormone (MSH) A hormone produced and secreted by the anterior pituitary with multiple activities, including regulating the behavior of pigment-containing cells in the skin of some vertebrates.

Melatonin A hormone that is secreted by the pineal gland and that is involved in the regulation of biological rhythms and sleep.

Membrane potential The difference in electrical charge (voltage) across a cell's plasma membrane due to the differential distribution of ions. Membrane potential affects the activity of excitable cells and the transmembrane movement of all charged substances.

Memory cells A clone of long-lived lymphocytes, formed during the primary immune response, that remain in a lymphoid organ until activated by exposure to the same antigen that triggered their formation. Activated memory cells mount the secondary immune response.

Menopause The cessation of ovulation and menstruation marking the end of a human female's reproductive years.

Menstrual cycle (men'-strū-ul) In humans and certain other primates, the periodic growth and shedding of the uterine lining that occurs in the absence of pregnancy.

Menstruation The shedding of portions of the endometrium during a uterine (menstrual) cycle.

Meristems (mār'-uh-stems) Plant tissues that remain embryonic as long as the plant lives, allowing for indeterminate growth.

Mesoderm (mez'-ō-derm) The middle primary germ layer in a triploblastic animal embryo; develops into the notochord, the lining of the coelom, muscles, skeleton, gonads, kidneys, and most of the circulatory system in species that have these structures.

Mesohyl (mez'-ō-hil) A gelatinous region between the two layers of cells of a sponge.

Mesophyll (mez'-ō-fil) Leaf cells specialized for photosynthesis. In C₃ and CAM plants, mesophyll cells are located between the upper and lower epidermis; in C₄ plants, they are located between the bundle-sheath cells and the epidermis.

Messenger RNA (mRNA) A type of RNA, synthesized using a DNA template, that attaches to ribosomes in the cytoplasm and specifies the primary structure of a protein. (In eukaryotes, the primary RNA transcript must undergo RNA processing to become mRNA.)

Metabolic pathway A series of chemical reactions that either builds a complex molecule (anabolic pathway) or breaks down a complex molecule to simpler molecules (catabolic pathway).

Metabolic rate The total amount of energy an animal uses in a unit of time.

Metabolism (muh-tab'-uh-lizm) The totality of an organism's chemical reactions, consisting of catabolic and anabolic pathways, which manage the material and energy resources of the organism.

Metagenomics The collection and sequencing of DNA from a group of species, usually an environmental sample of microorganisms. Computer software sorts partial sequences and assembles them into genome sequences of individual species making up the sample.

Metamonada A group of anaerobic, heterotrophic flagellated protists that are free-living or parasitic. This group includes well-known parasites of humans, including *Giardia intestinalis* and *Trichomonas vaginalis*.

Metamorphosis (met'-uh-mōr'-fuh-sis) A developmental transformation that turns an animal larva into either an adult or an adult-like stage that is not yet sexually mature.

Metanephridia (met'-uh-nuh-frid'-ē-uh) (singular, **metanephridium**) Excretory organs found in many invertebrates that typically consist of tubules connecting ciliated internal openings to external openings.

Metaphase The third stage of mitosis, in which the spindle is complete and the chromosomes, attached to microtubules at their kinetochores, are all aligned at the metaphase plate.

Metaphase plate An imaginary structure located at a plane midway between the two poles of a cell in metaphase on which the centromeres of all the duplicated chromosomes are located.

Metapopulation A group of spatially separated populations of one species that interact through immigration and emigration.

Metastasis (muh-tas'-tuh-sis) The spread of cancer cells to locations distant from their original site.

Methanogens (meth-an'-ō-jens) Organisms that produce methane as a waste product of the way they obtain energy. All known methanogens are in domain Archaea.

Methyl group A chemical group consisting of a carbon bonded to three hydrogen atoms. The methyl group may be attached to a carbon or to a different atom.

Microbiome The collection of microorganisms living in or on an organism's body, along with their genetic material.

Microclimate Climate patterns on a very fine scale, such as the specific climatic conditions underneath a log.

Microevolution Evolutionary change below the species level; change in the allele frequencies in a population over generations.

Microfilaments Cables composed of actin proteins in the cytoplasm of almost every eukaryotic cell, making up part of the cytoskeleton and acting alone or with myosin to cause cell contraction; also called actin filaments.

Micronutrients Essential elements that an organism needs in very small amounts.

Microphylls (mī'-krō-fils) Small, usually spine-shaped leaves supported by a single strand of vascular tissue, found only in lycophytes.

Microplastics Plastic particles less than 5 mm in size; microplastics have contaminated all of the world's oceans as well as freshwater and terrestrial ecosystems.

Micropyle A pore in the integuments of an ovule.

MicroRNAs (miRNAs) Small, single-stranded RNA molecules, generated from a double-stranded RNA precursor. The miRNA associates with one or more proteins in a complex that can degrade or prevent translation of an mRNA with a complementary sequence.

Microspores Spores from a heterosporous plant species that develop into male gametophytes.

Microsporidians Members of the fungal phylum Microsporidia, unicellular parasites of protists and animals.

Microtubules Hollow rods composed of tubulin proteins that make up part of the cytoskeleton in all eukaryotic cells and are found in cilia and flagella.

Microvilli (singular, **microvillus**) The many fine, finger-like projections of the epithelial

cells in the lumen of the small intestine that increase its surface area.

Midbrain One of three ancestral and embryonic regions of the vertebrate brain; in human children and adults, the part of the brainstem that receives and integrates sensory information and then transmits it to the cerebrum.

Middle ear One of three main regions of the vertebrate ear; in mammals, a chamber containing three small bones (the malleus, incus, and stapes) that convey vibrations from the eardrum to the oval window.

Middle lamella (luh-mel'-uh) In plants, a thin layer of adhesive extracellular material, primarily pectins, found between the primary walls of adjacent young cells.

Migration A regular, long-distance change in location.

Mineralcorticoids Steroid hormones such as aldosterone that regulate salt and water balance.

Minerals In nutrition, simple nutrients that are inorganic and therefore cannot be synthesized in the body.

Minimum viable population (MVP) The smallest population size at which a species is able to sustain its numbers and survive.

Miscarriage Early pregnancy loss (occurring in the first 20 weeks of gestation).

Mismatch repair The cellular process that uses specific enzymes to remove and replace incorrectly paired nucleotides.

Missense mutations Nucleotide-pair substitutions that result in a codon that codes for a different amino acid.

Mitochondria (mī'-tō-kon'-drē-uh) (singular, **mitochondrion**) Organelles in eukaryotic cells that serve as the site of cellular respiration; in a mitochondrion, oxygen is used to break down organic molecules and synthesize ATP.

Mitochondrial matrix The compartment of the mitochondrion enclosed by the inner membrane and containing enzymes and substrates for the citric acid cycle, as well as ribosomes and DNA.

Mitosis (mī-tō'-sis) A process of nuclear division in eukaryotic cells conventionally divided into five stages: prophase, prometaphase, metaphase, anaphase, and telophase. Mitosis conserves chromosome number by allocating replicated chromosomes equally to each of the daughter nuclei.

Mitotic phase (M) The phase of the cell cycle that includes mitosis and cytokinesis.

Mitotic spindle An assemblage of microtubules and associated proteins that is involved in the movement of chromosomes during mitosis.

Mixotrophs Organisms that are capable of both photosynthesis and heterotrophy.

Model organism A particular species chosen for research into broad biological principles

because it is representative of a larger group and usually easy to grow in a lab.

Molarity A common measure of solute concentration, referring to the number of moles of solute per liter of solution.

Molds Informal term for a fungi that grow as a filamentous fungus, producing haploid spores by mitosis and forming a visible mycelium.

Mole (mol) The number of grams of a substance that equals its molecular or atomic mass in daltons; a mole contains Avogadro's number of the molecules or atoms in question.

Molecular clock A method for estimating the time required for a given amount of evolutionary change, based on the observation that some regions of genomes evolve at constant rates.

Molecular mass The sum of the masses of all the atoms in a molecule; sometimes called molecular weight.

Molecule Two or more atoms held together by covalent bonds.

Molting A process in ecdysozoans in which the exoskeleton is shed at intervals, allowing growth by the production of a larger exoskeleton.

Monilophytes An informal name for members of the phylum Monilophyta, which includes ferns, horsetails, and whisk ferns and their relatives.

Monoclonal antibodies (mon'-ō-klōn'-ul) A preparation of antibodies that have been produced by a single clone of cultured cells and thus are all specific for the same epitope.

Monocots Members of a clade consisting of flowering plants that have one embryonic seed leaf, or cotyledon.

Monogamous (muh-nog'-uh-mus) Referring to a type of relationship in which one male mates with just one female.

Monohybrids Organisms that are heterozygous with respect to a single gene of interest. All the offspring from a cross between parents homozygous for different alleles are monohybrids. For example, parents of genotypes *AA* and *aa* produce a monohybrid of genotype *Aa*.

Monohybrid cross A cross between two organisms that are heterozygous for the character being followed (or the self-pollination of a heterozygous plant).

Monomers (mon'-uh-mers) Subunits that serve as the building block of a polymer.

Monophyletic (mon'-ō-fī-let'-ik) Pertaining to a group of taxa that consists of a common ancestor and all of its descendants. A monophyletic taxon is equivalent to a clade.

Monosaccharides (mon'-ō-sak'-uh-rīds) The simplest carbohydrates, active alone or serving as monomers to build disaccharides and polysaccharides. Also called simple sugars, monosaccharides have molecular formulas that are generally some multiple of CH_2O .

Monosomic Referring to a diploid cell that has only one copy of a particular chromosome instead of the normal two.

Monotremes Egg-laying mammals, such as the platypus or echidna. Like all mammals, monotremes have hair and produce milk, but they lack nipples.

Morphogenesis (mōr'-fō-jen'-uh-sis) The development of the form of an organism and its structures.

Morphogens Substances, such as Bicoid protein in *Drosophila*, that provide positional information in the form of a concentration gradient along an embryonic axis.

Morphological species concept Definition of a species in terms of measurable anatomical criteria.

Mosses Small, herbaceous, nonvascular plants that are members of the phylum Bryophyta.

Motor neurons Nerve cells that transmit signals from the brain or spinal cord to muscles or glands.

Motor proteins Proteins that interact with cytoskeletal elements and other cell components, producing movement of the whole cell or parts of the cell.

Motor system An efferent branch of the vertebrate peripheral nervous system composed of motor neurons that carry signals to skeletal muscles in response to external stimuli.

Motor unit A single motor neuron and all the muscle fibers it controls.

Movement corridor A series of small clumps or a narrow strip of quality habitat (usable by organisms) that connects otherwise isolated patches of quality habitat.

MPF Maturation-promoting factor (or M-phase-promoting factor); a protein complex required for a cell to progress from late interphase to mitosis. The active form consists of cyclin and a protein kinase.

Mucoromycetes Members of the fungal phylum Mucoromycota, characterized by the formation of a sturdy structure called a zygosporangium during sexual reproduction.

Mucus A viscous and slippery mixture of glycoproteins, cells, salts, and water that moistens and protects the membranes lining body cavities that open to the exterior.

Müllerian mimicry (myū-lār'-ē-un mim'-uh-krē) Reciprocal mimicry by two unpalatable species.

Multifactorial Referring to a phenotypic character that is influenced by multiple genes and environmental factors.

Multigene families Collections of genes with similar or identical sequences, presumably of common origin.

Multiple fruit A fruit derived from an entire inflorescence.

Multiplication rule A rule of probability stating that the probability of two or more independent events occurring together can be determined by multiplying their individual probabilities.

Muscle tissue Tissue consisting of long muscle cells that can contract, either on its own or when stimulated by nerve impulses.

Mutagens (myū'-tuh-jens) Chemical or physical agents that interact with DNA and can cause a mutation.

Mutations (myū-tā'-shuns) Changes in the nucleotide sequence of an organism's DNA or in the DNA or RNA of a virus.

Mutualism (myū'-chū-ul-izm) A positive-positive ecological interaction that benefits individuals of both interacting species.

Mycelium (mī-sē'-lē-um) (plural, **mycelia**) The densely branched network of hyphae in a fungus.

Mycorrhizae (singular, **mycorrhiza**) (mī'-kō-rī'-zē, mī'-kō-rī'-zuh) A mutualistic association of plant roots and fungus.

Mycosis (mī-kō'-sis) General term for a fungal infection.

Myelin sheath (mī'-uh-lin) Wrapped around the axon of a neuron, an insulating coat of cell membranes from Schwann cells or oligodendrocytes. It is interrupted by nodes of Ranvier, where action potentials are generated.

Myofibrils (mī'-ō-fī'-bril) Longitudinal bundles in a muscle cell (fiber) that contain thin filaments of actin and regulatory proteins and thick filaments of myosin.

Myoglobin (mī'-uh-glō'-bin) An oxygen-storing, pigmented protein in muscle cells.

Myosin (mī'-uh-sin) A type of motor protein that associates into filaments that interact with actin filaments to cause cell contraction.

Myriapods (mir'-ē-uh-pods') Terrestrial arthropods with many body segments and one or two pairs of legs per segment. Millipedes and centipedes are the two major groups of living myriapods.

NADH The reduced form of nicotinamide adenine dinucleotide that temporarily stores electrons during cellular respiration. NADH acts as an electron donor to the electron transport chain.

NADP⁺ The oxidized form of nicotinamide adenine dinucleotide phosphate, an electron carrier that can accept electrons, becoming NADPH. NADPH temporarily stores energized electrons produced during the light reactions.

NADPH The reduced form of nicotinamide adenine dinucleotide phosphate; temporarily stores energized electrons produced during the light reactions. NADPH acts as "reducing power" that can be passed along to an electron acceptor, reducing it.

Natural killer cells A type of white blood cell that can kill tumor cells and virus-infected cells as part of innate immunity.

Natural selection A process in which individuals that have certain inherited traits tend to

survive and reproduce at higher rates than other individuals because of those traits.

Negative feedback A form of regulation in which accumulation of an end product of a process slows the process; in physiology, a primary mechanism of homeostasis, whereby a change in a variable triggers a response that counteracts the initial change.

Negative pressure breathing A breathing system in which air is pulled into the lungs.

Nematocysts (nem'-uh-tuh-sists') In a cnidocyte of a cnidarian, capsule-like organelles containing a coiled thread that when discharged can penetrate the body wall of the prey.

Nephron (nef'-ron) The tubular excretory unit of the vertebrate kidney.

Neritic zone The water column in the relatively shallow region of the ocean overlying the continental shelf.

Nerve A fiber composed primarily of the bundled axons of neurons.

Nervous system In animals, the fast-acting internal system of communication involving sensory receptors, networks of nerve cells, and connections to muscles and glands that respond to nerve signals; functions in concert with the endocrine system to effect internal regulation and maintain homeostasis.

Nervous tissue tissue made up of neurons and supportive cells.

Net ecosystem production (NEP) The gross primary production of an ecosystem minus the energy used by all autotrophs and heterotrophs for respiration.

Net primary production (NPP) The gross primary production of an ecosystem minus the energy used by the producers for respiration.

Neural crest In vertebrates, a region located along the sides of the neural tube where it pinches off from the ectoderm. Neural crest cells migrate to various parts of the embryo and form pigment cells in the skin and parts of the skull, teeth, adrenal glands, and peripheral nervous system.

Neural tube A tube of infolded ectodermal cells that runs along the anterior-posterior axis of a vertebrate, just dorsal to the notochord. It will give rise to the central nervous system.

Neurohormones Molecules that are secreted by a neuron, travel in body fluids, and act on specific target cells, changing their functioning.

Neuron (nyūr'-on) A nerve cell; the fundamental unit of the nervous system, having structure and properties that allow it to conduct signals by taking advantage of the electrical charge across its plasma membrane.

Neuronal plasticity The capacity of a nervous system to change with experience.

Neuropeptides Relatively short chains of amino acids that serve as a neurotransmitter.

Neurotransmitters Molecules that are released from the synaptic terminal of a neuron at a chemical synapse, diffuse across the synaptic cleft, and bind to the postsynaptic cell, triggering a response.

Neutral variation Genetic variation that does not provide a selective advantage or disadvantage.

Neutrons Subatomic particles having no electrical charge (electrically neutral), found in the nucleus of an atom.

Neutrophils The most abundant type of white blood cell. Neutrophils are phagocytic and tend to self-destruct as they destroy foreign invaders, limiting their life span to a few days.

Nitric oxide (NO) A gas produced by many types of cells that functions as a local regulator and as a neurotransmitter.

Nitrogen cycle The natural process by which nitrogen, either from the atmosphere or from decomposed organic material, is converted by soil bacteria to compounds assimilated by plants. This incorporated nitrogen is then taken in by other organisms and subsequently released, acted on by bacteria, and made available again to the nonliving environment.

Nitrogen fixation The conversion of atmospheric nitrogen to ammonia. Biological nitrogen fixation is carried out by certain prokaryotic organisms, some of which have mutualistic relationships with plants.

No-till agriculture A plowing technique that minimally disturbs the soil, thereby reducing soil loss.

Nociceptors (nō'-si-sep'-tur) Sensory receptors that respond to noxious or painful stimuli; also called pain receptors.

Nodes Points along the stem of a plant at which leaves are attached.

Nodes of Ranvier (ron'-vē-ā') Gaps in the myelin sheath of certain axons where an action potential may be generated. In saltatory conduction, an action potential is regenerated at each node, appearing to "jump" along the axon from node to node.

Nodules A swelling on the root of a legume. Nodules are composed of plant cells that contain nitrogen-fixing bacteria of the genus *Rhizobium*.

Noncompetitive inhibitors Substances that reduce the activity of an enzyme by binding to a location remote from the active site, changing the enzyme's shape so that the active site no longer effectively catalyzes the conversion of substrate to product.

Nondisjunction An error in meiosis or mitosis in which members of a pair of homologous chromosomes or a pair of sister chromatids fail to separate properly from each other.

Nonequilibrium model A model that maintains that communities change constantly after being buffeted by disturbances.

Nonpolar covalent bond A type of covalent bond in which electrons are shared equally between two atoms of similar electronegativity.

Nonsense mutation A mutation that changes an amino acid codon to one of the three stop codons, resulting in a shorter and usually non-functional protein.

Norepinephrine (nor-ep'-i-nef'-rin) A catecholamine that is chemically and functionally similar to epinephrine and acts as a hormone or neurotransmitter; also called noradrenaline.

Northern coniferous forest A terrestrial biome characterized by long, cold winters and dominated by cone-bearing trees.

Notochord (nō'-tuh-kord') A longitudinal, flexible rod made of tightly packed mesodermal cells that runs along the anterior-posterior axis of a chordate in the dorsal part of the body.

Nuclear envelope In a eukaryotic cell, the double membrane that surrounds the nucleus, perforated with pores that regulate traffic with the cytoplasm. The outer membrane is continuous with the endoplasmic reticulum.

Nuclear lamina A netlike array of protein filaments that lines the inner surface of the nuclear envelope and helps maintain the shape of the nucleus.

Nucleariids Members of a group of unicellular, amoeboid protists that are more closely related to fungi than they are to other protists.

Nuclease An enzyme that cuts DNA or RNA, either removing one or a few bases or hydrolyzing the DNA or RNA completely into its component nucleotides.

Nucleic acid hybridization (nū-klā'-ik) The base pairing of one strand of a nucleic acid to the complementary sequence on a strand from another nucleic acid molecule.

Nucleic acid probe (nū'-klā'-ik) In DNA technology, a labeled single-stranded nucleic acid molecule used to locate a specific nucleotide sequence in a nucleic acid sample. Molecules of the probe hydrogen-bond to the complementary sequence wherever it occurs; radioactive, fluorescent, or other labeling of the probe allows its location to be detected.

Nucleic acids (nū-klā'-ik) Polymers (polynucleotides) consisting of many nucleotide monomers; serve as a blueprint for proteins and, through the actions of proteins, for all cellular activities. The two types are DNA and RNA.

Nucleoid (nū'-klē-oyd) A non-membrane-enclosed region in a prokaryotic cell where its chromosome is located.

Nucleolus (nū-klē'-ō-lus) (plural, **nucleoli**) A specialized structure in the nucleus, consisting of chromosomal regions containing ribosomal RNA (rRNA) genes along with ribosomal proteins imported from the cytoplasm; site of rRNA synthesis and ribosomal subunit assembly. *See also* ribosome.

Nucleosomes (nū'-klē-ō-sōms') The basic, bead-like units of DNA packing in eukaryotes, each consisting of a segment of DNA wound around a protein core composed of two copies of each of four types of histone.

Nucleotide excision repair (nū'-klē-ō-tīd') A repair system that removes and then correctly replaces a damaged segment of DNA using the undamaged strand as a guide.

Nucleotide-pair substitution (nū'-klē-ō-tīd') A type of point mutation in which one nucleotide in a DNA strand and its partner in the complementary strand are replaced by another pair of nucleotides.

Nucleotides (nū'-klē-ō-tīds') The building blocks of a nucleic acid, each consisting of a five-carbon sugar covalently bonded to a nitrogenous base and one to three phosphate groups.

Nucleus (1) An atom's central core, containing protons and neutrons. (2) The organelle of a eukaryotic cell that contains the genetic material in the form of chromosomes, made up of chromatin. (3) A cluster of neurons.

Nutrition The process by which an organism takes in and makes use of food substances.

Obligate aerobes (ob'-lig-et uh-rōbs) Organisms that require oxygen for cellular respiration and cannot live without it.

Obligate anaerobes (ob'-lig-et an'-uh-rōbs) Organisms that carry out only fermentation or anaerobic respiration. Such organisms cannot use oxygen and in fact may be poisoned by it.

Ocean acidification The process by which the pH of the ocean is lowered (made more acidic) when excess carbon dioxide dissolves in seawater and forms carbonic acid.

Oceanic zone The water column in most of the ocean's waters beyond the edge of the continental shelf, above the deep ocean floor.

Odorants Molecules that can be detected by sensory receptors of the olfactory system.

Okazaki fragments (ō'-kah-zah'-kē) Short segments of DNA synthesized away from the replication fork on a template strand during DNA replication. Many such segments are joined together to make up the lagging strand of newly synthesized DNA.

Olfaction The sense of smell.

Oligodendrocytes A type of glial cell that forms insulating myelin sheaths around the axons of neurons in the central nervous system.

Ommatidia (ōm'-uh-tīd'-ē-uh) (singular, **ommatidium**) One of the facets of the compound eye of arthropods and some polychaete worms.

Omnivores Animals that regularly eat animals as well as plants or algae.

Oncogenes (on'-kō-jēns) Genes found in viral or cellular genomes that are involved in triggering molecular events that can lead to cancer.

Oocyte (ō'-uh-sīt) A cell in the female reproductive system that differentiates to form an egg.

Oogenesis (ō'-uh-jen'-uh-sis) The process in the ovary that results in the production of female gametes.

Open circulatory system A circulatory system in which fluid called hemolymph bathes the tissues and organs directly and there is no distinction between the circulating fluid and the interstitial fluid.

Operator In bacterial and phage DNA, a sequence of nucleotides near the start of an operon to which an active repressor can attach. The binding of the repressor prevents RNA polymerase from attaching to the promoter and transcribing the genes of the operon.

Operculum (ō-per'-kyuh-lum) In aquatic osteichthyans, a protective bony flap that covers and protects the gills.

Operon (op'-er-on) A unit of genetic function found in bacteria and phages, consisting of a promoter, an operator, and a coordinately regulated cluster of genes whose products function in a common pathway.

Opisthokonts (uh-pis'-thuh-konts') An extremely diverse clade of eukaryotes that includes fungi, animals, and several closely related groups of protists.

Opposable thumb A thumb that can touch the ventral surface (fingerprint side) of the fingertip of all four fingers of the same hand with its own ventral surface.

Opsin A membrane protein bound to a light-absorbing pigment molecule.

Optimal foraging model The basis for analyzing behavior as a compromise between feeding costs and feeding benefits.

Oral cavity The mouth of an animal.

Orbital The three-dimensional space where an electron is found 90 percent of the time.

Orders In Linnaean classification, the taxonomic category above the level of family.

Organ A specialized center of body function composed of several different types of tissues.

Organelles (ōr-guh-nel') Any of several membrane-enclosed structures with specialized functions, suspended in the cytosol of eukaryotic cells.

Organic chemistry The study of carbon compounds (organic compounds).

Organic compounds Chemical compounds that contain carbon.

Organism An individual living thing, consisting of one or more cells.

Organ of Corti (kor'-tē) The actual hearing organ of the vertebrate ear, located in the floor of the cochlear duct in the inner ear; contains the receptor cells (hair cells) of the ear.

Organogenesis (ōr-gan'-ō-jen'-uh-sis) The process in which organ rudiments develop from the three germ layers after gastrulation.

Organ system A group of organs that work together in performing vital body functions.

Origin of replication Site where the replication of a DNA molecule begins, consisting of a specific sequence of nucleotides.

Orthologous genes Homologous genes that are found in different species because of speciation.

Osculum (os'-kyuh-lum) A large opening in a sponge that connects the spongocoel to the environment.

Osmoconformer An animal that is isoosmotic with its environment.

Osmolarity (oz'-mō-lār'-uh-tē) Solute concentration expressed as molarity.

Osmoregulation Regulation of solute concentrations and water balance by a cell or organism.

Osmoregulator An animal that controls its internal osmolarity independent of the external environment.

Osmosis (oz-mō-sis) The diffusion of free water across a selectively permeable membrane.

Osteichthyans (os'-tē-ik'-thē-an) A member of a vertebrate clade with jaws and mostly bony skeletons.

Outer ear One of the three main regions of the ear in reptiles (including birds) and mammals; made up of the auditory canal and, in many birds and mammals, the pinna.

Outgroup A species or group of species from an evolutionary lineage that is known to have diverged before the lineage that contains the group of species being studied. An outgroup is selected so that its members are closely related to the group of species being studied, but not as closely related as any study-group members are to each other.

Oval window In the vertebrate ear, a membrane-covered gap in the skull bone, through which sound waves pass from the middle ear to the inner ear.

Ovarian cycle (ō-vār'-ē-un) The cyclic recurrence of the follicular phase, ovulation, and the luteal phase in the mammalian ovary, regulated by hormones.

Ovary (ō'-vuh-rē) (1) In flowers, the portion of a carpel in which the egg-containing ovules develop. (2) In animals, the structure that produces female gametes and reproductive hormones.

Oviduct (ō'-vuh-duct) A tube passing from the ovary to the vagina in invertebrates or to the uterus in vertebrates, where it is also called a fallopian tube.

Oviparous (ō-vip'-uh-rus) Referring to a type of development in which young hatch from eggs laid outside the female's body.

Ovoviviparous (ō'-vō-vī-vip'-uh-rus) Referring to a type of development in which young hatch from eggs that are retained in the uterus.

Ovulation The release of an egg from an ovary. In humans, an ovarian follicle releases an egg during each uterine (menstrual) cycle.

Ovule (o'-vyūl) A structure that develops within the ovary of a seed plant and contains the female gametophyte.

Oxidation The complete or partial loss of electrons from a substance involved in a redox reaction.

Oxidative phosphorylation (fos'-fōr-uh-lā'-shun) The production of ATP using energy derived from the redox reactions of an electron transport chain; the third major stage of cellular respiration.

Oxidizing agent The electron acceptor in a redox reaction.

Oxytocin (ok'-si-tō'-sen) A hormone produced by the hypothalamus and released from the posterior pituitary. It induces contractions of the uterine muscles during labor and causes the mammary glands to eject milk during nursing.

P53 gene A tumor-suppressor gene that codes for a specific transcription factor that promotes the synthesis of proteins that inhibit the cell cycle.

Paedomorphosis (pē'-duh-mōr'-fuh-sis) The retention in an adult organism of the juvenile features of its evolutionary ancestors.

Pain receptors Sensory receptors that respond to noxious or painful stimuli; also called nociceptors.

Paleoanthropology The study of human origins and evolution.

Paleontology (pā'-lē-un-tol'-ō-jē) The scientific study of fossils.

Pancreas (pan'-krē-us) A gland with exocrine and endocrine tissues. The exocrine portion functions in digestion, secreting enzymes and an alkaline solution into the small intestine via a duct; the ductless endocrine portion functions in homeostasis, secreting the hormones insulin and glucagon into the blood.

Pancrustaceans A diverse arthropod clade that includes lobsters, crabs, barnacles and other crustaceans, as well as insects and their six-legged terrestrial relatives.

Pandemic A global epidemic.

Pangaea (pan-jē'-uh) The supercontinent that formed near the end of the Paleozoic era, when plate movements brought all the landmasses of Earth together.

Paracrine Referring to a secreted molecule that acts on a neighboring cell.

Paralogous genes Homologous genes that are found in the same genome as a result of gene duplication.

Paraphyletic (pār'-uh-fi-let'-ik) Pertaining to a group of taxa that consists of a common ancestor and some, but not all, of its descendants.

Parasite (pār'-uh-sīt) An organism that feeds on the cell contents, tissues, or body fluids of another species (the host) while in or on the host organism. Parasites harm but usually do not kill their host.

Parasitism (pār'-uh-sit-izm) A positive-negative ecological interaction in which one organism, the parasite, benefits by feeding upon another organism, the host, which is harmed; some parasites live within the host (feeding on its tissues), while others feed on the host's external surface.

Parasympathetic division A division of the autonomic nervous system; generally enhances body activities that gain and conserve energy, such as digestion and reduced heart rate.

Parathyroid glands Four small endocrine glands, embedded in the surface of the thyroid gland, that secrete parathyroid hormone.

Parathyroid hormone (PTH) A hormone secreted by the parathyroid glands that raises blood calcium level by promoting calcium release from bone and calcium retention by the kidneys.

Parenchyma cells (puh-ren'-ki-muh) A relatively unspecialized plant cell type that carries out most of the metabolism, synthesizes and stores organic products, and develops into a more differentiated cell type.

Parental types Offspring with a phenotype that matches one of the true-breeding parental (P generation) phenotypes; also refers to the phenotype itself.

Parkinson's disease A progressive brain disease characterized by difficulty in initiating movements, slowness of movement, and rigidity.

Parthenogenesis (par'-thuh-nō'-jen'-uh-sis) A form of asexual reproduction in which females produce offspring from unfertilized eggs.

Partial pressure The pressure exerted by a particular gas in a mixture of gases (for instance, the pressure exerted by oxygen in air).

Passive immunity Short-term immunity conferred by the transfer of antibodies, as occurs in the transfer of maternal antibodies to a fetus or nursing infant.

Passive transport The diffusion of a substance across a biological membrane with no expenditure of energy.

Pathogen-associated molecular pattern (PAMP) A set of molecules characteristic of a group of pathogens that can be recognized by an animal's immune system.

Pathogens Organisms or viruses that cause disease.

Pattern formation The development of a multicellular organism's spatial organization, the arrangement of organs and tissues in their characteristic places in three-dimensional space.

Peat Extensive deposits of partially decayed organic material often formed primarily from the wetland moss *Sphagnum*.

Pedigree A diagram of a family tree with conventional symbols, showing the occurrence of heritable characters in parents and offspring over multiple generations.

Pelagic zone The open-water component of aquatic biomes.

Penis The copulatory structure of male mammals.

PEP carboxylase An enzyme that adds carbon dioxide to phosphoenolpyruvate (PEP) to form oxaloacetate in mesophyll cells of C_4 plants. It acts prior to photosynthesis.

Pepsin An enzyme present in gastric juice that begins the hydrolysis of proteins.

Pepsinogen The inactive form of pepsin secreted by chief cells located in gastric pits of the stomach.

Peptide bond The covalent bond between the carboxyl group on one amino acid and the amino group on another, formed by a dehydration reaction.

Peptidoglycan (pep'-tid-ō-glī'-kan) A type of polymer in bacterial cell walls consisting of modified sugars cross-linked by short polypeptides.

Perception The interpretation of sensory system input by the brain.

Pericycle The outermost layer in the vascular cylinder, from which lateral roots arise.

Periderm (pār'-uh-derm') The protective coat that replaces the epidermis in woody plants during secondary growth, formed of the cork and cork cambium.

Peripheral nervous system (PNS) The sensory and motor neurons that connect to the central nervous system.

Peripheral proteins Proteins loosely bound to the surface of a membrane or to part of an integral protein and not embedded in the lipid bilayer.

Peristalsis (pār'-uh-stal'-sis) (1) Alternating waves of contraction and relaxation in the smooth muscles lining the alimentary canal that push food along the canal. (2) A type of movement on land produced by rhythmic waves of muscle contractions passing from front to back, as in many annelids.

Peristome (pār'-uh-stōme') A ring of interlocking, tooth-like structures on the upper part of a moss capsule (sporangium), often specialized for gradual spore discharge.

Peroxisome (puh-rok'-suh-sōm') An organelle containing enzymes that transfer hydrogen atoms from various substrates to oxygen, producing and then degrading hydrogen peroxide.

Personalized medicine A type of medical care in which each person's specific genetic profile can provide information about diseases or conditions for which the person is especially at risk and help make health-care decisions.

Petals Modified leaves of a flowering plant; the often colorful parts of a flower that advertise it to insects and other pollinators.

Petiole (pet'-ē-ōl) The stalk of a leaf, which joins the leaf to a node of the stem.

P generation The true-breeding (homozygous) parent individuals from which F1 hybrid offspring are derived in studies of inheritance. (P stands for parental.)

pH A measure of hydrogen ion concentration equal to the negative log of the concentration of hydrogen ions and ranging in value from 0 to 14.

Phages (singular, **phage**) (fāj) A virus that infects bacteria; also called a bacteriophage.

Phagocytosis (fag'-ō-sī-tō'-sis) A type of endocytosis in which large particulate substances or small organisms are taken up by a cell. It is carried out by some protists and by certain immune cells of animals (in mammals, mainly macrophages, neutrophils, and dendritic cells).

Pharyngeal clefts (fuh-rin'-jē-ul) In chordate embryos, the grooves that separate a series of arches along the outer surface of the pharynx and may develop into a pharyngeal slit.

Pharyngeal slits (fuh-rin'-jē-ul) In chordate embryos, the narrow openings that form from the pharyngeal clefts and open into the pharynx, later developing into gill slits in many vertebrates.

Pharynx (fār'-inks) (1) An area in the vertebrate throat where air and food passages cross. (2) In flatworms, the muscular tube that protrudes from the ventral side of the worm and ends in the mouth.

Phase changes (1) Shifts from one developmental phase to another. (2) In plants, morphological changes that arise from a transition in shoot apical meristem activity.

Phenotype (fē'-nō-tīp) The observable physical and physiological traits of an organism, which are determined by its genetic makeup.

Phenotypic plasticity The ability of an individual genotype to produce different phenotypes under different environmental conditions.

Pheromones (fār'-uh-mōns) In animals and fungi, small molecules released into the environment that function in communication between members of the same species. In animals, a pheromone acts much like a hormone in influencing physiology and behavior.

Phloem (flō-em) Vascular plant tissue consisting of living cells arranged into elongated tubes that transport sugar and other organic nutrients throughout the plant.

Phloem sap (flō'-em) The sugar-rich solution carried through a plant's sieve tubes.

Phosphate group A chemical group consisting of a phosphorus atom bonded to four oxygen atoms; important in energy transfer.

Phosphodiester linkage In DNA or RNA, the covalent bonds joining the sugars of two

adjacent nucleotides via the phosphate group between them.

Phospholipid (fos'-fō-lip'-id) A lipid made up of glycerol joined to two fatty acids and a phosphate group. The hydrocarbon chains of the fatty acids act as nonpolar, hydrophobic tails, while the rest of the molecule acts as a polar, hydrophilic head. Phospholipids form bilayers that function as biological membranes.

Phosphorylated intermediate (fos'-fōr-uh-lā'-ted) A molecule (often a reactant) with a phosphate group covalently bound to it, making it more reactive (less stable) than the unphosphorylated molecule.

Phosphorylation cascade (fos'-fōr-uh-lā'-shun) A series of chemical reactions during cell signaling mediated by enzymes (kinases), in which each kinase in turn phosphorylates and activates another, ultimately leading to phosphorylation of many proteins.

Photic zone (fō'-tic) The narrow top layer of an ocean or lake, where light penetrates sufficiently for photosynthesis to occur.

Photoautotroph (fō'-tō-ot'-ō-trōf) An organism that harnesses light energy to drive the synthesis of organic compounds from carbon dioxide.

Photoheterotroph (fō'-tō-het'-er-ō-trōf) An organism that uses light to generate ATP but must obtain carbon in organic form.

Photomorphogenesis Effects of light on plant morphology.

Photons (fō'-tons) Quanta, or discrete quantities, of light energy that behave as if they were a particle.

Photoperiodism (fō'-tō-pēr'-ē-ō-dizm) A physiological response to photoperiod, the interval in a 24-hour period during which an organism is exposed to light. An example of photoperiodism is flowering.

Photophosphorylation (fō'-tō-fos'-fōr-uh-lā'-shun) The process of generating ATP from ADP and phosphate by means of chemiosmosis, using a proton-motive force generated across the thylakoid membrane of the chloroplast or the membrane of certain prokaryotes during the light reactions of photosynthesis.

Photoreceptors Electromagnetic receptors that detect the radiation known as visible light.

Photorespiration A metabolic pathway that consumes oxygen and ATP, releases carbon dioxide, and decreases photosynthetic output. Photorespiration generally occurs on hot, dry, bright days, when the stomata close and the oxygen to carbon dioxide ratio in the leaf increases, favoring the binding of oxygen rather than carbon dioxide by rubisco.

Photosynthesis (fō'-tō-sin'-thi-sis) The conversion of light energy to chemical energy that is stored in sugars or other organic compounds; occurs in plants, algae, and certain prokaryotes.

Photosystem A light-capturing unit located in the thylakoid membrane of the chloroplast or in the membrane of some prokaryotes, consisting of a reaction-center complex surrounded by numerous light-harvesting complexes. There are two types of photosystems, I and II; they absorb light best at different wavelengths.

Photosystem I (PS I) A light-capturing unit in a chloroplast's thylakoid membrane or in the membrane of some prokaryotes; it has two molecules of P700 chlorophyll *a* at its reaction center.

Photosystem II (PS II) One of two light-capturing units in a chloroplast's thylakoid membrane or in the membrane of some prokaryotes; it has two molecules of P680 chlorophyll *a* at its reaction center.

Phototropism (fō'-tō-trō'-pizm) The bending of a plant or other organism in response to light, either toward the source of light (positive phototropism) or away from it (negative phototropism).

Phyla (fī'-luh) (singular, **phylum**) In Linnaean classification, the taxonomic category above class.

Phylogenetic tree A branching diagram that represents a hypothesis about the evolutionary history of a group of organisms.

Phylogeny (fī-loj'-uh-nē) The evolutionary history of a species or group of related species.

Physiology The processes and functions of an organism.

Phytochromes (fī'-tuh-krōm) Plant pigments that absorb mostly red and far-red light and regulate many plant responses, such as seed germination and shade avoidance.

Phytoremediation An emerging technology that seeks to reclaim contaminated areas by taking advantage of some plant species' ability to extract heavy metals and other pollutants from the soil and to concentrate them in easily harvested portions of the plant.

Pili (singular, **pilus**) (pī'-lī, pī'-lus) In bacteria, a structure that links one cell to another at the start of conjugation; also called a sex pilus or conjugation pilus.

Pineal gland (pī'-nē-ul) A small gland on the dorsal surface of the vertebrate forebrain that secretes the hormone melatonin.

Pistil A single carpel (a simple pistil) or a group of fused carpels (a compound pistil).

Pith Ground tissue that is internal to the vascular tissue in a stem; in many monocot roots, parenchyma cells that form the central core of the vascular cylinder.

Pituitary gland (puh-tū'-uh-tār'-ē) An endocrine gland at the base of the hypothalamus; consists of a posterior lobe, which stores and releases two hormones produced by the hypothalamus, and an anterior lobe, which produces and secretes many hormones that regulate diverse body functions.

Placenta (pluh-sen'-tuh) A structure in the uterus of a pregnant eutherian mammal that

nourishes the fetus with the maternal blood supply; formed from the uterine lining and embryonic membranes.

Placoderms Members of an extinct group of fishlike vertebrates that had jaws and were enclosed in a tough outer armor.

Planarians Free-living flatworms found in ponds and streams.

Plasma (plaz'-muh) The liquid matrix of blood in which the blood cells are suspended.

Plasma membrane (plaz'-muh) The membrane at the boundary of every cell that acts as a selective barrier, regulating the cell's chemical composition.

Plasmids (plaz'-mids) Small, circular, double-stranded DNA molecules that carry accessory genes separate from those of a bacterial chromosome; in DNA cloning, plasmids are used as vectors carrying up to about 10,000 base pairs (10 kb) of DNA. Plasmids are also found in some eukaryotes, such as yeasts.

Plasmodesmata (singular, **plasmodesma**) An open channel through the cell wall that connects the cytoplasm of adjacent plant cells, allowing water, small solutes, and some larger molecules to pass between the cells.

Plasmogamy (plaz-moh'-guh-mē) In fungi, the fusion of the cytoplasm of cells from two individuals; occurs as one stage of sexual reproduction, followed later by karyogamy.

Plasmolysis (plaz-mol'-uh-sis) A phenomenon in walled cells in which the cytoplasm shrivels and the plasma membrane pulls away from the cell wall; occurs when the cell loses water to a hypertonic environment.

Plastids A family of closely related organelles, including chloroplasts, chromoplasts, and amyloplasts. Plastids are found in cells of photosynthetic eukaryotes.

Plate tectonics The theory that the continents are part of great plates of Earth's crust that float on the hot, underlying portion of the mantle. Movements in the mantle cause the continents to move slowly over time.

Platelets Pinched-off cytoplasmic fragments of a specialized bone marrow cell. Platelets circulate in the blood and are important in blood clotting.

Pleiotropy (plī'-o-truh-pē) The ability of a single gene to have multiple effects.

Pluripotent Describing a cell that can give rise to many, but not all, parts of an organism.

Point mutations Changes in a single nucleotide pair of a gene.

Polar covalent bond A covalent bond between atoms that differ in electronegativity. The shared electrons are pulled closer to the more electronegative atom, making it slightly negative and the other atom slightly positive.

Polar covalent bonds Covalent bonds between atoms that differ in electronegativity. The

shared electrons are pulled closer to the more electronegative atom, making it slightly negative and the other atom slightly positive.

Polar molecule A molecule (such as water) with an uneven distribution of charges in different regions of the molecule.

Pollen grain In seed plants, a structure consisting of the male gametophyte enclosed within a pollen wall.

Pollen tube A tube that forms after germination of the pollen grain and that functions in the delivery of sperm to the ovule.

Pollination (pol'-uh-nā'-shun) The transfer of pollen to the part of a seed plant containing the ovules, a process required for fertilization.

Poly-A tail A sequence of 50 to 250 adenine nucleotides added onto the 3' end of a pre-mRNA molecule.

Polygamous Referring to a type of relationship in which an individual of one sex mates with several of the other.

Polygenic inheritance (pol'-ē-jen'-ik) An additive effect of two or more genes on a single phenotypic character.

Polymer (pol'-uh-mer) A long molecule consisting of many similar or identical monomers linked together by covalent bonds.

Polymerase chain reaction (PCR) (puh-lim'-uh-rās) A technique for amplifying DNA *in vitro* by incubating it with specific primers, a heat-resistant DNA polymerase, and nucleotides.

Polynucleotides (pol'-ē-nū'-klē-ō-tīds) Polymers consisting of many nucleotide monomers in a chain. The nucleotides can be those of DNA or RNA.

Polypeptide (pol'-ē-pep'-tīd) A polymer of many amino acids linked together by peptide bonds.

Polyphyletic (pol'-ē-fī-lef'-ik) Pertaining to a group of taxa that includes distantly related organisms but does not include their most recent common ancestor.

Polyploidy (pol'-ē-ploy'-dē) A chromosomal alteration in which the organism possesses more than two complete chromosome sets. It is the result of an accident of cell division.

Polyps Sessile variants of the cnidarian body plan. The alternate form is the medusa.

Polyribosomes (pol'-ē-rī'-buh-sōms') A group of several ribosomes attached to, and translating, the same messenger RNA molecule; also called polysomes.

Polysaccharides (pol'-ē-sak'-uh-rīds) Polymers of many monosaccharides, formed by dehydration reactions.

Polysomes A group of several ribosomes attached to, and translating, the same messenger RNA molecule; also called polyribosomes.

Polyspermy The fertilization of an egg by more than one sperm.

Polytomy In a phylogenetic tree, a branch point from which more than two descendant lineages emerge. A polytomy indicates that the sequence of divergence and evolutionary relationships between the descendant lineages are unclear.

Pons A portion of the brain that participates in certain automatic, homeostatic functions, such as regulating the breathing centers in the medulla.

Population A group of individuals of the same species that live in the same area and interbreed, producing fertile offspring.

Population dynamics The study of how complex interactions between biotic and abiotic factors influence variations in population size.

Positional information Molecular cues that control pattern formation in an animal or plant embryonic structure by indicating a cell's location relative to the organism's body axes. These cues elicit a response by genes that regulate development.

Positive feedback A form of regulation in which an end product of a process speeds up that process; in physiology, a control mechanism in which a change in a variable triggers a response that reinforces or amplifies the change.

Positive interactions Positive-positive or positive-neutral ecological interactions between individuals of two species in which at least one individual benefits and neither is harmed; positive interactions include mutualism and commensalism.

Positive pressure breathing A breathing system in which air is forced into the lungs.

Posterior pituitary An extension of the hypothalamus composed of nervous tissue that secretes oxytocin and antidiuretic hormone made in the hypothalamus; a temporary storage site for these hormones.

Postzygotic barriers (pōst'-zī-got'-ik) Reproductive barriers that prevent hybrid zygotes produced by two different species from developing into viable, fertile adults.

Potential energy The energy that matter possesses as a result of its location or spatial arrangement (structure).

Predation An interaction in which an individual of one species, the predator, kills and eats an individual of the other species, the prey.

Pregnancy The condition of carrying one or more embryos in the uterus; also called gestation.

Pressure potential(represented by psi sub-script P) A component of water potential that consists of the physical pressure on a solution, which can be positive, zero, or negative.

Prezygotic barriers (prē'-zī-got'-ik) Reproductive barriers that impede mating between species or hinders fertilization if inter-specific mating is attempted.

Primary cell wall In plants, a relatively thin and flexible layer that surrounds the plasma membrane of a young cell.

Primary consumers Herbivores; organisms that eat plants or other autotrophs.

Primary electron acceptor In the thylakoid membrane of a chloroplast or in the membrane of some prokaryotes, a specialized molecule that shares the reaction-center complex with a pair of chlorophyll *a* molecules and that accepts an electron from them.

Primary growth Growth produced by apical meristems, lengthening stems and roots.

Primary immune response The initial adaptive immune response to an antigen, which appears after a lag of about 10 to 17 days.

Primary meristems The three meristematic derivatives (protoderm, procambium, and ground meristem) of an apical meristem.

Primary producers Autotrophs, usually photosynthetic organisms, but in some cases chemosynthetic. Collectively, autotrophs make up the trophic level of an ecosystem that ultimately supports all other levels.

Primary production The amount of light energy converted to chemical energy (organic compounds) by the autotrophs in an ecosystem during a given time period.

Primary structure The level of protein structure referring to the specific linear sequence of amino acids.

Primary succession A type of ecological succession that occurs in an area where there were originally no organisms present and where soil has not yet formed.

Primary transcript An initial RNA transcript from any gene; also called pre-mRNA when transcribed from a protein-coding gene.

Primase An enzyme that joins RNA nucleotides to make a primer during DNA replication, using the parental DNA strand as a template.

Primer A short polynucleotide with a free 3' end, bound by complementary base pairing to the template strand and elongated with DNA nucleotides during DNA replication.

Prions Infectious agents that are misfolded versions of normal cellular proteins. Prions appear to increase in number by converting correctly folded versions of the protein to more prions.

Problem solving The cognitive activity of devising a method to proceed from one state to another in the face of real or apparent obstacles.

Producers Organisms that produce organic compounds from carbon dioxide by harnessing light energy (in photosynthesis) or by oxidizing inorganic chemicals (in chemosynthetic reactions carried out by some prokaryotic organisms).

Production efficiency The percentage of energy stored in assimilated food that is not used for respiration or eliminated as waste.

Products Materials resulting from a chemical reaction.

Progesterone A steroid hormone that contributes to the menstrual cycle and prepares the uterus for pregnancy; the major progestin in mammals.

Prokaryotes Single-celled organisms of the domain Bacteria or Archaea.

Prokaryotic cell (prō'-kār'-ē-ot'-ik) A type of cell lacking a membrane-enclosed nucleus and membrane-enclosed organelles. Organisms with prokaryotic cells (bacteria and archaea) are called prokaryotes.

Prolactin A hormone produced and secreted by the anterior pituitary with a great diversity of effects in different vertebrate species. In mammals, it stimulates growth of and milk production by the mammary glands.

Prometaphase The second stage of mitosis, in which the nuclear envelope fragments and the spindle microtubules attach to the kinetochores of the chromosomes.

Promoter A specific nucleotide sequence in the DNA of a gene that binds RNA polymerase, positioning it to start transcribing RNA at the appropriate place.

Prophage (prō'-fāj) A phage genome that has been inserted into a specific site on a bacterial chromosome.

Prophase The first stage of mitosis, in which the chromatin condenses into discrete chromosomes visible with a light microscope, the mitotic spindle begins to form, and the nucleolus disappears but the nucleus remains intact.

Prostaglandins (pros'-tuh-glan'-dins) A group of modified fatty acids that are secreted by virtually all tissues and that perform a wide variety of functions as local regulators.

Prostate gland (pros'-tāt) A gland in human males that secretes an acid-neutralizing component of semen.

Protease (prō'-tē-āz) An enzyme that digests proteins by hydrolysis.

Protein (prō'-tēn) A biologically functional molecule consisting of one or more polypeptides folded and coiled into a specific three-dimensional structure.

Protein kinase (prō'-tēn kī'-nās) An enzyme that transfers phosphate groups from ATP to a protein, thus phosphorylating the protein.

Protein phosphatases (prō'-tēn fos'-fuh-tās) Enzymes that remove phosphate groups from (dephosphorylate) proteins, often functioning to reverse the effect of a protein kinase.

Proteoglycans (prō'-tē-ō-gli'-kans) Large molecules each consisting of a small core protein with many carbohydrate chains attached, found in the extracellular matrix of animal cells. A proteoglycan may consist of up to 95 percent carbohydrate.

Proteome The entire set of proteins expressed by a given cell, tissue, or organism.

Proteomics (prō'-tē-ō'-miks) The systematic study of sets of proteins and their properties, including their abundance, chemical modifications, and interactions.

Protists An informal term applied to eukaryotes that are not plants, animals, or fungi. Most protists are unicellular, though some are colonial or multicellular.

Proto-oncogenes (prō'-tō-on'-kō-jēn) A normal cellular gene that has the potential to become an oncogene.

Protocells An abiotic precursor of a living cell that had a membrane-like structure and that maintained an internal chemistry different from that of its surroundings.

Proton pump (prō'-ton) An active transport protein in a cell membrane that uses ATP to transport hydrogen ions out of a cell against their concentration gradient, generating a membrane potential in the process.

Proton-motive force (prō'-ton) The potential energy stored in the form of a proton electrochemical gradient, generated by the pumping of hydrogen ions across a biological membrane during chemiosmosis.

Protonema (prō'-tuh-nē'-muh) (plural, **protone-mata**) A mass of green, branched, one-cell-thick filaments produced by germinating moss spores.

Protonephridia (prō'-tō-nuh-frid'-ē-uh) (singular, **protonephridium**) An excretory system, such as the flame bulb system of flatworms, consisting of a network of tubules lacking internal openings.

Protons (prō'-tons) Subatomic particles with a single positive electrical charge, found in the nucleus of an atom.

Protoplast The living part of a plant cell, which also includes the plasma membrane.

Protostome development In animals, a developmental mode distinguished by the development of the mouth from the blastopore; often also characterized by spiral cleavage and by the body cavity forming when solid masses of mesoderm split.

Provirus A viral genome that is permanently inserted into a host genome.

Proximal tubule In the vertebrate kidney, the portion of a nephron immediately downstream from Bowman's capsule that conveys and helps refine filtrate.

Pseudogenes (sū'-dō-jēn) DNA segments that are very similar to a real gene but do not yield a functional product; DNA segments that formerly functioned as genes but have become inactivated in a particular species because of mutation.

Pseudopodia (sū'-dō-pō'-dē-uh) (singular, **pseudopodium**) Cellular extensions of amoeboid cells used in moving and feeding.

P site One of a ribosome's three binding sites for tRNA during translation. The P site holds the tRNA carrying the growing polypeptide chain. (P stands for peptidyl tRNA.)

Pterosaurs Winged reptiles that lived during the Mesozoic era.

Pulmonary circuit The branch of the circulatory system that pumps oxygen-poor blood from the right side of the heart into the capillary beds of the lungs, where oxygen can be picked and carbon dioxide removed.

Pulse The rhythmic bulging of the artery walls with each heartbeat.

Punctuated equilibria In the fossil record, long periods of apparent stasis, in which a species undergoes little or no morphological change, interrupted by relatively brief periods of sudden change.

Punnett square A diagram used in the study of inheritance to show the predicted genotypic results of random fertilization in genetic crosses between individuals of known genotype.

Pupil The opening in the iris, which admits light into the interior of the vertebrate eye. Muscles in the iris regulate its size.

Purines (pyū'-rēns) Two types of nitrogenous bases found in nucleotides, characterized by a six-membered ring fused to a five-membered ring. Adenine (A) and guanine (G) are purines.

Pyrimidine (puh-rim'-uh-dēn) One of two types of nitrogenous bases found in nucleotides, characterized by a six-membered ring. Cytosine (C), thymine (T), and uracil (U) are pyrimidines.

Quantitative characters Heritable features that vary continuously over a range rather than in an either-or fashion.

Radial cleavage A type of embryonic development in deuterostomes in which the planes of cell division that transform the zygote into a ball of cells are either parallel or perpendicular to the vertical axis of the embryo, thereby aligning tiers of cells one above the other.

Radicle An embryonic root of a plant.

Radioactive isotope An isotope (an atomic form of a chemical element) that is unstable; the nucleus decays spontaneously, giving off detectable particles and energy.

Radiolarians A group of protists, mostly marine, with a shell generally made of silica and pseudopodia that radiate from the central body.

Radiometric dating A method for determining the absolute age of rocks and fossils, based on the half-life of radioactive isotopes.

Radula A straplike scraping organ used by many molluscs during feeding.

Ras gene A gene that codes for Ras, a G protein that relays a growth signal from a growth factor receptor on the plasma membrane to a cascade

of protein kinases, ultimately resulting in stimulation of the cell cycle.

Ratites (rat'-it) Members of the group of flightless birds.

Ray-finned fishes Members of the clade Actinopterygii, aquatic osteichthyans with fins supported by long, flexible rays, including tuna, bass, and herring.

Reabsorption In excretory systems, the recovery of solutes and water from filtrate.

Reactants Starting materials in a chemical reaction.

Reaction-center complex A complex of proteins associated with a special pair of chlorophyll *a* molecules and a primary electron acceptor. Located centrally in a photosystem, this complex triggers the light reactions of photosynthesis. Excited by light energy, the pair of chlorophylls donates an electron to the primary electron acceptor, which passes an electron to an electron transport chain.

Reading frame On an mRNA, the triplet grouping of ribonucleotides used by the translation machinery during polypeptide synthesis.

Receptacle The base of a flower; the part of the stem that is the site of attachment of the floral organs.

Receptor potential A graded potential occurring in a receptor cell.

Receptor tyrosine kinase (RTK) A receptor protein spanning the plasma membrane, the cytoplasmic (intracellular) part of which can catalyze the transfer of a phosphate group from ATP to a tyrosine on another protein. Receptor tyrosine kinases often respond to the binding of a signaling molecule by dimerizing and then phosphorylating a tyrosine on the cytoplasmic portion of the other receptor in the dimer.

Recessive allele An allele whose phenotypic effect is not observed in a heterozygote.

Reciprocal altruism Altruistic behavior between unrelated individuals, whereby the altruistic individual benefits in the future when the beneficiary reciprocates.

Recombinant chromosomes Chromosomes that result when crossing over combines DNA from two parents into a single chromosome.

Recombinant DNA molecule A DNA molecule made *in vitro* with segments from different sources.

Recombinant types Offspring whose phenotype differs from that of the true-breeding P generation parents; also refers to the phenotype itself.

Rectum The terminal portion of the large intestine, where the feces are stored prior to elimination.

Red algae A group of photosynthetic protists, named for their color, which results from a red pigment that masks the green of chlorophyll. Most red algae are multicellular and marine.

Redox reactions (rē'-doks) Chemical reactions involving the complete or partial transfer of one or more electrons from one reactant to another; short for **reduction-oxidation** reaction.

Reducing agent The electron donor in a redox reaction.

Reduction The complete or partial addition of electrons to a substance involved in a redox reaction.

Reference genome A complete sequence that researchers agree best represents the genome of a given species, arrived at by sequencing multiple individuals.

Reflexes Automatic reactions to a stimulus, mediated by the spinal cord or lower brain.

Refractory period (rē-frakt'-ōr-ē) A period, immediately following a response to stimulation, during which a cell or organ is unresponsive to further stimulation.

Regulator An animal for which mechanisms of homeostasis moderate internal changes in a particular variable in the face of external fluctuation of that variable.

Regulatory gene A gene that codes for a protein, such as a repressor, that controls the transcription of another gene or group of genes.

Reinforcement In evolutionary biology, a process in which natural selection strengthens prezygotic barriers to reproduction, thus reducing the chances of hybrid formation. Such a process is likely to occur only if hybrid offspring are less fit than members of the parent species.

Relative abundance The proportional abundance of different species in a community.

Relative fitness The contribution an individual makes to the gene pool of the next generation, relative to the contributions of other individuals in the population.

Renal cortex The outer portion of the vertebrate kidney.

Renal medulla The inner portion of the vertebrate kidney, beneath the renal cortex.

Renal pelvis The funnel-shaped chamber that receives processed filtrate from the vertebrate kidney's collecting ducts and is drained by the ureter.

Renin-angiotensin-aldosterone system (RAAS) A hormone cascade pathway that helps regulate blood pressure and blood volume.

Repetitive DNA Nucleotide sequences, usually noncoding, that are present in many copies in a eukaryotic genome. The repeated units may be short and arranged tandemly (in series) or long and dispersed in the genome.

Replication fork A Y-shaped region on a replicating DNA molecule where the parental strands are being unwound and new strands are being synthesized.

Repressor A protein that inhibits gene transcription. In prokaryotes, repressors bind to the DNA in or near the promoter. In eukaryotes, repressors may bind to control elements within enhancers, to activators, or to other proteins in a way that blocks activators from binding to DNA.

Reproductive isolation The existence of biological factors (barriers) that impede members of two species from producing viable, fertile offspring.

Reptile A member of the clade of amniotes that includes tuataras, lizards and snakes, turtles, crocodilians, and birds.

Reservoirs In biogeochemical cycles, locations of a chemical element, consisting of either organic or inorganic materials that are either available for direct use by organisms or unavailable as nutrients.

Residual volume The amount of air that remains in the lungs after forceful exhalation.

Resource partitioning The division of environmental resources by coexisting species such that the niche of each species differs by one or more significant factors from the niches of all coexisting species.

Respiratory pigments Proteins that transport oxygen in blood or hemolymph.

Response (1) In cellular communication, the change in a specific cellular activity brought about by a transduced signal from outside the cell. (2) In feedback regulation, a physiological activity triggered by a change in a variable.

Resting potential The membrane potential characteristic of a nonconducting excitable cell, with the inside of the cell more negative than the outside.

Restriction enzymes Endonucleases (a type of enzyme) that recognize and cut DNA molecules foreign to a bacterium (such as phage genomes). The enzyme cuts at specific nucleotide sequences (restriction sites).

Restriction fragments DNA segments that result from the cutting of DNA by a restriction enzyme.

Restriction site A specific sequence on a DNA strand that is recognized and cut by a restriction enzyme.

Retina (ret'-i-nuh) The innermost layer of the vertebrate eye, containing photoreceptor cells (rods and cones) and neurons; transmits images formed by the lens to the brain via the optic nerve.

Retinal The light-absorbing pigment in rods and cones of the vertebrate eye.

Retrotransposons (re'-trō-trans-pō'-zon) Transposable elements that move within a genome by means of an RNA intermediate, a transcript of the retrotransposon DNA.

Retroviruses (re'-trō-vī'-rus) RNA viruses that replicate by transcribing their RNA into DNA and then inserting the DNA into a cellular

chromosome; an important class of cancer-causing viruses.

Reverse transcriptase (tran-skrip'-tās) An enzyme encoded by certain viruses (retroviruses) that uses RNA as a template for DNA synthesis.

Reverse transcriptase polymerase chain reaction (RT-PCR) A technique for determining expression of a particular gene. It uses reverse transcriptase and DNA polymerase to synthesize cDNA from all the mRNA in a sample and then subjects the cDNA to PCR amplification using primers specific for the gene of interest.

Rhizarians (rī-za'-rē-uhns) One of the three major subgroups for which the SAR eukaryotic supergroup is named. Many species in this clade are amoebas characterized by threadlike pseudopodia.

Rhizobacteria Soil bacteria whose population size is much enhanced in the rhizosphere, the soil region close to a plant's roots.

Rhizoids (rī'-zoyd) Long, tubular single cells or filaments of cells that anchor bryophytes to the ground. Unlike roots, rhizoids are not composed of tissues, lack specialized conducting cells, and do not play a primary role in water and mineral absorption.

Rhizosphere The soil region close to plant roots and characterized by a high level of microbiological activity.

Rhodopsin (rō-dop--sin) A visual pigment consisting of retinal and opsin. Upon absorbing light, the retinal changes shape and dissociates from the opsin.

Ribonucleic acid (RNA) (rī'-bō-nū-klā'-ik) A type of nucleic acid consisting of a polynucleotide made up of nucleotide monomers with a ribose sugar and the nitrogenous bases adenine (A), cytosine (C), guanine (G), and uracil (U); usually single-stranded; functions in protein synthesis, in gene regulation, and as the genome of some viruses.

Ribose The sugar component of RNA nucleotides.

Ribosomal RNAs (rRNA) (rī'-buh-sō'-mul) RNA molecules that, together with proteins, make up ribosomes; the most abundant type of RNA.

Ribosomes (rī'-buh-sōm) Complexes of rRNA and protein molecules that function as sites of protein synthesis in the cytoplasm; each ribosome consists of a large and a small subunit. In eukaryotic cells, each subunit is assembled in the nucleolus. *See also* nucleolus.

Ribozymes (rī'-buh-zīm) RNA molecules that function as an enzyme, such as an intron that catalyzes its own removal during RNA splicing.

RNA interference (RNAi) A mechanism for silencing the expression of specific genes. In RNAi, double-stranded RNA molecules that match the sequence of a particular gene are processed into siRNAs that either block translation or trigger the degradation of the gene's messenger RNA. This happens naturally in some cells and can be carried out in laboratory experiments as well.

RNA polymerase An enzyme that links ribonucleotides into a growing RNA chain during transcription, based on complementary binding to nucleotides on a DNA template strand.

RNA processing Modification of RNA primary transcripts, including splicing out of introns, joining together of exons, and alteration of the 5' and 3' ends.

RNA sequencing (RNA-seq) (RNA-sēk) A method of analyzing large sets of RNAs that involves making cDNAs and sequencing them.

RNA splicing After synthesis of a eukaryotic primary RNA transcript, the removal of portions of the transcript (introns) that will not be included in the mRNA and the joining together of the remaining portions (exons).

Rods Rodlike cells in the retina of the vertebrate eye, sensitive to low light intensity.

Root An organ in vascular plants that anchors the plant and enables it to absorb water and minerals from the soil.

Root cap A cone of cells at the tip of a plant root that protects the apical meristem.

Root hairs Tiny extensions of a root epidermal cell, growing just behind the root tip and increasing surface area for absorption of water and minerals.

Root pressure Pressure exerted in the roots of plants as the result of osmosis, causing exudation from cut stems and guttation of water from leaves.

Root system All of a plant's roots, which anchor it in the soil, absorb and transport minerals and water, and store food.

Rooted Describing a phylogenetic tree that contains a branch point (often, the one farthest to the left) representing the most recent common ancestor of all taxa in the tree.

Rough ER That portion of the endoplasmic reticulum with ribosomes attached.

Round window In the mammalian ear, the point of contact where vibrations of the stapes create a traveling series of pressure waves in the fluid of the cochlea.

R plasmids Bacterial plasmids carrying genes that confer resistance to certain antibiotics.

r-selection Selection for life history traits that maximize reproductive success in uncrowded environments.

Rubisco (rū-bis'-kō) Ribulose biphosphate (RuBP) carboxylase-oxygenase, the enzyme that normally catalyzes the first step of the Calvin cycle (the addition of carbon dioxide to RuBP). When excess oxygen is present or carbon dioxide levels are low, rubisco can bind oxygen, resulting in photorespiration.

Ruminants (rūh'-muh-nents) Cud-chewing animals, such as cows or sheep, with multiple stomach compartments specialized for an herbivorous diet.

Sacculle (sack'-yū-uhl) In the vertebrate ear, a chamber in the vestibule behind the oval window that participates in the sense of balance.

Salicylic acid (sal'-i-sil'-ik) A signaling molecule in plants that may be partially responsible for activating systemic acquired resistance to pathogens.

Salivary glands Glands associated with the oral cavity that secrete substances that lubricate food and begin the process of chemical digestion.

Saltatory conduction (sol'-tuh-tōr'-ē) Rapid transmission of a nerve impulse along an axon, resulting from the action potential jumping from one node of Ranvier to another, skipping the myelin-sheathed regions of membrane.

Salts Compounds resulting from the formation of an ionic bond; also called ionic compounds.

SAR A supergroup of eukaryotes proposed in a current hypothesis of the evolutionary history of eukaryotes that includes a large, extremely diverse collection of protists from three major subgroups: stramenopiles, alveolates, and rhizarians.

Sarcomeres (sar'-kō-mēr) The fundamental, repeating units of striated muscle, delimited by the Z lines.

Sarcoplasmic reticulum (SR) (sar'-kō-plaz'-mikh-tik'-yū-lum) A specialized endoplasmic reticulum that regulates the calcium concentration in the cytosol of muscle cells.

SARS-CoV-2 (Severe Acute Respiratory Syndrome-CoronaVirus-2) The virus that causes COVID-19.

Saturated fatty acid A fatty acid in which all carbons in the hydrocarbon tail are connected by single bonds, thus maximizing the number of hydrogen atoms that are attached to the carbon skeleton.

Savannas A category of grassland biome located in the tropics, characterized by scattered individual trees and large herbivores and maintained by occasional fires and drought.

Scaffolding protein A type of large relay protein to which several other relay proteins are simultaneously attached, increasing the efficiency of signal transduction.

Scanning electron microscope (SEM) A microscope that uses an electron beam to scan the surface of a sample, coated with metal atoms, to study details of its topography.

Scatter plot A graph in which each piece of data is represented by a point. A scatter plot is used when the data for all variables are numerical and continuous.

Schizophrenia (skit'-suh-frē'-nē-uh) A severe mental disturbance characterized by psychotic episodes in which patients have a distorted perception of reality.

Schwann cells A type of glial cell that forms insulating myelin sheaths around the axons of neurons in the peripheral nervous system.

Science An approach to understanding the natural world.

Scion (st'-un) The twig grafted onto the stock when making a graft.

Sclerenchyma cells (skluh-ren'-kim-uh) A rigid, supportive plant cell type usually lacking a protoplast and possessing thick secondary walls strengthened by lignin at maturity.

Scrotum A pouch of skin outside the abdomen that houses the testes; functions in maintaining the testes at the lower temperature required for spermatogenesis.

Second law of thermodynamics The principle stating that every energy transfer or transformation increases the entropy of the universe. Usable forms of energy are at least partly converted to heat.

Second messengers Molecules that relay messages in a cell from a receptor to a target where an action within the cell takes place.

Secondary cell wall In plant cells, a strong and durable matrix that is often deposited in several laminated layers around the plasma membrane and provides protection and support.

Secondary consumers Carnivores that eat herbivores.

Secondary endosymbiosis A process in eukaryotic evolution in which a heterotrophic eukaryotic cell engulfed a photosynthetic eukaryotic cell, which survived in a symbiotic relationship inside the heterotrophic cell.

Secondary growth Growth produced by lateral meristems, thickening the roots and shoots of woody plants.

Secondary immune response The adaptive immune response elicited on second or subsequent exposures to a particular antigen. The secondary immune response is more rapid, of greater magnitude, and of longer duration than the primary immune response.

Secondary oocyte (ō'-uh-sīt) An oocyte that has completed meiosis I.

Secondary production The amount of chemical energy in consumers' food that is converted to their own new biomass during a given time period.

Secondary structure Regions of repetitive coiling or folding of the polypeptide backbone of a protein due to hydrogen bonding between constituents of the backbone (not the side chains).

Secondary succession A type of succession that occurs where an existing community has been cleared by some disturbance that leaves the soil or substrate intact.

Secretion (1) The discharge of molecules synthesized by a cell. (2) The active transport of wastes and certain other solutes from the body fluid into the filtrate in an excretory system.

Seed An adaptation of some terrestrial plants consisting of an embryo packaged along with a store of food within a protective coat.

Seed coat A tough outer covering of a seed, formed from the outer coat of an ovule. In a flowering plant, the seed coat encloses and protects the embryo and endosperm.

Seedless vascular plants An informal name for plants that have vascular tissue but lack seeds. Seedless vascular plants form a paraphyletic group that includes the phyla Lycopphyta (club mosses and their relatives) and Monilophyta (ferns and their relatives).

Selective permeability A property of biological membranes that allows them to regulate the passage of substances across them.

Self-incompatibility The ability of a seed plant to reject its own pollen and sometimes the pollen of closely related individuals.

Semelparity (seh'-mel-pār'-i-tē) Reproduction in which an organism produces all of its offspring in a single event; also called big-bang reproduction.

Semen (sē'-mun) The fluid that is ejaculated by the male during orgasm; contains sperm and secretions from several glands of the male reproductive tract.

Semicircular canals Three-part chamber of the inner ear that functions in maintaining equilibrium.

Semiconservative model Type of DNA replication in which the replicated double helix consists of one old strand, derived from the parental molecule, and one newly made strand.

Semilunar valve A valve located at each exit of the heart, where the aorta leaves the left ventricle and the pulmonary artery leaves the right ventricle.

Seminal vesicles (sem'-i-nul ves'-i-kul) In males, glands that secrete a fluid component of semen that lubricates and nourishes sperm.

Seminiferous tubules (sem'-i-nif'-er-us) Highly coiled tubes in the testis in which sperm are produced.

Senescence (se-nēs'-ens) The programmed death of certain cells or organs or the entire organism.

Sensor In homeostasis, a receptor that detects a stimulus.

Sensory adaptation The tendency of sensory neurons to become less sensitive when they are stimulated repeatedly.

Sensory neurons Nerve cells that receive information from the internal or external environment and transmit signals to the central nervous system.

Sensory reception The detection of a stimulus by sensory cells.

Sensory receptor A specialized structure or cell that responds to a stimulus from an animal's internal or external environment.

Sensory transduction The conversion of stimulus energy to a change in the membrane potential of a sensory receptor cell.

Sepals (sē'-puls) Modified leaves in angiosperms that help enclose and protect a flower bud before it opens.

Septa (singular, **septum**) One of the cross-walls that divide a fungal hypha into cells. Septa generally have pores large enough to allow ribosomes, mitochondria, and even nuclei to flow from cell to cell.

Serial endosymbiosis A hypothesis for the origin of eukaryotes consisting of a sequence of endosymbiotic events in which mitochondria, chloroplasts, and perhaps other cellular structures were derived from small prokaryotic organisms that had been engulfed by larger cells.

Set point In homeostasis in animals, a value maintained for a particular variable, such as body temperature or solute concentration.

Seta (sē'-tuh) (plural, **setae**) The elongated stalk of a bryophyte sporophyte.

Sex chromosomes Chromosomes responsible for determining the sex of an individual; in humans, the X and Y chromosomes.

Sex-linked gene A gene located on either sex chromosome. Most sex-linked genes are on the X chromosome and show distinctive patterns of inheritance; there are very few genes on the Y chromosome.

Sexual dimorphism (dī-mōr'-fizm) Differences between the secondary sex characteristics of males and females of the same species.

Sexual reproduction Reproduction arising from fusion of two gametes.

Sexual selection A process in which individuals with certain inherited characteristics are more likely than other individuals of the same sex to obtain mates.

Shannon diversity index An index of community diversity symbolized by H and represented by the equation $H = -(p_A \ln p_A + p_B \ln p_B + p_C \ln p_C + \dots)$, where A, B, C... are species, p is the relative abundance of each species, and $\ln$ is the natural logarithm.

Shared ancestral character A character, shared by members of a particular clade, that originated in an ancestor that is not a member of that clade.

Shared derived character An evolutionary novelty that is unique to a particular clade.

Shoot system The aerial portion of a plant body, consisting of stems, leaves, and (in angiosperms) flowers.

Short tandem repeats (STRs) Simple sequence DNA containing multiple tandemly repeated units of two to five nucleotides. Variations in STRs act as genetic markers in STR analysis, used to prepare genetic profiles.

Short-day plant A plant that flowers (usually in late summer, fall, or winter) only when the light period is shorter than a critical length.

Short-term memory The ability to hold information, anticipations, or goals for a time and then release them if they become irrelevant.

Sickle-cell disease A recessively inherited human blood disease in which a single nucleotide change in the gene causes hemoglobin to aggregate, changing red blood cell shape and causing multiple symptoms in affected individuals.

Sieve plates The end walls in a sieve-tube element, which facilitate the flow of phloem sap in angiosperm sieve tubes.

Sieve-tube elements A living cell that conducts sugars and other organic nutrients in the phloem of angiosperms; also called a sieve-tube member. Connected end to end, they form sieve tubes.

Sign stimulus An external sensory cue that triggers a fixed action pattern by an animal.

Signal Any kind of information sent from one organism to another, or from one place in an organism to another place.

Signal peptide A sequence of about 20 amino acids at or near the leading (amino) end of a polypeptide that targets it to the endoplasmic reticulum or other organelles in a eukaryotic cell.

Signal reception In cellular communication, the first step of a signaling pathway in which a signaling molecule is detected by a receptor molecule on or in the cell.

Signal-recognition particle (SRP) A protein-RNA complex that recognizes a signal peptide as it emerges from a ribosome and helps direct the ribosome to the endoplasmic reticulum (ER) by binding to a receptor protein on the ER.

Signal transduction The linkage of a mechanical, chemical, or electromagnetic stimulus to a specific cellular response.

Signal transduction pathway A series of steps linking a mechanical, chemical, or electrical stimulus to a specific cellular response.

Silent mutation A nucleotide-pair substitution that has no observable effect on the phenotype; for example, within a gene, a mutation that results in a codon that codes for the same amino acid.

Simple fruit A fruit derived from a single carpel or several fused carpels.

Simple sequence DNA A DNA sequence that contains many copies of tandemly repeated short sequences.

Single bond A single covalent bond; the sharing of a pair of valence electrons by two atoms.

Single circulation A circulatory system consisting of a single pump and circuit, in which blood passes from the sites of gas exchange to the rest of the body before returning to the heart.

Single nucleotide polymorphism (SNP) (snip) A single base-pair site in a genome where nucleotide variation is found in at least 1 percent of the population.

Single-lens eyes The camera-like eyes found in some jellies, polychaete worms, spiders, and many molluscs.

Single-strand binding proteins Proteins that bind to the unpaired DNA strands during DNA replication, stabilizing them and holding them apart while they serve as templates for the synthesis of complementary strands of DNA.

Sinoatrial node (SA) (sī'-nō-ā'-trē-uhl) A region in the right atrium of the heart that sets the rate and timing at which all cardiac muscle cells contract; the pacemaker.

Sister chromatids Two copies of a duplicated chromosome attached to each other by proteins at the centromere and, sometimes, along the arms. While joined, two sister chromatids make up one chromosome. Chromatids are eventually separated during mitosis or meiosis II.

Sister taxa Groups of organisms that share an immediate common ancestor and hence are each other's closest relatives.

Skeletal muscle A type of striated muscle that is generally responsible for the voluntary movements of the body.

Sliding-filament model The idea that muscle contraction is based on the movement of thin (actin) filaments along thick (myosin) filaments, shortening the sarcomere, the basic unit of muscle organization.

Slow-twitch fibers Muscle fibers that can sustain long contractions.

Small interfering RNAs (siRNAs) Small, single-stranded RNA molecules generated by cellular machinery from a long, linear, double-stranded RNA molecule. The siRNA associates with one or more proteins in a complex that can degrade or prevent translation of an mRNA with a complementary sequence.

Small intestine The longest section of the alimentary canal, so named because of its small diameter compared with that of the large intestine; the principal site of the enzymatic hydrolysis of food macromolecules and the absorption of nutrients.

Smooth ER That portion of the endoplasmic reticulum that is free of ribosomes.

Smooth muscle A type of muscle lacking the striations of skeletal and cardiac muscle because of the uniform distribution of myosin filaments in the cells; responsible for involuntary body activities.

Social learning Modification of behavior through the observation of other individuals.

Sodium-potassium pump A transport protein in the plasma membrane of animal cells that actively transports sodium out of the cell and potassium into the cell.

Soil horizons Soil layers with physical characteristics that differ from those of the layers above or beneath them.

Solute (sol'-yūt) A substance that is dissolved in a solution.

Solute potential (represented by psi subscript S) A component of water potential that is proportional to the molarity of a solution and that measures the effect of solutes on the direction of water movement; also called osmotic potential, it can be either zero or negative.

Solution A liquid that is a homogeneous mixture of two or more substances.

Solvent The dissolving agent of a solution. Water is the most versatile solvent known.

Somatic cell (sō-mat'-ik) Any cell in a multicellular organism except a sperm or egg or their precursors.

Somites A series of blocks of mesoderm that exist in pairs just lateral to the notochord in a vertebrate embryo.

Soredia (suh-rē'-dē-uh) (singular, **soredium**) In lichens, small clusters of fungal hyphae with embedded algae.

Sori (singular, **sorus**) A cluster of sporangia on a fern sporophyll. Sori may be arranged in various patterns, such as parallel lines or dots, which are useful in fern identification.

Spatial learning The establishment of a memory that reflects the environment's spatial structure.

Speciation (spē'-sē-ā'-shun) An evolutionary process in which one species splits into two or more species.

Species (spē'-sēz) A population or group of populations whose members have the potential to interbreed in nature and produce viable, fertile offspring but do not produce viable, fertile offspring with members of other such groups.

Species-area curve (spē'-sēz) The biodiversity pattern that shows that the larger the geographic area of a community is, the more species it has.

Species diversity (spē'-sēz) The number and relative abundance of species in a biological community.

Species richness (spē'-sēz) The number of species in a biological community.

Specific heat The amount of heat that must be absorbed or lost for 1 g of a substance to change its temperature by 1 degree C.

Spectrophotometer An instrument that measures the proportions of light of different wavelengths absorbed and transmitted by a pigment solution.

Sperm The male gamete.

Spermathecae (sper'-muh-thē'-kuay) (singular, **spermatheca**) In many insects, a sac in the female reproductive system where sperm are stored.

Spermatogenesis (sper-ma'-tō-gen'-uh-sis) The continuous and prolific production of mature sperm in the testis.

Spermatogonium (sper-ma'-tō-gō'-nē-um) (plural, **spermatogonia**) A cell that divides mitotically to form spermatocytes.

S phase The synthesis phase of the cell cycle; the portion of interphase during which DNA is replicated.

Sphincter (sfink'-ter) A ringlike band of muscle fibers that controls the size of an opening in the body, such as the passage between the esophagus and the stomach.

Spiral cleavage A type of embryonic development in protostomes in which the planes of cell division that transform the zygote into a ball of cells are diagonal to the vertical axis of the embryo. As a result, the cells of each tier sit in the grooves between cells of adjacent tiers.

Spliceosome (splī'-sō-sōm) A large complex made up of proteins and RNA molecules that splices RNA by interacting with the ends of an RNA intron, releasing the intron and joining the two adjacent exons.

Spongocoel (spon'-jō-sēl) The central cavity of a sponge.

Spontaneous process A process that occurs without an overall input of energy; a process that is energetically favorable.

Sporangia (spōr-an'-jē-uh) (singular, **sporangium**) A multicellular organ in fungi and plants in which meiosis occurs and haploid cells develop.

Sporangium (spōr-an'-jē-um) (plural, **sporangia**) A multicellular organ in fungi and plants in which meiosis occurs and haploid cells develop.

Spore (1) In the life cycle of a plant or alga undergoing alternation of generations, haploid cells produced in the sporophyte by meiosis. A spore can divide by mitosis to develop into a multicellular haploid individual, the gametophyte, without fusing with another cell. (2) In fungi, haploid cells, produced either sexually or asexually, that produce a mycelium after germination.

Sporophylls (spō'-ruh-fils) Modified leaves that bear sporangia and hence are specialized for reproduction.

Sporophytes (spō'-ruh-fit) In organisms (plants and some algae) that have alternation of generations, the multicellular diploid form that results from the union of gametes. Meiosis in the sporophyte produces haploid spores that develop into gametophytes.

Sporopollenin (spōr-uh-pol'-eh-nin) A durable polymer that covers exposed zygotes of charophyte algae and forms the walls of plant spores, preventing them from drying out.

Stability In evolutionary biology, a term referring to a hybrid zone in which hybrids continue to be produced; this causes the hybrid zone to be "stable" in the sense of persisting over time.

Stabilizing selection Natural selection in which intermediate phenotypes survive or reproduce more successfully than do extreme phenotypes.

Stamens (stā'-mens) The pollen-producing reproductive organs of a flower, consisting of an anther and a filament.

Standard deviation A measure of the variation found in a set of data points.

Standard metabolic rate (SMR) Metabolic rate of a resting, fasting, and nonstressed ectotherm at a particular temperature.

Starch A storage polysaccharide in plants, consisting entirely of glucose monomers joined by glycosidic linkages.

Start point In transcription, the nucleotide position on the promoter where RNA polymerase begins synthesis of RNA.

Statocysts (stat'-uh-sist') A type of mechanoreceptor that functions in equilibrium in invertebrates by use of statoliths, which stimulate hair cells in relation to gravity.

Statoliths (stat'-uh-liths') (1) In plants, specialized plastids that contain dense starch grains and may play a role in detecting gravity. (2) In invertebrates, dense particles that settle in response to gravity and are found in sensory organs that function in equilibrium.

Stele (stēl) The vascular tissue of a stem or root.

Stem A vascular plant organ consisting of an alternating system of nodes and internodes that support the leaves and reproductive structures.

Stem cell Any relatively unspecialized cell that can produce, during a single division, two identical daughter cells or two more specialized daughter cells that can undergo further differentiation, or one cell of each type.

Steroids Lipids characterized by a carbon skeleton consisting of four fused rings with various chemical groups attached.

Sticky end A single-stranded end of a double-stranded restriction fragment.

Stigma (plural, **stigmata**) The sticky part of a flower's carpel, which receives pollen grains.

Stimulus In feedback regulation, a fluctuation in a variable that triggers a response.

Stipe A stemlike structure of a seaweed.

Stock The plant that provides the root system when making a graft.

Stomach An organ of the digestive system that stores food and performs preliminary steps of digestion.

Stomata (singular, **stoma**) Microscopic pores surrounded by guard cells in the epidermis of leaves and stems that allow gas exchange between the environment and the interior of the plant.

Stramenopiles (strah'-men-ō'-pē-lēs) One of the three major subgroups for which the SAR eukaryotic supergroup is named. This clade arose by secondary endosymbiosis and includes diatoms and brown algae.

Strata (strah'-tuh) (singular, **stratum**) Rock layers formed when new layers of sediment cover older ones and compress them.

Strigolactones A class of plant hormones that inhibit shoot branching, trigger the germination of parasitic plant seeds, and stimulate the association of plant roots with mycorrhizal fungi.

Strobili (singular, **strobilus**) The technical term for a cluster of sporophylls known commonly as a cone, found in most gymnosperms and some seedless vascular plants.

Stroke The death of nervous tissue in the brain, usually resulting from rupture or blockage of arteries in the neck or head.

Stroke volume The volume of blood pumped by a heart ventricle in a single contraction.

Stroma (strō'-muh) The dense fluid within the chloroplast surrounding the thylakoid membrane and containing ribosomes and DNA; involved in the synthesis of organic molecules from carbon dioxide and water.

Stromatolites Layered rock that results from the activities of prokaryotes that bind thin films of sediment together.

Structural isomers Two or more compounds that have the same molecular formula but differ in the covalent arrangements of their atoms.

Style The stalk of a flower's carpel, with the ovary at the base and the stigma at the top.

Substrate The reactant on which an enzyme works.

Substrate feeding Method of ingestion in which an animal lives in or on its food source, eating its way through the food.

Substrate-level phosphorylation (fos'-fōr-uh-lā'-shun) The enzyme-catalyzed formation of ATP by direct transfer of a phosphate group to ADP from an intermediate substrate in catabolism.

Sugar sink A plant organ that is a net consumer or storer of sugar. Growing roots, buds, stems, and fruits are examples of sugar sinks supplied by phloem.

Sugar source A plant organ in which sugar is being produced by either photosynthesis or the breakdown of starch. Mature leaves are the primary sugar sources of plants.

Sulfhydryl group A chemical group consisting of a sulfur atom bonded to a hydrogen atom.

Summation A phenomenon of neural integration in which the membrane potential of the postsynaptic cell is determined by the combined effect of EPSPs or IPSPs produced in rapid succession at one synapse or simultaneously at different synapses.

Suprachiasmatic nucleus (SCN) (sūp'-ruh-kē'-as-ma-tik) A group of neurons in the hypothalamus of mammals that functions as a biological clock.

Surface tension A measure of how difficult it is to stretch or break the surface of a liquid. Water has a high surface tension because of the hydrogen bonding of surface molecules.

Surfactant A substance secreted by alveoli that decreases surface tension in the fluid that coats the alveoli.

Survivorship curve A plot of the number of members of a cohort that are still alive at each age; one way to represent age-specific mortality.

Suspension feeder An animal that feeds by removing suspended food particles from the surrounding medium by a capture, trapping, or filtration mechanism.

Sustainable agriculture Long-term productive farming methods that are environmentally safe.

Sustainable development Development that meets the needs of people today without limiting the ability of future generations to meet their needs.

Swim bladder In aquatic osteichthyans, an air sac that enables the animal to control its buoyancy in the water.

Symbiont (sim'-bē-ont) The smaller participant in a symbiotic relationship, living in or on the host.

Symbiosis An ecological relationship between organisms of two different species that live together in direct and intimate contact.

Sympathetic division A division of the autonomic nervous system; generally increases energy expenditure and prepares the body for action.

Sympatric speciation (sim-pat'-rik) The formation of new species in populations that live in the same geographic area.

Symplast In plants, the continuum of cytosol connected by plasmodesmata between cells.

Synapse (sin'-aps) The junction where a neuron communicates with another cell across a narrow gap via a neurotransmitter or an electrical coupling.

Synapsids (si-nap'-sid) Members of an amniote clade distinguished by having a single hole on each side of the skull. Synapsids include the mammals.

Synapsis (si-nap'-sis) The pairing and physical connection of one duplicated chromosome to its homolog during prophase I of meiosis.

Synaptonemal complex (si-nap'-tuh-nē'-muhl) A zipper-like structure composed of proteins, which connects a chromosome to its homolog tightly along their lengths during part of prophase I of meiosis.

Systematics A scientific discipline focused on classifying organisms and determining their evolutionary relationships.

Systemic acquired resistance A defensive response in infected plants that helps protect healthy tissue from pathogenic invasion.

Systemic circuit The branch of the circulatory system that supplies oxygenated blood to and carries deoxygenated blood away from organs and tissues throughout the body.

Systems biology An approach to studying biology that aims to model the dynamic behavior of whole biological systems based on a study of the interactions among the system's parts.

Systole (sis'-tō-lē) The stage of the cardiac cycle in which a heart chamber contracts and pumps blood.

Systolic pressure Blood pressure in the arteries during contraction of the ventricles.

Taproot A main vertical root that develops from an embryonic root and gives rise to lateral (branch) roots.

Tastants Any chemicals that stimulate the sensory receptors in a taste bud.

Taste buds A collection of modified epithelial cells on the tongue or in the mouth that are receptors for taste in mammals.

TATA box A DNA sequence in eukaryotic promoters crucial in forming the transcription initiation complex.

Taxis (tak'-sis) An oriented movement toward or away from a stimulus.

Taxon (plural, **taxa**) A named taxonomic unit at any given level of classification.

Tay-Sachs disease A human genetic disease caused by a recessive allele for a dysfunctional enzyme, leading to accumulation of certain lipids in the brain. Seizures, blindness, and degeneration of motor and mental performance usually become manifest a few months after birth, followed by death within a few years.

T cells The class of lymphocytes that mature in the thymus; they include both effector cells for the cell-mediated immune response and helper cells required for both branches of adaptive immunity.

Technology The application of scientific knowledge for a specific purpose, often involving industry or commerce but also including uses in basic research.

Telomeres (tel'-uh-mēr) The tandemly repetitive DNA at the end of a eukaryotic chromosome's DNA molecule. Telomeres protect the organism's genes from being eroded during successive rounds of replication; Nucleotide sequences, usually noncoding, that are present in many copies in a eukaryotic genome. The repeated units may be short and arranged tandemly (in series) or long and dispersed in the genome.

Telophase The fifth and final stage of mitosis, in which daughter nuclei are forming and cytokinesis has typically begun.

Temperate broadleaf forest A biome located throughout midlatitude regions where there is sufficient moisture to support the growth of large, broadleaf deciduous trees.

Temperate grasslands A category of terrestrial biome that exists at midlatitude regions and is dominated by grasses and forbs.

Temperate phages Phages that are capable of replicating by either a lytic or lysogenic cycle.

Temperature A measure in degrees of the average kinetic energy (thermal energy) of the atoms and molecules in a body of matter.

Template strand The DNA strand that provides the pattern, or template, for ordering, by complementary base pairing, the sequence of nucleotides in an RNA transcript.

Tendon A fibrous connective tissue that attaches muscle to bone.

Terminator In bacteria, a sequence of nucleotides in DNA that marks the end of a gene and signals RNA polymerase to release the newly made RNA molecule and detach from the DNA.

Territoriality A behavior in which an animal defends a bounded physical space against encroachment by other individuals, usually of its own species.

Tertiary consumers (ter'-shē-ār'-ē) Carnivores that eat other carnivores.

Tertiary structure (ter'-shē-ār'-ē) The overall shape of a protein molecule due to interactions of amino acid side chains, including hydrophobic interactions, ionic bonds, hydrogen bonds, and disulfide bridges.

Testcross Breeding an organism of unknown genotype with a homozygous recessive individual to determine the unknown genotype. The ratio of phenotypes in the offspring reveals the unknown genotype.

Testes (singular, **testis**) The male reproductive organ, or gonad, in which sperm and reproductive hormones are produced.

Testosterone A steroid hormone required for development of the male reproductive system, spermatogenesis, and male secondary sex characteristics; the major androgen in mammals.

Tests In foram protists, porous shells that consist of a single piece of organic material hardened with calcium carbonate.

Tetanus (tet'-uh-nus) The maximal, sustained contraction of a skeletal muscle, caused by a very high frequency of action potentials elicited by continual stimulation.

Tetrapods A vertebrate clade whose members have limbs with digits. Tetrapods include mammals, amphibians, and birds and other reptiles.

Thalamus (thal'-uh-mus) An integrating center of the vertebrate forebrain. Neurons with cell bodies in the thalamus relay neural input to specific areas in the cerebral cortex and regulate what information goes to the cerebral cortex.

Theory An explanation that is broader in scope than a hypothesis, generates new hypotheses, and is supported by a large body of evidence.

Thermal energy Kinetic energy due to the random motion of atoms and molecules; energy in its most random form; Thermal energy in transfer from one body of matter to another.

Thermocline A narrow stratum of abrupt temperature change in the ocean and in many temperate-zone lakes.

Thermodynamics (ther'-mō-dī-nam'-iks) The study of energy transformations that occur in a collection of matter; The principle of conservation of energy: Energy can be transferred and transformed, but it cannot be created or destroyed and the principle stating that every energy transfer or transformation increases the entropy of the universe. Usable forms of energy are at least partly converted to heat.

Thermoreceptors Receptors stimulated by either heat or cold.

Thermoregulation The maintenance of internal body temperature within a tolerable range.

Theropods Members of a group of dinosaurs that were bipedal carnivores.

Thick filaments Filaments composed of staggered arrays of myosin molecules; a component of myofibrils in muscle fibers.

Thigmomorphogenesis (thig'-mō-mor'-phō-gen'-uh-sis) A response in plants to chronic mechanical stimulation, resulting from increased ethylene production. An example is thickening stems in response to strong winds.

Thigmotropism (thig-mō'-truh-pizm) A directional growth of a plant in response to touch.

Thin filaments Filaments consisting of two strands of actin and two strands of regulatory protein coiled around one another; a component of myofibrils in muscle fibers.

Threatened species A species that is considered likely to become endangered in the foreseeable future.

Threshold The potential that an excitable cell membrane must reach for an action potential to be initiated.

Thrombus (plural, **thrombi**) A fibrin-containing clot that forms in a blood vessel and blocks the flow of blood.

Thylakoids (thī'-luh-koyds) Flattened, membranous sacs inside a chloroplast. Thylakoids often exist in stacks called grana that are interconnected; their membranes contain molecular "machinery" used to convert light energy to chemical energy.

Thymus (thī'-mus) A small organ in the thoracic cavity of vertebrates where maturation of T cells is completed.

Thyroid gland An endocrine gland, located on the ventral surface of the trachea, that secretes two iodine-containing hormones, triiodothyronine (T subscript 3) and thyroxine (T subscript 4), as well as calcitonin.

Thyroid hormone Either of two iodine-containing hormones (triiodothyronine and thyroxine) that are secreted by the thyroid gland and that help regulate metabolism, development, and maturation in vertebrates.

Tidal volume The volume of air a mammal inhales and exhales with each breath.

Tight junctions A type of intercellular junction between animal cells that prevents the leakage of material through the space between cells.

Tissue system One or more tissues organized into a functional unit connecting the organs of a plant.

Tissues Integrated groups of cells with a common structure, function, or both.

Tolerance In ecological succession, the situation in which early-arriving species neither help nor hinder the establishment of later-arriving species.

Toll-like receptor (TLR) A membrane receptor on a phagocytic white blood cell that recognizes fragments of molecules common to a set of pathogens.

Tonicity The ability of a solution surrounding a cell to cause that cell to gain or lose water.

Top-down control A situation in which the abundance of organisms at each trophic level is controlled by the abundance of consumers at higher trophic levels; thus, predators limit herbivores, and herbivores limit plants.

Topoisomerase A protein that breaks, swivels, and rejoins DNA strands. During DNA replication, topoisomerase helps to relieve strain in the double helix ahead of the replication fork.

Topsoil A mixture of particles derived from rock, living organisms, and decaying organic material (humus).

Torpor A physiological state in which activity is low and metabolism decreases.

Totipotent (tō'-tuh-pōt'-ent) Describing a cell that can give rise to all parts of the embryo and adult, as well as extraembryonic membranes in species that have them.

Trace elements Elements indispensable for life but required in extremely small amounts.

Trachea (trā'-kē-uh) The portion of the respiratory tract that passes from the larynx to the bronchi; also called the windpipe.

Tracheal system In insects, a system of branched, air-filled tubes that extends throughout the body and carries oxygen directly to cells.

Tracheids (trā'-kē-ids) Long, tapered water-conducting cells found in the xylem of nearly all vascular plants. Functioning tracheids are no longer living.

Trait One of two or more detectable variants in a genetic character.

Trans fats An unsaturated fat, formed artificially during hydrogenation of oils, containing one or more *trans* double bonds.

Transcription The synthesis of RNA using a DNA template.

Transcription factors Regulatory proteins that bind to DNA and affect transcription of specific genes.

Transcription initiation complex The completed assembly of transcription factors and RNA polymerase bound to a promoter.

Transcription unit A region of DNA that is transcribed into an RNA molecule.

Transduction A process in which phages (viruses) carry bacterial DNA from one bacterial cell to another. When these two cells are members of different species, transduction results in horizontal gene transfer; A series of steps linking a mechanical, chemical, or electrical stimulus to a specific cellular response.

Transfer RNA (tRNA) An RNA molecule that functions as a translator between nucleic acid and protein languages by picking up a specific amino acid and carrying it to the ribosome, where the tRNA recognizes the appropriate codon in the mRNA.

Transformation (1) The process by which a cell in culture acquires the ability to divide indefinitely, similar to the division of cancer cells. (2) A change in genotype and phenotype due to the assimilation of external DNA by a cell. When the external DNA is from a member of a different species, transformation results in horizontal gene transfer.

Transgene A gene that has been transferred naturally or by a genetic engineering technique from one organism to another.

Transgenic Pertaining to an organism whose genome contains DNA introduced from another organism of the same or a different species.

Translation The synthesis of a polypeptide using the genetic information encoded in an mRNA molecule. There is a change of "language" from nucleotides to amino acids.

Translocation (1) An aberration in chromosome structure resulting from attachment of a chromosomal fragment to a nonhomologous chromosome. (2) During protein synthesis, the third stage in the elongation cycle, when the RNA carrying the growing polypeptide moves from the A site to the P site on the ribosome. (3) The transport of organic nutrients in the phloem of vascular plants.

Transmembrane proteins A type of integral proteins that span the entire membrane.

Transmission electron microscope (TEM) A microscope that passes an electron beam through very thin sections stained with metal atoms and is primarily used to study the internal structure of cells.

Transpiration The evaporative loss of water from a plant.

Transport epithelia (singular, **transport epithelium**) One or more layers of specialized epithelial cells that carry out and regulate solute movement.

Transport proteins Transmembrane proteins that help a certain substance or class of closely related substances to cross the membrane.

Transport vesicles Small membranous sacs in a eukaryotic cell's cytoplasm carrying molecules produced by the cell.

Transposable elements Segments of DNA that can move within the genome of a cell by means of a DNA or RNA intermediate; also called transposable genetic elements.

Transposons Transposable elements that move within a genome by means of a DNA intermediate.

Transverse tubules (T tubules) Infoldings of the plasma membrane of skeletal muscle cells.

Trichomes In plants, epidermal cells that are highly specialized, often hairlike outgrowths on a plant shoot.

Triple response A plant growth maneuver in response to mechanical stress, involving slowing of stem elongation thickening of the stem, and a curvature that causes the stem to start growing horizontally.

Triplet code A genetic information system in which a series of three-nucleotide-long words specifies a sequence of amino acids for a polypeptide chain.

Triploblastic Possessing three germ layers: the endoderm, mesoderm and ectoderm. All bilaterian animals are triploblastic.

Trisomic Referring to a diploid cell that has three copies of a particular chromosome instead of the normal two.

Trochophore larva (trō'-kuh-för) Distinctive larval stage observed in some lophotrochozoan animals, including some annelids and molluscs.

Trophic efficiency The percentage of production transferred from one trophic level to the next higher trophic level.

Trophic level The position an organism occupies in a food chain.

Trophic structure The different feeding relationships in an ecosystem, which determine the route of energy flow and the pattern of chemical cycling.

Trophoblast The outer epithelium of a mammalian blastocyst. It forms the fetal part of the placenta, supporting embryonic development but not forming part of the embryo proper.

Tropism A growth response that results in the curvature of whole plant organs toward or away from stimuli due to differential rates of cell elongation.

Tropomyosin The regulatory protein that blocks the myosin-binding sites on actin molecules.

Troponin complex The regulatory proteins that control the position of tropomyosin on the thin filament.

True-breeding Referring to organisms that produce offspring of the same variety over many generations of self-pollination.

Tubal ligation A means of sterilization in which the two oviducts (fallopian tubes) are tied closed and a segment of each is removed to prevent eggs from reaching the uterus.

Tube feet The many extensions of an echinoderm's water vascular system. Tube feet function in locomotion and feeding.

Tumor-suppressor genes Genes whose protein product inhibits cell division, thereby preventing the uncontrolled cell growth that contributes to cancer.

Tundra A terrestrial biome at the extreme limits of plant growth. At the northernmost limits, it is called arctic tundra, and at high altitudes, where plant forms are limited to low shrubby or matlike vegetation, it is called alpine tundra.

Tunicates Members of the clade Urochordata, sessile marine chordates that lack a backbone.

Turgid (ter'-jid) Swollen or distended, as in plant cells. (A walled cell becomes turgid if it has a lower water potential than its surroundings, resulting in entry of water.)

Turgor pressure The force directed against a plant cell wall after the influx of water and swelling of the cell due to osmosis.

Turnover The mixing of waters as a result of changing water-temperature profiles in a lake.

Tympanic membrane Another name for the eardrum, the membrane between the outer and middle ear.

Unsaturated fatty acid A fatty acid that has one or more double bonds between carbons in the hydrocarbon tail. Such bonding reduces the number of hydrogen atoms attached to the carbon skeleton.

Urban ecology The study of organisms and their environment in urban and suburban settings.

Urea A soluble nitrogenous waste produced in the liver by a metabolic cycle that combines ammonia with carbon dioxide.

Ureter (yū-rē'-ter) A duct leading from the kidney to the urinary bladder.

Urethra (yū-rē'-thruh) A tube that releases urine from the mammalian body near the vagina in females and through the penis in males; also serves in males as the exit tube for the reproductive system.

Uric acid A product of protein and purine metabolism and the major nitrogenous waste product of insects, land snails, and many reptiles. Uric acid is relatively nontoxic and largely insoluble in water.

Urinary bladder The pouch where urine is stored prior to elimination.

Uterine cycle The cyclic changes in the endometrium (uterine lining) of mammals that occur in the absence of pregnancy. In certain primates, including humans, the uterine cycle is a menstrual cycle.

Uterus A female organ where eggs are fertilized and/or development of the young occurs.

Utricle (yū'-trick-uhl) In the vertebrate ear, a chamber in the vestibule behind the oval window that opens into the three semicircular canals.

Vaccine A harmless variant or derivative of a pathogen that stimulates a host's immune system to mount defenses against the pathogen.

Vacuoles (vak'-yū-ōls') Membrane-bounded vesicles whose specialized function varies in different kinds of cells.

Vagina Part of the female reproductive system between the uterus and the outside opening; the birth canal in mammals. During copulation, the vagina accommodates the male's penis and receives sperm.

Valence The bonding capacity of a given atom; the number of covalent bonds that an atom can form, which usually equals the number of unpaired electrons in its outermost (valence) shell.

Valence electrons Subatomic particles having no electrical charge (electrically neutral), found in the nucleus of an atom.

Valence shell The outermost energy shell of an atom, containing the valence electrons involved in the chemical reactions of that atom.

Van der Waals interactions Weak attractions between molecules or parts of molecules that result from transient local partial charges.

Variables Factors that vary in an experiment.

Variation Differences between members of the same species.

Vas deferens In mammals, the tube in the male reproductive system in which sperm travel from the epididymis to the urethra.

Vasa recta The capillary system in the kidney that serves the loop of Henle.

Vascular cambium A cylinder of meristematic tissue in woody plants that adds layers of secondary vascular tissue called secondary xylem (wood) and secondary phloem.

Vascular plants Plants with vascular tissue. Vascular plants include all living plant species except liverworts, mosses, and hornworts.

Vascular tissue Plant tissue consisting of cells joined into tubes that transport water and nutrients throughout the plant body.

Vasectomy The cutting and sealing of each vas deferens to prevent sperm from entering the urethra.

Vasoconstriction A decrease in the diameter of blood vessels caused by contraction of smooth muscles in the vessel walls.

Vasodilation An increase in the diameter of blood vessels caused by relaxation of smooth muscles in the vessel walls.

Vasopressin (an'-tī-dī-yū-ret'-ik) A peptide hormone, also called vasopressin, that promotes water retention by the kidneys. Produced in the hypothalamus and released from the posterior pituitary, ADH also functions in the brain.

Vector An organism that transmits pathogens from one host to another.

Vegetal pole The point at the end of an egg in the hemisphere where most yolk is concentrated; opposite of animal pole.

Vegetative propagation Asexual reproduction in plants that is facilitated or induced by humans.

Vegetative reproduction Asexual reproduction in plants.

Veins (1) In animals, vessels that carry blood toward the heart. (2) In plants, vascular bundles in a leaf.

Ventilation The flow of air or water over a respiratory surface.

Ventral In an animal with bilateral symmetry, pertaining to the underside (in most animals) or front (in animals with upright posture) of the body.

Ventricles (ven-'tri-kul) (1) Heart chambers that pump blood out of the heart. (2) Spaces in the vertebrate brain, filled with cerebrospinal fluid.

Venules (ven-'yūl) Vessels that convey blood between a capillary bed and a vein.

Vernalization The use of cold treatment to induce a plant to flower.

Vertebrates Chordate animals with vertebrae, the series of bones that make up the backbone.

Vesicles (ves-'i-kuls) Membrane-bound sacs in or outside a cell.

Vessel A continuous water-conducting micro-pipe found in most angiosperms and a few nonflowering vascular plants.

Vessel elements Short, wide, water-conducting cells found in the xylem of most angiosperms and a few nonflowering vascular plants. Dead at maturity, vessel elements are aligned end to end to form micropipes called vessels.

Vestigial structures Features of an organism that are historical remnants of structures that served a function in the organism's ancestors.

Villi (singular, **villus**) (1) A finger-like projection of the inner surface of the small intestine. (2) A finger-like projection of the chorion of the mammalian placenta. Large numbers of villi increase the surface areas of these organs.

Viral envelopes In viruses, membranes, derived from membranes of the host cell, that cloak the capsid, which in turn encloses a viral genome.

Virulent phage A phage that replicates only by a lytic cycle.

Virus An infectious particle incapable of replicating outside of a cell, consisting of an RNA or DNA genome surrounded by a protein coat (capsid) and, for some viruses, a membranous envelope.

Visceral mass (1) The portion of a bryophyte sporophyte that gathers sugars, amino acids, water, and minerals from the parent gametophyte via transfer cells. (2) One of the three main parts of a mollusc; a muscular structure usually used for movement.

Visible light That portion of the electromagnetic spectrum that can be detected as various colors by the human eye, ranging in wavelength from about 380 nm to about 740 nm.

Vital capacity The maximum volume of air that a mammal can inhale and exhale with each breath.

Vitamins Organic molecules that are required in the diet in very small amounts. Many vitamins serve as coenzymes or parts of coenzymes.

Viviparous (vī-vip'-uh-rus) Referring to a type of development in which the young are born alive after having been nourished in the uterus by blood from the placenta.

Voltage-gated ion channel A specialized ion channel that opens or closes in response to changes in membrane potential.

Vulva Collective term for the female external genitalia.

Water potential (represented by the Greek letter psi) The physical property predicting the direction in which water will flow, governed by solute concentration and applied pressure.

Water vascular system A network of hydraulic canals unique to echinoderms that branches into extensions called tube feet, which function in locomotion and feeding.

Wavelength The distance between crests of waves, such as those of the electromagnetic spectrum.

Wetlands Habitats that are inundated by water at least some of the time and that support plants adapted to water-saturated soil.

White matter Tracts of axons within the CNS.

Whole-genome shotgun approach Procedure for genome sequencing in which the genome is randomly cut into many overlapping short segments that are sequenced; computer software then assembles the complete sequence.

Wild type The phenotype most commonly observed in natural populations; also refers to the individual with that phenotype.

Wilting The drooping of leaves and stems as a result of plant cells becoming flaccid.

Wobble Flexibility in the base-pairing rules in which the nucleotide at the 5' end of a tRNA anticodon can form hydrogen bonds with more than one kind of base in the third position (3' end) of a codon.

X-linked genes Genes located on the X chromosome; such genes show a distinctive pattern of inheritance.

X-ray crystallography A technique used to study the three-dimensional structure of molecules. It depends on the diffraction of an X-ray beam by the individual atoms of a crystallized molecule.

Xerophytes (zir'-ō-fits') Plants adapted to an arid climate.

Xylem (zī-lum) Vascular plant tissue consisting mainly of tubular dead cells that conduct most of the water and minerals upward from the roots to the rest of the plant.

Xylem sap (zī'-lum) The dilute solution of water and minerals carried through vessels and tracheids.

Yeasts Single-celled fungi. Yeasts reproduce asexually by binary fission or by the pinching of small buds off a parent cell. Many fungal species can grow both as yeasts and as a network of filaments; relatively few species grow only as yeasts.

Yolk Nutrients stored in an egg.

Zona pellucida The extracellular matrix surrounding a mammalian egg.

Zone of polarizing activity (ZPA) A block of mesoderm located just under the ectoderm where the posterior side of a limb bud is attached to the body; required for proper pattern formation along the anterior-posterior axis of the limb.

Zoned reserve An extensive region that includes areas relatively undisturbed by humans surrounded by areas that have been changed by human activity and are used for economic gain.

Zoonotic pathogens Disease-causing agents that are transmitted to humans from other animals.

Zoopagomycetes Members of the fungal phylum Zoopagomycota, multicellular parasites or commensal symbionts of animals; sexual reproduction, where known, involves the formation of a sturdy structure called a zygosporangium.

Zoospores Flagellated spores found in chytrid fungi and some protists.

Zygosporangium (zī'-guh-spōr-an'-jē-um) (plural, **zygosporangia**) In zygomycete fungi, a sturdy multinucleate structure in which karyogamy and meiosis occur.

Zygote (zī'-gōt) The diploid cell produced by the union of haploid gametes during fertilization; a fertilized egg.

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